# Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator

B. Müller Mario Poneder Alexander Osika Kai Xue Daniel Ari Friedman

November 7, 2026

**Paper release:** r2062 **Released:** November 7, 2026

**Author affiliations:** Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

**Abstract**

Observer Patch Holography proposes that particles are reproducible features of the records of one specified observer, repair and memory process, and that stable massive matter consists of localized defects that repair cannot erase. We give the particle content of that process and compare it with measurement. Complete-response premises fix the Standard Model gauge type, and a declared lift gives one chiral generation. On a specified interacting cube, exact enclosures give 228 charged lines visible to single-fermion probes and 138 composite lines invisible to them, and a compiled instrument detects the composite response. The same localized modes carry all twelve native currents and a pixel-dependent charged-lepton matrix realized by complete memory dynamics. A compact opposite-winding texture has finite energy and four isolated modes, and its quartic extension has exact interacting poles. The specified mass branch gives twelve comparison values. Charged leptons lie within 71 parts per million under a pole premise and a target-informed law; the W, Z, Higgs and top lie within one percent; five lighter quarks miss by 23 to 209 percent. All twelve lie outside their printed intervals. We prove the exclusions this comparison supports and list what a source-selected spectrum must supply.

## 1 The particle question

Observer Patch Holography (OPH) proposes a self-referential complete history: observers infer the specification of their world from shared records and realize it through operations within that same history. The executable version specifies a carrier Hilbert space, accessible algebras, repair instruments, a scheduler, memory, feedback and preparation services [2, 6]. These are permitted design choices. Operational agreement concerns accepted shared data; state selection and endogenous implementation of response transformations are additional stated principles, recorded with the exact theorem premises in the common registry [30].

The source specification is a tuple

$$\mathcal{S} = (\mathcal{H}, \mathcal{A}, \{\Phi_{e,r}\}, \sigma, \mathcal{M}, \mathcal{F}, \mathcal{P}),$$

where $\Phi_{e,r}$ is the operation on seam $e$ with retained outcome $r$, $\sigma$ selects operations, $\mathcal{M}$ is memory, $\mathcal{F}$ is feedback, and $\mathcal{P}$ is the preparation domain. A particle assertion concerns the joint and

conditional laws of actual preparations, interventions and detector outputs of $\mathcal{S}$. Two equal spectra do not identify the states, currents, clocks or instruments that produced them, and Section 7 proves this in several exact forms.

**Hypothesis 1.1** (Matter as localized defects)**.** *Stable massive matter admits a localized excitation of a continuing source reference, with an invariant preserved by the admitted dynamics. Its creation, transport, interactions and detection are operations of the same source. Its total excess energy and propagation use one energy and clock dictionary. Unstable species may be metastable or internal excitations; massless radiation need not carry a defect invariant.*

The hypothesis does not ask for a distinct conserved topological charge on every mass eigenstate. Flavor mixing and decay can connect internal states of a defect family, and a protected label cannot change at an admitted vertex without another field or record carrying the compensating charge. Section 2.5 gives the exact constraint.

## 1.1 Main results

Table 1 states what this paper establishes. Three kinds of statement occur and we keep them apart. Some results follow from the unchanged source with every outcome retained. Some are exact on a chosen design, whose inputs are stated. Some compare a specified branch with measurement, and these are conditional postdictions.

Table 1: Main results and the inputs each one consumes.

| **Result** | **Inputs** |
|---|---|
| Internal charges of one chiral generation, three Higgs lines and the conditional quadratic mass table (Section 2) | Complete response, inner rotations, spin lift, three family copies, canonical kinetic forms |
| Complete transfer spectrum of the native carrier, every compact core among 4096 port sets, a positive-excess phase excitation and a positive distance from every quadratic (Section 3.1) | Uniform seam selection and the fair coin; no fitted coefficient |
| Certified charged spectra of the interacting cube, 228 linear lines and 138 composite lines, and a compiled composite detector (Section 3.2) | Couplings $c = 1/8$, $U = 1$; imported Lieb uniqueness; pulse services |
| Complete classification of conserved quadratics (Theorem 3.1) | Open connected box, positive hopping and repulsion |
| A complete memory realization of the charged coefficients and certified pixel feedback roots (Section 4) | Target-informed mass law; declared probes, gain and scale design |
| One localized algebra for currents and masses, a compact finite-energy pair, a gauge-covariant paired band, exact interacting poles and a controlled work operator (Section 5) | Chosen textures, coherent pairing service and quartic coupling |
| Twelve comparison values against the 2026 Particle Data Group tables (Section 6) | Target-informed branch, external unit, supersymmetric running proxy, imposed criticality |
| Exact exclusions of mass attachments and record-based selection rules (Section 7) | Each states its own domain |

No result in this paper derives a measured particle mass, and the construction is a precise conditional particle model. Section 8 lists what a source-selected spectrum must supply.

# Contents


**1 The particle question** **1**
1.1 Main results . . . 2

**2 Internal quantum numbers** **4**
2.1 Complete response and the compact Lie algebra . . . 4
2.2 The matter module and its family factor . . . 5
2.3 The conditional field action . . . 5
2.4 Fermion operators, gauge links and the regulator boundary . . . 6
2.5 Flavor, neutrinos and conserved defect labels . . . 7

**3 The spectrum of the native source** **7**
3.1 Transfer modes, compact cores and a phase excitation . . . 8
3.2 The charged spectrum of the interacting cube . . . 8
3.3 Complete conserved quadratics . . . 9
3.4 One-body massless and massive sectors . . . 10

**4 The charged mass operator** **11**
4.1 Pixel capacity and the mass law . . . 11
4.2 A pixel mass law in native memory dynamics . . . 11
4.3 Memory correlations beyond the mean unitary . . . 13
4.4 Complete records, capacity and recovery . . . 13
4.5 Flavor from current-gated memory . . . 14
4.6 Pixel feedback with one charged-fermion mass prescription . . . 18

**5 Localizing the same current and mass operators** **19**
5.1 Topology, total energy and the radial source . . . 19
5.2 One local carrier for defect states, currents and mass coefficients . . . 20
5.3 Compact opposite-degree textures and their isolated modes . . . 22
5.4 Protected texture records and controlled band transport . . . 23
5.5 Gauge-covariant paired compact bands . . . 24
5.6 Native record response . . . 24
5.7 An observable native contact mode . . . 32
5.8 Connected contact response . . . 33
5.9 Positive-energy correlations from open records . . . 34
5.10 Controlled transfer state and time . . . 37
5.11 Local preparation and readout on one attempt clock . . . 38
5.12 Complete local records at growing capacity . . . 39
5.13 Rooted source refinement . . . 40
5.14 Native clock observables . . . 41
5.15 Protected charge and its native reference . . . 44
5.16 A quartic interaction on the complete compact band . . . 52
5.17 Work on the moving compact interacting band . . . 55
5.18 A finite gauge–Higgs response with complete detector records . . . 56
5.19 Compensated probe uncertainty and collision records . . . 58

**6 Comparison with the measured spectrum** **60**
6.1 Observables and conventions . . . 60
6.2 The twelve comparison values . . . 60
6.3 Charged leptons . . . 61
6.4 Quarks . . . 62
6.5 Weak bosons, Higgs and top . . . 62
6.6 Massless and neutral sectors . . . 62

**7 What records and symmetries exclude** **63**
7.1 Coherent phases and family averaging . . . 63
7.2 The passive family metric . . . 63
7.3 Neutral hierarchy . . . 64
7.4 Residues, references and mass readouts . . . 64
7.5 Comparing capacities . . . 65

**8 The catalogue and what a source-selected spectrum must supply** **65**
8.1 What the displayed source assigns . . . 65
8.2 The required attachments . . . 66
8.3 Interaction, chiral continuum and responsive geometry . . . 68

**9 Finite selection tests for particle readout** **68**
9.1 Complete interacting records on the specified finite source . . . 70
9.2 The full scalar ground and its occupation records . . . 71
9.3 Gauge-coordinate cutoffs and recorded evolution . . . 72
9.4 Electric sign response and a smooth interacting action . . . 74
9.5 Position and momentum on the referenced field . . . 75
9.6 A propagation estimate uniform in physical refinement . . . 76
9.7 An interacting thermal preparation with complete flux records . . . 76
9.8 Gap-free low-energy preparation and signed detector records . . . 77
9.9 The common quadratic gauge characteristic . . . 78
9.10 Spatial strain and pressure from the chosen action . . . 79
9.11 Finite holonomy and shared event meaning . . . 80
9.12 Energy conservation and the geometric premises . . . 81

# 2 Internal quantum numbers

## 2.1 Complete response and the compact Lie algebra

The carrier is the oriented icosahedral boundary, with twelve ports, thirty seams and twenty faces. Its real port module under the proper rotation group $A_5$ decomposes as

$$V_{12} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \qquad \dim V_{12}^{A_5} = 1.$$

Assume a faithful real-linear response map $D : V_{12} \to \mathfrak{u}(\mathcal{K})$ whose image is the complete connected response algebra and is closed under commutators, with proper carrier rotations acting through the same connected inner response group up to operationally silent centralizer factors. The gauge paper [5] proves that the compact classification then gives

$$\mathfrak{g} \simeq \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3), \tag{1}$$

because inner rotations fix a centre of dimension at most one and a centre-free $\mathfrak{su}(2)^4$ cannot have a one-dimensional fixed space, which leaves the semisimple split $11 = 3 + 8$. It also constructs an explicit current law on the protected native carrier realizing $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Twelve ports alone do not imply (1), and carrier rechartings are distinct from physical spatial rotations of a relativistic scattering theory. This paper consumes the result and its twelve current matrices.

### 2.2 The matter module and its family factor

A declared spin lift to five internal coordinates and the nonvacuum even exterior module give

$$M_{15} = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5, \qquad \dim M_{15} = 10 + 5 = 15,$$

the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ branching [9]. The source-current lift preserves all brackets and has an exact intertwiner with this module. Its simultaneous primitive-charge and color/weak Casimir readouts give the following all-left convention.

| Label | Color | Weak | $6Y$ | Components |
|---|---|---|---|---|
| $Q$ | $\mathbf{3}$ | $\mathbf{2}$ | 1 | 6 |
| $u^c$ | $\overline{\mathbf{3}}$ | $\mathbf{1}$ | $-4$ | 3 |
| $d^c$ | $\overline{\mathbf{3}}$ | $\mathbf{1}$ | 2 | 3 |
| $L$ | $\mathbf{1}$ | $\mathbf{2}$ | $-3$ | 2 |
| $e^c$ | $\mathbf{1}$ | $\mathbf{1}$ | 6 | 1 |

The labels abbreviate computed finite quantum numbers; their identification with laboratory Weyl fields requires the physical current, spin and kinetic maps. The gauge paper [5] gives the anomaly scan, which selects this module among exterior masks up to conjugation, the $\mathbb{Z}_6$ global form and the conditional family theorem. For the particle construction we explicitly choose $M_{15} \otimes \mathbb{C}^3$, with gauge currents acting trivially on $\mathbb{C}^3$. A two-dimensional radial kernel or a winding number three is no proof of three generations, and equivalent triplet representations on spatial, color and family factors do not identify those factors.

Exchange records have a separate finite route. On the native ordered-cover transport, let $p_h(x)$ be the permutation parity accumulated by branch $h$. A coherent ordering bit gives

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I \otimes X, \qquad [K_h, Q] = 0.$$

An odd exchange loop acts as $X$ on the bit, and the corresponding colored fermionic lift acts as $-$SWAP. The sectors survive complete record-adaptive native schedules in this specified memory design; dephasing or resetting the bit changes that conclusion. These are exchange-history sectors, distinct from the spatially localized texture charge of Section 5.

### 2.3 The conditional field action

Choose one complex weak doublet $H$ of hypercharge $1/2$, canonical positive kinetic forms, physical Weyl statistics and a Lorentz-covariant local field interpretation in $3 + 1$ dimensions. Restrict the polynomial action to operator dimension at most four, modulo integration by parts. The invariant grammar then gives the gauge kinetic terms, scalar kinetic term and potential, fermion kinetic terms and exactly three charged Yukawa lines. Independent gauge couplings, topological theta

terms and an additive vacuum-energy constant are also free. In the four-component convention the charged Yukawa part is

$$\mathcal{L}_Y = -\overline{Q}_L \widetilde{H} Y_u u_R - \overline{Q}_L H Y_d d_R - \overline{L}_L H Y_e e_R + \text{h.c.}$$

Each $Y_a$ is a complex $3 \times 3$ matrix. The native invariant-line computation solves all twelve current equations for both Higgs components and gives one line for each allowed origin $(6Y_1, 6Y_2) = (-4, 1), (1, 2), (-3, 6)$. Tensoring these lines with the gauge-trivial family factor leaves exactly the three matrix coefficients. Canonical kinetic normalization must precede any reading of their singular values as action masses [5, 26, 25].

For the explicitly chosen broken potential

$$V(H) = \lambda_H (H^\dagger H - v^2/2)^2, \qquad \lambda_H > 0,$$

the vacuum shell has $|H| = v/\sqrt{2}$. Taking $H = (0, v/\sqrt{2})$ gives the following conditional quadratic table.

| Field directions | Action mass or coefficient |
|---|---|
| Three charged leptons | $m_{e,i} = v\, s_i(Y_e)/\sqrt{2}$ |
| Three up and three down flavors | $m_{a,i} = v\, s_i(Y_a)/\sqrt{2}$, each in three colors |
| Three minimal left neutrino fields | Zero in the dimension-four charged grammar |
| Charged weak vectors | $m_W = g_2 v/2$ |
| Neutral massive weak vector | $m_Z = v\sqrt{g_2^2 + g_Y^2}/2$ |
| Electromagnetic direction | Zero quadratic mass |
| Eight color directions | Zero quadratic mass; no free physical gluon claim |
| One radial scalar | $m_h = \sqrt{2\lambda_H}\, v$ |

The neutral vector matrix follows from the coefficient of $\frac{1}{2} A^T M^2 A$ in $|DH|^2$; this factor of two matters. The three zero scalar Hessian directions are gauge-orbit directions. Neither the sign of the broken potential nor its quantum vacuum is selected by the finite representation.

For a supplied Weyl principal symbol and the antisymmetric pairing $\epsilon_{\text{spin}} \otimes \overline{K}$, squaring the complete Nambu symbol cancels its momentum–mass cross terms, so

$$E_j(\mathbf{p})^2 = c^2 |\mathbf{p}|^2 + s_j(K)^2.$$

This is a theorem of that supplied relativistic action. The spatial source of Section 5 has additive bands instead, and the formula does not transfer to it by matching $K$ alone.

### 2.4 Fermion operators, gauge links and the regulator boundary

The internal permutation $P_{15}$ lifts to $P_{90} = I_2 \otimes P_{15} \otimes I_3$ on the configured spin, internal and family factors. Its exterior lift obeys, on every occupation sector,

$$\Gamma(P_{90}) a^\dagger(f) \Gamma(P_{90})^* = a^\dagger(P_{90} f), \qquad [d\Gamma(T_A), a^\dagger(f)] = a^\dagger(T_A f).$$

Thus the same twelve currents act on the full odd-CAR field; auxiliary even occupation masks do not supply its statistics. Exact source controls also intertwine the three Weyl axes and all complex

Higgs coefficients [1]. For the sine regulator the eight nodes have chiralities $\chi_b = -(-1)^{b_1+b_2+b_3}$. Opposite-winding particles carry the same internal representation, whereas holes carry its conjugate: $\mathrm{Tr}_{15}\, Y^5 = 5/9$ changes sign under conjugation. This fifth trace distinguishes representations; it is not an anomaly coefficient. The complete local free-field addition/removal law assigns weight 1/8 to each whole valley cell. A valley projector fails to commute with local hypercharge, so selecting such a detector branch does not remove mirrors from a local gauge theory.

There is also a finite interacting gauge attachment. On the specified sixty-state links, the hopping $\sum c_x^\dagger (ic\sigma_j/2) U_{xy} c_y + \text{h.c.}$ commutes with every full-Fock local Gauss generator. The same fifteen-state matrices have zero linear and symmetric cubic anomaly tensors, counting the internal Weyl representation once rather than its two spinor rows. These are exact finite identities. The quantum-link raising operators are nilpotent and cannot be replaced by identity backgrounds to import the free dispersion. A compatible gauge state, continuum scaling and mirror treatment are additional premises for an interacting chiral field interpretation.

### 2.5 Flavor, neutrinos and conserved defect labels

The canonical family-frame freedom is $\mathrm{U}(3)^5$, and it classifies every same-current, canonical-kinetic family embedding. It preserves the charged singular spectra and basis-invariant mixing data. In particular $V_{\mathrm{CKM}} = U_{uL}^\dagger U_{dL}$ is determined by the same left-handed weak current; relabeling families cannot repair a mass hierarchy or generate physical CP violation. A real symmetric coefficient class has $\mathrm{tr}[Y_u Y_u^\dagger, Y_d Y_d^\dagger]^3 = 0$ [24].

The native current also supplies one $LH$ invariant line, and its symmetric family lift enters the complete action with six free complex Weinberg coefficients. A supplied dimension-five operator $(L\cdot H)^T C_5 (L\cdot H)/(2\Lambda) + \text{h.c.}$ gives $M_\nu = v^2 C_5/(2\Lambda)$ and Takagi singular masses. A calibrated instance has one massless neutrino and two nonzero coefficients. The charged-lepton construction supplies neither $C_5$ nor its scale.

Conserved additive defect labels must respect the same vertices. If the charged-current mixing graph is connected, the relation

$$q(W^-) - q(e_\alpha) + q(\nu_i) = 0$$

for every nonzero entry forces one common neutrino label $N$ and one common charged-lepton label. A massive Majorana field adds $2N = 0$, which in an integer winding group implies $N = 0$. If a massless neutrino is excluded from defect assignments and has label zero, connected mixing forces the other neutrino labels to zero as well. Distinct protected winding numbers therefore cannot simply be assigned to the three mixed mass eigenstates. Internal excitations, nonadditive invariants and charged compensating vertices require their own source construction.

## 3 The spectrum of the native source

Before any mass law is chosen, the source has spectra of its own. This section collects them. No coefficient here is fitted to a particle, every energy is in the source's own dimensionless unit, and each result retains the complete state space on which it is stated.

The native source acts on twelve nonnegative integer port loads of total two, which form 78 configurations with an invariant shell of 66 binary ones. For an oriented seam $(a, b)$ with contrast $d = x_a - x_b$ and a fair placement coin $c$, repair moves $j = \frac{1}{2}[d - (2c-1)(|d| \bmod 2)]$ units from $a$ to $b$. Instrument C records $(c, j)$ and instrument E also records the seam-total parity; they share one classical update and differ in the coherence they retain [6].

### 3.1 Transfer modes, compact cores and a phase excitation

Keep all sixty-six binary total-two configurations. Uniform independent seam selection and fair placement give the complete attempt transfer $P = C/60$, with $2I/3 \le P \le I$, from all 3960 endpoint branches. Exact characteristic polynomial factorization, independently checked by sixty-six Newton traces, finds sixteen distinct eigenvalues and a unique constant ground vector. The largest nontrivial eigenvalue is $(55 + \sqrt{5})/60$.

A compact amplitude core means that both loads lie in a selected port set. The largest confined invariant subspace is the kernel of the outgoing block and all its products with powers of the internal block. Exhausting all 4096 subsets, with exact rank certificates, gives no nonzero such subspace on nine or fewer ports. Six ten-port cores support three modes, with transfer eigenvalues 4/5, 5/6, 13/15 and integrated normalized native autocorrelations 5, 6, 15/2 attempts. Admitted changes of seam weights destroy their confinement, and the full recorded instruments do not conserve their transfer sector. A common invariant amplitude subspace under every native seam uses all twelve ports.

The faithful reconstructed reference $\rho = P/\operatorname{Tr} P$, with $\operatorname{Tr} P = 56$, has a different localized excitation. Let $D = (-1)^{n_0}$ and $\rho' = D\rho D$. It has the same entropy, configuration probabilities and full eleven-port exterior as $\rho$. For the source-derived $H = -\log P$,

$$\frac{1}{1008} \le \operatorname{Tr}(H(\rho' - \rho)) \le \frac{1}{672}.$$

The bounds follow from $\|[P, D]\|_{\mathrm{HS}}^2 = 1/9$ and the positive spectral bounds on $P$. The native C and E unread channels contract the state contrast by 5/6 and 7/10 per uniform attempt, and complete tape-plus-exterior laws agree when compared with the equally evolved reference. The excitation is the reconstructed functional $A \mapsto \ell(DAD)$ and a decaying finite excitation; no native phase gate or coherent preparation is inferred from it.

Interaction is detectable without a fitted spectrum. Every quadratic operator in this fixed occupation dictionary has diagonal $K_{\{i,j\},\{i,j\}} = c + \epsilon_i + \epsilon_j$, regardless of its hopping terms, so its four-pair contrast on $01, 23, 02, 13$ vanishes. For $I - P$ the same contrast is $-1/15$, which puts it at norm distance at least $1/60$ from every such quadratic. A positive logarithm expansion with an explicit tail bounds the distance for $-\log P$ below by 0.01984562. The test retains every native state and permits arbitrary quadratic hoppings, on-site potentials and an energy offset. Arbitrary re-encodings lie outside it.

### 3.2 The charged spectrum of the interacting cube

The open eight-site Weyl/contact cube is the smallest interacting source with a complete charged spectrum. In its exact hidden-spin frame the source is

$$K = \mathrm{d}\Gamma(A \otimes I_2)/16 + \sum_x n_{x\uparrow} n_{x\downarrow} - N/2, \qquad A = A^T, \quad A^2 = 3I,$$

at $c = 1/8$, $U = 1$. All source matrices come from the full 65,536-state integer Hamiltonian. The reference is the unique half-filled ground, with hidden spin zero, eta zero and energy in $[-4.12405873337830, -4.12405873337828]$. Its uniqueness uses the theorem of Lieb on the half-filled Hubbard model [10] together with a full-number-sector bound. Hidden spin and eta are internal algebras, and neither carries a Lorentz-spin assignment.

**Lines visible to single-fermion probes.** Every local one-spin addition or removal measure has exactly 228 distinct positive energies, each with rigorously positive weight. The signed retarded measure has 456 atoms, each charge branch has total weight 1/2, and every eight-site residue matrix has visible rank four per hidden spin. The smallest weight is near $5.63 \times 10^{-21}$, and no threshold discards a line.

**Lines invisible to them.** Combine a local fermion removal with a neighboring hidden-spin flip,

$$C_x = \Big(\sum_{y\sim x} S_y^+\Big) c_{x\downarrow}, \qquad S_y^+ = c_{y\uparrow}^\dagger c_{y\downarrow}, \qquad \|C_x\| \le 3.$$

On the singlet reference the output has total hidden spin 3/2 and eta 1/2. The linear probes have hidden spin 1/2 and $K$ commutes with hidden spin, so the complete dynamical cyclic spaces of the two responses are orthogonal. Exact integer residuals certify all 138 spectral lines in the composite probe's reducing block, each with strictly positive weight, and direct calculations of $C^\dagger C$ and $C^\dagger[K, C]$ on the unreduced source independently enclose the total weight and energy moment.

Table 2: Certified charged lines of the centered cube, in source-clock energy units. The first six of each spectrum are shown; the complete tables retain every line and weight.

| Line | Linear probe | Relative to first | Composite probe | Relative to first |
|---|---|---|---|---|
| 0 | 0.361682711806 | 1.000000000 | 0.402468515770 | 1.000000000 |
| 1 | 0.402293044056 | 1.112281652 | 0.422089756328 | 1.048752237 |
| 2 | 0.415174263333 | 1.147896346 | 0.438990742922 | 1.090745551 |
| 3 | 0.438196496809 | 1.211549467 | 0.479938217500 | 1.192486365 |
| 4 | 0.462192207579 | 1.277894111 | 0.485785522474 | 1.207014967 |
| 5 | 0.474429777956 | 1.311729210 | 0.494035650369 | 1.227513783 |
| Largest | 3.886434754951 | | 2.974756508168 | |
| Conditional mean | 0.49955163275 | | 0.52537938729 | |

Each composite residue matrix is twice its local weight times one of the source's two kinetic-band projectors, of exact rank four, with 69 lines on each band. Spin and eta ladders give 2,208 visible composite states per charge sector. Their orthogonal union with the linear probes covers 4,032 of the 11,440 states of that number sector, and the block therefore does not exhaust every charged composite. Independent source equations and all 192 spatial energy moments retain the contact contribution and the kinetic/contact interference. The finite spatial projectors supply no continuous momentum dispersion.

The composite response can be read. A compiled pulse instrument implements the success Kraus operator $C_0/3$ in 1,008 source pulses on the authenticated hidden-frame ground preparation, retaining all 7,776 raw selector records, native aborts and coherent field-reference amplitudes. Its success probability is approximately 0.08261623248916, one ninth of the certified composite spectral weight. It detects the total response; no clock resolving the 138 energies is executed.

These lines give one clean exclusion. The visible energy spread is below eleven, so no three distinct lines with one positive unit reproduce the charged-lepton mass ratios. That excludes this finite gap dictionary, and it leaves other sources, refinements and unseen composites untouched.

### 3.3 Complete conserved quadratics

The labels of these spectra come from a complete classification.

**Theorem 3.1** (Complete quadratic conservation)**.** *On a finite connected open rectangular box with $V$ sites and two fermion modes per site, let*

$$H_\mu = \sum_{\langle x,y=x+e_j\rangle} c_x^\dagger(\tfrac{ic}{2}\sigma_j)c_y + \text{h.c.} + U\sum_x n_{x\uparrow}n_{x\downarrow} - \mu N, \qquad c, U > 0.$$

*Modulo scalars, the real space of self-adjoint even conserved operators of degree at most two is spanned by $N - V$ and $S_1, S_2, S_3$ when $\mu \neq U/2$, and by these together with $\eta_x, \eta_y$ when $\mu = U/2$, where*

$$S_a = \tfrac{1}{2}\sum_x (-1)^{\sum_{j\neq a} x_j} c_x^\dagger \sigma_a c_x, \qquad \eta_+ = \sum_x (-1)^{x_1+x_2+x_3} c_{x\uparrow}^\dagger c_{x\downarrow}^\dagger.$$

*The traceless conserved Lie algebra is hidden $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ for $H_0$ and hidden $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ for the centered generator $K = H_{U/2}$.*

*Proof.* Commutators with $H_\mu$ preserve the number change of each term, so the number-preserving and anomalous parts separate. The one-particle restriction gives $[h, q] = 0$ for a one-body matrix $q$, hence kinetic commutation on the whole Fock algebra and commutation with the doublon count. An input with one spectator and its doublon output isolate each intersite coefficient and force it to vanish. The kinetic condition $q_y = \sigma_j q_x \sigma_j$ then transports one Hermitian root matrix over the connected box, which gives the four stated coefficients. For a pair creator, comparing the vacuum-to-pair entry with its spectator entry gives $U\Delta_{pq} = 0$ on distinct sites. Onsite terms obey $(U - 2\mu)f_x = 0$, and at $\mu = U/2$ the kinetic condition forces the alternating profile, whose real and imaginary parts are $\eta_x$ and $\eta_y$ [11]. □

Exact full-CAR coefficient ranks and independent occupation-matrix checks verify the finite instances, and centering changes the conserved pairing algebra as well as the charged energies. Higher-degree currents, Lorentz stress and an interacting chiral continuum are outside the theorem. The radial defect source has its own classification: every conserved sum of onsite even operators is eta-pairing SU(2), and on the mass-64 source every center Pauli current has leakage norm above 1/64 outside the low quartet. The complete native color, weak and hypercharge action therefore cannot be inherited from that onsite-even algebra.

### 3.4 One-body massless and massive sectors

The same source has exact one-body sectors. On the periodic cube of eight sites per axis a complete spectral detector resolves all 1,024 one-particle modes, including sixteen zeros and six pairs of signed shells, with every pointer and abort outcome retained. A stationary one-particle state cannot occupy only six sites, and the infinite lattice has no square-summable stationary one-particle state.

With centered clock $H-(U/2)N$ and $c = v/a$, the continuum is the free massless field in all eight Weyl valleys, $E = v|\mathbf{p}|$, throughout $0 \leq U$, $U = o(c^{2/7})$ [6]. Supplied quadratic mass coefficients give two Dirac copies at each rest energy $|m_0 \pm |\mathbf{m}||$, with continuum dispersion $E^2 = v^2|\mathbf{p}|^2 + M^2$ and a proved finite-spacing error. This uniform family has at most two distinct positive rest energies, so a direct assignment cannot reproduce three distinct charged-lepton masses. The ordered-spin continuum has nonzero positive-energy bands at every small momentum. Complete energy–symmetry readouts bound the original response near a specified dispersion, with aliases, clock leakage and all failures retained. Selection-field and zero-field histories have a same-input comparison bound; a finite prepared state is not identified with the infinite ordered vacuum. Relaxed inverse susceptibility determines an infinitesimal-response cost, not a particle mass. A uniform nonvanishing shrinking-tube bound, Lorentz covariance and surviving continuum interactions are unproved. Matching a light speed fixes length unit $= cT$; determining $T$ requires an independently identified same-generator clock.

# 4 The charged mass operator

## 4.1 Pixel capacity and the mass law

The pixel variable is the dimensionless capacity ratio $P = a_{\mathrm{cell}}/\ell_P^2$, distinct from a projector $P_j$ in the memory controller. The declared self-reference relation reads

$$P = \varphi + \sqrt{\pi}\,\alpha, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

In the heat/running construction the inner response supplying $\alpha$ depends on $P$ through a unified scale, a heat-sum equation, low-energy couplings and charged masses, so the loop has the form

$$P \longrightarrow (\alpha_U, v, \text{field census}) \longrightarrow (M_f, J_f) \longrightarrow A_{\mathrm{response}} \longrightarrow \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{response}}}.$$

The same masses and current normalization appear on both sides. A fixed point of a declared response proves consistency of that map, and its identification with the physical electromagnetic response is a further claim.

The controller is a native family memory. It stores one of twelve port addresses, each with a rank-one family projector $P_j$. A C event rotates the port label, a joint event then applies the rotation of E, and other records reset the address to port two. The fair-coin rates of the C and E events are $p = 1/12$ and $q = 1/6$, and all 128 raw record labels are retained. Declared current probes lower these rates to the values that follow.

The mass law used here is a chosen response architecture with icosahedral registers and determinant normalization. Its charged shape uses a three-phase Koide-type parametrization [7, 8]. The architecture has target-informed provenance: its numerical comparisons are conditional postdictions even when its execution reads no measured masses. The quark exponents, scalar quartic, neutrino operator and physical flavor mixing are not determined from the premises A1–A3 of the registry [30]. Independent validation requires consequences not used to choose it. The operator realization below applies to the displayed $\beta = 4$ branch and to the normalized coefficients of the explicit $\beta = 7$ branch.

## 4.2 A pixel mass law in native memory dynamics

A chosen simulator specification can realize the conditional pixel-dependent charged mass law on the native gauge-family carrier. The construction uses the twelve-port reset controller and the complete native current instruments. Its design inputs are independent left and right probe durations, a rule enabling feedback on a successful C event, and a common feedback amplitude. The conditional mass-law architecture is target-informed; evaluating that architecture without a charged-mass input does not establish a prospective prediction.

The selected response architecture gives

$$\kappa = \frac{\alpha_U/50}{1+\alpha_U/31+\alpha_U^2/310}, \qquad \chi = \frac{\alpha_U^2/512}{1-\alpha_U/77}, \tag{2}$$

$$\zeta = \frac{(P-\varphi)^2/(21\pi)}{1-\alpha_U/27-\alpha_U^2/135}, \qquad r_j = 1+\sqrt{2}e^{-\chi}\cos\left(\frac{2}{9}+\zeta+\frac{2\pi j}{3}\right), \tag{3}$$

where $\varphi = (1+\sqrt{5})/2$ and $j = 0, 1, 2$. Its charged coefficients, ordered increasingly, are

$$t_i = \frac{m_i}{v} = \frac{e^{\kappa} r_i^2}{\left(2\,6^{14}\prod_{j=0}^{2} r_j^2\right)^{1/3}}, \qquad \frac{v}{E_*} = P^{-1/2}\exp\left(-\frac{2\pi}{4\alpha_U}\right). \tag{4}$$

The register dimensions and determinant exponent in these formulas are part of the selected architecture. The scale $E_*$ is a source coordinate; assigning it an energy in laboratory units requires a clock calibration.

Let $t_i > 0$ be these three charged coefficients and $\lambda_i = t_i / \sum_j t_j$. For the native projectors $P_2, P_3, P_{10}$, direct multiplication gives $\operatorname{tr}(P_i P_j) = 1/5$ for distinct indices and the spanning-vector Gram determinant $d = 2/5 - 2\sqrt{5}/25$. Consequently the matrix $Y = uP_2 + v_1 P_3 + wP_{10}$ has elementary eigenvalue invariants

$$\operatorname{tr} Y = u + v_1 + w, \qquad e_2(Y) = \frac{4}{5}(uv_1 + uw + v_1 w), \qquad \det Y = duv_1 w. \tag{5}$$

Choose its three positive weights as the roots of

$$z^3 - z^2 + \frac{5}{4} e_2(\lambda) z - \frac{e_3(\lambda)}{d} = 0, \qquad w < u < v_1. \tag{6}$$

It follows that $\operatorname{spec} Y = \{\lambda_i\}$.

The complete C-enabled memory coefficient, with $a = p(1-q)$, $b = pq$ and $D = 1 - a^2 + ab$, is

$$Z_C = \frac{(1-a)P_2 + a(1-a)P_3 + abP_{10}}{D} - (1-p)P_2. \tag{7}$$

It has trace $p$; disabled records have not been removed or normalized away. Put $A = v_1(v_1 + w)$ and $B = v_1(1 - v_1 + w) + (v_1 + w)w$. The inverse equations

$$-Ap^2 + Bp - w = 0, \qquad q = 1 - \frac{v_1}{1 - p(v_1 + w)} \tag{8}$$

give the exact identity $Z_C = pY$. This follows by substituting the second equation and reducing each of the three coefficient differences by the quadratic in the first equation.

At each of the supplied forward and measured-endpoint pixel coordinates, rational Taylor enclosures and polynomial endpoint signs give exactly three roots of (6), and exactly two solutions of (8) satisfying $0 < p < 1/12$, $0 < q < 1/6$. For the forward coordinate the first solution is approximately $p = 0.01915526454$, $q = 0.05474497127$. The native charged-lepton support Casimirs are $9/8$ and $3$, so the complete probes implement these rates with the declared durations

$$\theta_L = \frac{\arcsin\sqrt{12p}}{\sqrt{9/8}}, \qquad \theta_R = \frac{\arcsin\sqrt{6q}}{\sqrt{3}}. \tag{9}$$

Both probes retain all thirteen outcomes. Their product retains all $169 \times 128$ current and repair labels per old memory input. The covariant binary grouping retains 512 labels per input and all twelve old inputs.

Choose $g = \sum_i t_i / p$. Then $gZ_C$ has exactly the three coefficients $t_i$. Its diagonalizing unitary acts on the gauge-trivial family factor, commutes with every native gauge current, and preserves the canonical kinetic form. Thus this is an operator realization of the conditional mass law, up to the allowed family frame. It does not assert an additional alternating-group intertwiner. The source-coordinate enclosures also give $\sum_i t_i < 1$ by bounding the cube of that sum.

The coefficient is the generator of a controlled limit of the full correlated memory. Apply $\exp(-ihgX \otimes P_{\text{new}})$ on enabled records and identity on disabled records. Let $\Psi_h$ be this complete classical/quantum channel. Its classical transition satisfies $T = (1-p)R + pS$, where $R$ resets to port two and $S$ is stochastic. For every normalized initial classical/quantum state $z_0$, including an arbitrary untouched finite reference, the reduced state after $N$ steps of size $h = \tau/N$ obeys

$$\left\| \sum_{\text{memory}} \Psi_{\tau/N}^N(z_0) - \mathcal{U}_\tau\left( \sum_{\text{memory}} z_0 \right) \right\|_1 \le \frac{4g\tau}{N(1-p)} + \left( \frac{8}{1-p} + 4 \right) \frac{g^2\tau^2}{N}, \tag{10}$$

where $\mathcal{U}_\tau$ is generated by $gX \otimes Z_C$. For a stationary product preparation the first term vanishes.

To prove the bound, let $\Pi z = \pi \sum z$ and $e_k = (I - \Pi) z_k$. The sum of trace norms on memory blocks gives $\|Te\|_1 \le p\|e\|_1$ on zero-sum blocks and $\|\Psi_h - T\| \le 2gh$. Hence $\|e_k\|_1 \le 2p^k + 4gh/(1-p)$. The reduced stationary step and the mean unitary step have the same first derivative; their two Taylor remainders sum to at most $4g^2h^2$. The reduced error due to $e_k$ is at most $2gh\|e_k\|_1$. Summing these errors proves (10).

This limit is in dimensionless simulator time. A finite unread channel need not equal its limiting unitary. The feedback pulse service is an explicit design choice; compilation into a smaller native gate alphabet is a separate requirement. The forward pixel coordinate has no complete physical endpoint proof, and the measured-endpoint coordinate inherits measured-alpha calibration. No physical pole-mass identification, renormalization map, operational energy scale, or complete quark and neutrino spectrum follows from this finite realization.

### 4.3 Memory correlations beyond the mean unitary

The native stationary memory has three distinct projector rays, because $P_9 = P_{10}$. Summing their complete temporal correlations gives a controlled correction to the mean evolution. Let $T$ be the reduced column-stochastic transition, $E$ its enabled-pulse part, $\pi$ its stationary column, $\Pi = \pi \mathbf{1}^T$ and $w = E\pi$. The reset identity contracts zero-sum memory vectors by $p < 1$, so the group inverse $\mathcal{R} = (I - T + \Pi)^{-1} - \Pi$ sums every lag. The one-sided covariance and its positive symmetric part are

$$K = \tfrac{1}{2}(\operatorname{diag} w - ww^T) + E\mathcal{R} \operatorname{diag} w, \qquad D = K + K^T \succeq 0,$$

where positivity follows from the limiting covariance of accumulated pulse counts. For $H_i = gX \otimes P_i$ and $L_i = -i[H_i, \cdot]$, the reduced generator is $G_0 + hG_1$, with $G_0 = \sum_i w_i L_i$, $G_1 = \sum_{ij} K_{ij} L_i L_j$. Its antisymmetric coefficient gives the coherent ordering potential $H_{\mathrm{ord}} = -(i/2) \sum_{ij} K_{ij} [H_i, H_j]$; the symmetric part gives the dissipator $-\frac{1}{2} \sum_{ij} D_{ij} [H_i, [H_j, \cdot]]$. Both belong to the same correction, and $H_{\mathrm{ord}} = I_2 \otimes A$ is a family potential with a different auxiliary grading from a scalar mass.

For stationary product preparation, any untouched reference, $h = t/N$ and fixed $t$, the corrected completely positive semigroup has an explicit rational $O(h^2)$ finite-time diamond-norm bound on the unread channel. Complete record tapes and correlated initial memory need their own terms.

### 4.4 Complete records, capacity and recovery

The native twelve-address family memory admits an exact smaller encoding. Every raw branch preserves the recurrent set $S = \{2, 3, 9, 10\}$. Let $E_S$ embed four memory addresses into these source addresses. For all 128 raw labels, all twelve original old-memory labels and every real pulse size, the corresponding reduced branch $B_y$ obeys

$$K_y (E_S \otimes I) = (E_S \otimes I) B_y.$$

Multiplication proves equality of all joint records and terminal work states for any initially encoded state, including entanglement with an arbitrary reference. Reset to address 2 and stationary memory qualify. Address 0 admits one fully retained startup event, landing at 2 or 9, followed by exact recurrent encoding. These maps encode one complete law, and physical independence of capacity would need more.

The native matching code obeys the ideal recovery identity $R_y K_y E = \sqrt{w_y} E$, with $\sum_y w_y = 1$, for every retained branch, on both matching codes, both repair laws and all thirty seams. Specified logical evolution between corrected repairs is therefore preserved even conditional on the records, and faithful recovery cannot shift its poles.

Changing feedback duty changes the evolution. The native always-duty and C-duty generators have ratios approximately $1 : 195.117 : 3579.826$ and $1 : 53.588 : 271.403$. Their normalized quadratic traces differ exactly, so neither a family relabeling nor a common clock rescales one into the other. Dividing by event probability changes only the common scale. The specified nine-transition preparation and two-clock inverse have stationary-kernel error at most $1/2579890176$, which controls a first-order kernel approximation.

### 4.5 Flavor from current-gated memory

Reading a specified memory's angle, reset rule, duty policy and feedback law determines its stationary coefficients, and these are source-derived for that specification. The declared native current gates the complete repair record differently in the up, down and charged-lepton sectors, and the static mean-action service has certified full-action spectra and nontrivial real quark mixing. For $\theta = \pi/5$, duty "always" and unit Higgs norm, the dimensionless coefficients are as follows.

| Channel | Three ordered coefficients | | |
|---|---|---|---|
| Up | 0.00015764215 | 0.04279127122 | 0.95705108663 |
| Down | 0.00014014156 | 0.04369077654 | 0.95616908190 |
| Charged lepton | 0.00003179436 | 0.02148604675 | 0.97848215888 |

Every coefficient in the retained real-projector class has $\mathrm{Tr}[H_u, H_d]^3 = 0$, and complex family basis changes cannot create charged-current CP violation. All four default duty policies fail direct identification of the charged-lepton ratios: under a common positive normalization and relative corrections bounded by $\delta$, the unconditional and C-duty cases require $\delta \geq 0.79167$ and $\delta \geq 0.59931$, and E and CE duties have an exact zero coefficient. Other controllers and source angles lie outside this comparison.

The correlated memory has a controlled realization of these same mean coefficients. For duty $D \in \{1, C, E, CE\}$, retain every path and apply $e^{-ih(H_0+\chi_D(r)V_{F(m,r)})}$, with $\|V_m\| \leq L$ and $h = t/N$. The reset probability $1 - p$ gives the reference-safe reduced bound

$$\delta_N \leq \frac{4pLt + 4(1+p)L^2t^2}{N(1-p)}.$$

The complete archive differs by at most $\sqrt{2\delta_N}$ in unhalved trace distance from the *same* record law with the mean unitary on the target. Initial correlations and all fine outcomes are retained. The interaction-picture estimate contains no $\|H_0\|$ factor. The physical CAR pairing lift has norm at most 14 per cell at unit neutral Higgs amplitude, using directed tensor ranks $3, 3, 1$. Its record-conditioned controller is specified wiring. The finite-volume bound does not identify target pulse time with auxiliary gate overhead, prepare a vacuum, or give a volume-uniform limit.

This limit preserves the four charged-lepton coefficients: their action Koide quotients $\sum m_i/(\sum \sqrt{m_i})^2$ lie near $0.76758, 0.57975, 0.95024, 0.77544$, respectively, and exclude $2/3$. A separate coherent source construction supplies a different action. For the regular three-cycle $R$, its fully retained analyzer has block $C/\sqrt{3}$, where

$$C = \sqrt{10/72}\, I + \sqrt{5/72}(e^{i\phi}R + e^{-i\phi}R^2), \qquad Y = C^2/3.$$

In the positive chamber $|\phi| \leq \pi/12$, the action singular values of $Y$ satisfy Koide exactly. A projection kick with fresh ancillary flags realizes the logical mean $X \otimes Y$; keeping all eight outcomes

at every step gives complete-output error at most $9t^2/(4N)$. The full exterior lift of the analyzer acts on charged ancillary CAR copies. Vacuum compression of its conjugated charged-Higgs kick is exactly the physical pairing with coefficient $Y$ on all ninety-mode occupations. Committing classical vacuum/nonvacuum flags and retaining the entire charged quantum archive gives unhalved trace error at most $108|v_H|^2t^2/N$. The fresh modes, analyzer and pulse clock are specified apparatus; their overhead is retained. This coherent controller is distinct from the four memory duties and leaves $\phi$ explicit. Ordinary Yukawa masses are singular values of $Y$; the square in $Y = C^2/3$ belongs to this particular source construction.

An additional neutral loss flag gives a complete attenuation controller. Before the Fourier analyzer it multiplies both rotation-branch amplitudes by $r \in [0, 1]$ and sends their remaining amplitudes to retained loss records. The sixteen-outcome instrument has

$$C_r = \sqrt{10/72}I + r\sqrt{5/72}(e^{i\phi}R + e^{-i\phi}R^2), \qquad Y_r = C_r^2/3, \qquad p_{\text{loss}} = 5(1 - r^2)/36.$$

The full-CAR compression theorem applies to this enlarged apparatus. In the positive chamber its quotient is $Q = (1 + r^2)/3$; the exact controls $r = 1, 3/5, 0$ give $2/3, 34/75, 1/3$. All outcomes are retained without renormalizing the selected effect. Thus $r = e^{-\chi}$ is constructible, but the unchanged primitive records do not select that control. Its native logical analyzer and complete weak round are executed; a full many-fermion native tape is not claimed (`model/particle_candidate_atlas/BARE_EFFECTIVE_CONTROLLER.md`).

With the same native Weyl kinetic symbol and a chosen filled-negative-energy Gaussian reference, the quadratic action has three charged Dirac family branches

$$E_j(\mathbf{p})^2 = c^2|\mathbf{p}|^2 + |v_H|^2\sigma_j(Y)^2.$$

The lattice sine symbol converges uniformly on each compact momentum set, with Hamiltonian error at most $ca^2P^3/2$ for $|p_j| \le P$; a positive charged mass floor controls the reference-projector error. All eight valleys and their mirror currents remain. These are family-resolved free branches, not three lower thresholds of the full electric-charge sector. The complete phase score reads their unordered coefficients, while the ordinary measured repair maps are phase-blind. The full-CAR and free-field proofs are in `model/charged_lepton_response/`.

The same Clifford proof accepts every positive Hermitian family matrix. Writing $V(Y)$ for the native charged pairing, real linearity gives $V(gZ_C) = g\sum_r w_r V(P_r)$ and $\|V(P_r)\| = 2$ for a rank-one family projector. Complete C-duty feedback therefore supplies the entire pixel-dependent coefficient to this full-Fock free-field map. For a finite region of $V$ cells the correlated-history theorem uses $L = 2Vg|v_H|$ and $p_{\text{joint}} = 1-(1-p)^V$, with all original records. This establishes conditional free masses for the deformed coefficient without assuming exact Koide balance. The charged-only free action also has massless spectators: three zero-momentum down-type color modes form a global color singlet of electric charge $-1$ and zero energy. This countercontrol imposes no local Gauss constraint and identifies no hadron; it excludes reading the three family masses as the three lowest unrestricted electric-sector thresholds (`model/particle_candidate_atlas/CHARGED_FIELD_ADMISSION.md`).

A finite interacting extension keeps the scalar quantum. Its carrier is a vacuum plus the conjugate Higgs doublet. The actual charged-Higgs tensor defines a CAR-even vertex that commutes with all twelve local Gauss generators. Exact occupation arithmetic on its complete 192-state active scalar/fermion carrier gives rank-one vertex norm 2. The same complete feedback theorem therefore applies with the common quantum gauge/Higgs Hamiltonian. This is a finite gauge-invariant interacting action limit; vacuum selection and the interacting chiral continuum do not follow from it (`model/particle_candidate_atlas/INTERACTING_CHIRAL_JOIN.md`).

The zero-phase program has ratios $1 : 1 : (34 + 24\sqrt{2})$. A separate calibrated test uses only $m_\mu/m_e = 206.7682827$ to set $\tan(\phi/2) = 55785269584552807/500000000000000000$. The source then gives $m_\tau/m_e \simeq 3477.4416363$, within one quoted standard uncertainty of the held-out CODATA 2022 value $3477.23 \pm 0.23$. With the additional electron unit 0.51099895069 MeV, the conditional value is 1776.9690272 MeV, compared with $1776.86 \pm 0.12$ MeV. These correlated comparisons are not a joint significance test or exact central agreement. The phase is fitted and the default is unchanged. The same conditional value is 1.7769690272 GeV; the separate PDG2026 tau average is $1.77693 \pm 0.00009$ GeV. Agreement with these individual intervals does not require equality to their central estimates.

A distinct three-input test fits all three charged central masses with the complete attenuator. Put $x = (\sqrt{m_\tau}, \sqrt{m_e}, \sqrt{m_\mu})$, $A = \sum_j x_j/3$ and $B = \sum_j x_j e^{-2\pi i j/3}/3$. Choosing $r = \sqrt{2}|B|/A$, $z = B/|B|$ and an overall energy factor $s = 3A^2/a^2$, with $a = \sqrt{10/72}$, gives $s\,\mathrm{spec}(C_r^2/3) = \{m_e, m_\mu, m_\tau\}$ by inverse Fourier transformation. The quoted central values admit $r \simeq 0.99999669509849$, $\arg z \simeq 0.22222476189459$ and $s \simeq 6.7791801525$ GeV, with all sixteen records retained. Here $r^2 = 3Q - 1 < 1$, so attenuation suffices. The exact radical construction and its rational controller approximation have an output enclosure error below $10^{-36}$ GeV. All three measured masses select this controller; that numerical bound is neither a physical error budget nor a held-out prediction. The factor $s$ is no identification of the physical Higgs expectation value, and interacting threshold or resonance matching is an additional premise (`model/particle_candidate_atlas/CALIBRATED_TRIPLET.md`).

The separate pixel-dependent response law also has a complete C-duty realization: two admissible controllers obey $\mathrm{spec}(gZ_C) = \{m_j/v\}$, including every disabled record. The mass modes diagonalize the entire coefficient, not its nonorthogonal port-projector constituents. Its retained capacity coordinates give the rounded ratios

| Conditional source branch | $m_2/m_1$ | $m_3/m_1$ |
|---|---|---|
| Forward coordinates | 206.768300595181 | 3477.365536596023 |
| Measured-$\alpha$ endpoint | 206.768282739123 | 3477.365262268937 |
| Supersymmetric proxy, $\beta = 4$ | 206.768297922262 | 3477.365495531309 |
| Minimal census, $\beta = 7$ | 206.782621310399 | 3477.644439943646 |

Only the measured-$\alpha$ endpoint lies inside both quoted individual CODATA2022 ratio intervals. It inherits measured-$\alpha$ input; the architecture is target-informed. The two closure roots depend on their specified census, scale and response kernel. Their amplitude changes with the phase, giving $Q = (1 + e^{-2\chi})/3 < 2/3$. Transferring only their phase to the balanced analyzer changes the action. Pixel capacity is a declared response coordinate, without an identity to total cell count or hardware memory. The interval calculations and ancestry are in `model/charged_lepton_response/P_DEPENDENCE.md`.

The two shapes can describe bare and effective coefficients. A physical common-source identification requires a derived family-dependent correction. The response architecture stipulates $\chi = (u^2/512)/(1 - u/77)$. At a common phase, set $B = (I + T)^2$ and $D = (I + e^{-\chi}T)^2$, where $T = (e^{i\delta}R + e^{-i\delta}R^2)/\sqrt{2}$. When both root matrices are positive, the supplied kinetic choice $Z_L = Z_R = BD^{-1}$ whitens $B$ to $D$ exactly. This solves an inverse matching problem; it does not calculate a source self-energy. Passive field changes preserve canonically normalized masses. A common-source identification requires independent kinetic and mass-response data and must distinguish $\delta(P, u)$ from the fitted phase (`model/particle_candidate_atlas/COMPATIBILITY.md`).

Reversing the native half-turn order conjugates the coherent phase and preserves its unordered masses. The all-lag ordering potential is exactly $I \otimes i\kappa_{\mathrm{ord}}(R - R^{-1})$, with

$\kappa_{\mathrm{ord}} = 603(1+\sqrt{5})/6666724$, accompanied by a dissipative correction. It commutes with the coherent coefficient, so its conjugation cannot select that phase. An exact inverse memory realization establishes a specified action, without selecting its control law or physical clock. The actual default all-lag diffusion also leaves the circulant algebra; it is not the displayed scalar attenuation. At finite pulse angle $h$, disabled and enabled records alone give full-CAR vacuum purity loss at least $2w_0w_3(1-\cos^4 h)$ in the unread working channel. For $h \notin \pi\mathbb{Z}$, this excludes an exact unitary mass-only replacement on that register. It neither discards the full archive nor excludes a physical interaction correction in a different source model (`model/particle_candidate_atlas/NATIVE_RESPONSE.md`). The eighteen-channel candidate dictionary retains the internal source weights, these distinct charged-action families and the separate eleven protected objects. It assigns no physical spin, mass, width or defect-to-species map (`model/charged_lepton_response/CANDIDATE_MAP.md`).

A conditional Standard Model completion makes the physical target explicit. Supply a local Lorentzian chiral gauge action on three copies of the fifteen-state module, a complex Higgs doublet, positive kinetic forms and a symmetry-breaking vacuum. Canonical normalization then gives the fermion action masses from the singular values of the normalized Yukawa matrices. For $\langle H\rangle = (0, v/\sqrt{2})$, the tree relations are $m_f = v\sigma_f(Y_f)/\sqrt{2}$, $m_W = g_2 v/2$, $m_Z = v\sqrt{g_2^2+g_1^2}/2$ and $m_h^2 = 2\lambda v^2$. The unbroken electromagnetic and color gauge fields have zero gauge-boson mass parameters. Supplying fermionic CAR and the chiral spacetime action assigns spin 1/2 and Fermi statistics to matter, spin 1 to gauge fields and spin 0 to the Higgs. Conjugate representations specify antiparticles; these assignments require more than the internal weights.

Measured masses enter this completion as reference inputs to a specified renormalized matching prescription. They cannot all be inserted into the tree formulas as though pole, resonance and running masses shared one definition. Independent Yukawa matrices and mixing parameters can reproduce the charged spectrum without imposing exact Koide on measured masses. Retaining the coherent Koide coefficient instead requires an independently derived interaction correction. The minimal chiral module has no neutrino mass term; a declared Weinberg operator or additional neutral singlets supplies a neutrino-mass extension. Absolute neutrino masses and the Dirac or Majorana choice are explicit undetermined data. Thus a complete conditional field census is compatible with the observed particle assignments, while its source-selected action, state, clock and defect map are unproved. This is a phenomenological effective-action completion, not a nonperturbative continuum existence theorem (`model/particle_candidate_atlas/CONDITIONAL_STANDARD_MODEL.md`).

For an interacting comparison, independent kinetic forms $Z_L, Z_R$ and a mass correction $E$ require bounds in the same state and clock. If $\|Z_{L,R}-I\| \le \epsilon_{L,R} < 1$ and $\|E\| \le \eta\sigma_{\min}(M_0)$, then each singular value of $Z_L^{-1/2}(M_0+E)Z_R^{-1/2}$ lies between its original value times $(1-\eta)/\sqrt{(1+\epsilon_L)(1+\epsilon_R)}$ and $(1+\eta)/\sqrt{(1-\epsilon_L)(1-\epsilon_R)}$. Exact allowed corrections to this source coefficient violate Koide. No interacting correction is computed here. A stable charged infraparticle may have a lower spectral boundary without a mass-shell atom; unstable leptons require a resonance prescription. The interacting gauge vacuum, mirror treatment and observable matching are separate from the free theorem.

The numerical discrepancy has a sharp conditional matching floor. If $r_0 = 1$ and $[L_i, U_i]$ are the reference ratio intervals, the smallest required multiplicative spread is

$$D_{\min} = \max\left(1, \frac{\max_i(L_i/r_i)}{\min_j(U_j/r_j)}\right).$$

The corresponding additive singular-value optimization has minimum $\eta = \max(0, \max_{ij}(L_j r_i - U_i r_j)/(L_j+U_i))$ after optimizing one common unit. Exact diagonal controls attain these bounds. As-

signing the entire change to componentwise observable matching, the forward pixel branch requires a relative correction of at least $3.21 \times 10^{-8}$ for the stated ratio windows, far above its numerical enclosure width. This is a necessary correction size, not a calculated physical correction or a rejection of every interacting completion (`model/particle_candidate_atlas/MEASURED_MASS_JOIN.md`).

Capacity self-consistency does not select the attenuation law. Holding the inner heat, scale and coupling equations fixed, the supplied change $\chi(u) \mapsto \chi(u) + 10^{-14}$ preserves a root in each of two certified closure intervals. The original strict self-map margins exceed the map perturbation bound $4 \times 10^{-16}$. The two attenuation and Koide-quotient ranges are disjoint, and the complete C-duty inverse admits both. This compares different specified laws; uniqueness for any fixed original law is unaffected (`model/particle_candidate_atlas/CAPACITY_NON_SELECTION.md`).

### 4.6 Pixel feedback with one charged-fermion mass prescription

The pixel feedback must use the same mass coefficients as the declared field and memory operators. Let $u = \alpha_U$ and retain

$$M_U/E_* = e^{-2\pi} P^{1/6}, \qquad v/E_* = P^{-1/2} e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi} \log(M_U/m_Z).$$

Here $E_*$ is a declared energy unit. The same conditional three-lepton coefficients are inserted into the charged-fermion integral, together with all six declared quark coefficients, including the finite top contribution. With the prescribed factor $1 - 3\alpha_3/\pi$ on the quark terms, the resulting inverse response $A(P)$ defines $P = \phi + \sqrt{\pi}/A(P)$. This kernel supplies no charged-vector or charged-scalar terms, confining hadronic spectral measure, threshold matching, or derivation of the native current's physical Thomson limit.

There is an exact obstruction to substituting the minimal Standard Model census while retaining $\beta = 4$. Assume $P \geq 1$, $M_U \geq m_Z$, positive couplings and threshold-free one-loop coefficients $b = (41/10, -19/6, -7)$. Set $x = \alpha_2^{-1}$ and $y = \alpha_3^{-1}$. Eliminating the unified coupling and scale ratio gives $0 < y \leq x$ and

$$\frac{\pi(18x + 5y)}{46} = 2\pi - \frac{2}{3} \log P + \frac{1}{2} \log\left[\pi \left(\frac{1}{x} + \frac{69}{333x - 218y}\right)\right].$$

The logarithm's argument is at most $8\pi/(5x)$. The function $9\pi x/23 - 2\pi - \frac{1}{2}\log[8\pi/(5x)]$ is increasing and exceeds one at $x = 6$, so $x < 6$ and both nonabelian couplings exceed $1/6$. Their heat-sum denominators are at least one; direct outward summation with infinite-tail bounds places their combined numerators below $1/80$. The required pixel entropy is $P/4 \geq 1/4$, a contradiction. This excludes the stated scale and census combination, independently of the readout kernel.

Two other declared maps have certified local fixed points. The supersymmetric running proxy with $\beta = 4$ gives $P \simeq 1.630971514159316$ and $A \simeq 137.000994766342$. The minimal census with the explicit scale choice $\beta = |b_3| = 7$ gives $P \simeq 1.628533369611833$ and $A \simeq 168.815082928453$. The first requires extra supersymmetric fields absent from the native quadratic carrier. The second supplies a scale choice, without identifying the Higgs scale physically with color transmutation. For each map, verified implicit-root signs, infinite heat tails and interval automatic differentiation give a strict self-map and $|g'| < 0.00734$ on the reported coupling interval. Uniqueness is local to that interval.

Rational interval inversion over both complete root enclosures admits both native memory solutions, with $0 < p < 1/12$ and $0 < q < 1/6$. The gain $g = \sum_j (m_j/v)/p$ preserves the unconditional duty probability, so the same full-record memory coefficient has the required three singular energies.

Independent simultaneous-root and quadrature checks accompany the proof. These are conditional consistent feedback calculations; neither is a complete physical particle or fine-structure prediction.

The beta coefficients in the feedback calculation are obtained from the actual internal-current traces with the intended chiral field multiplicities. This assumes that those are the surviving continuum fields, and it does not count or remove spatial regulator valleys or doublers. The minimal-census root and the supersymmetric proxy are different scale designs, not two calibrations of one proved physical theory. The absence of a positive solution for the minimal census with $\beta = 4$ is a mathematical inconsistency of that specified combination; it is not an exclusion of other admitted OPH designs.

The three increasing coefficients returned by these closed conditional maps are dimensionless singular coefficients $m_i/v$.

| Scale design | $m_1/v$ | $m_2/v$ | $m_3/v$ |
|---|---|---|---|
| Supersymmetric proxy, $\beta = 4$ | $2.07203042\times10^{-6}$ | $4.28430204\times10^{-4}$ | $7.20520710\times10^{-3}$ |
| Minimal census, $\beta = 7$ | $2.07103435\times10^{-6}$ | $4.28253912\times10^{-4}$ | $7.20232110\times10^{-3}$ |

The interval certificate encloses these values and both admissible memory inverses over each complete root box, so the table and the controller refer to the same conditional source coordinates.

# 5 Localizing the same current and mass operators

## 5.1 Topology, total energy and the radial source

The seed of the defect hypothesis is a precise finite frustration statement. For a finite connected graph, abelian edge offsets $b_e$ and positive weights $w_e$, define

$$\Phi_b(x) = \sum_{e:u\to v} w_e \, \mathbf{1}[x_v - x_u \neq b_e].$$

If a cycle has nonzero signed offset sum, then $\min_x \Phi_b(x) > 0$: the possible costs form a finite set of positive-weight subset sums, and a zero cost would solve every edge equation and force zero cycle sums [3]. This is frustration at fixed compatibility data. Protection of $b$ against an update that changes its class, localization of its energy and a rest mass all need more.

For a one-cochain $b$ on the full carrier complex, face curvature is $F = \delta b$ and $b(\partial c) = F(c)$ for every two-chain $c$. The icosahedral boundary has trivial first homology, and the native load difference $b_{uv} = n_v - n_u$ is exact, so $F = 0$ and all loop sums vanish. An actual defect invariant therefore requires additional transport or field data.

Consider the two-spin cubic hopping operator with forward link $i\sigma_j/2$ and its Hermitian reverse. A declared radial onsite texture is

$$B_x = M \sum_{j=1}^{3} (-1)^{x_j} \frac{x_j}{|x|} \sigma_j, \qquad B_0 = 0.$$

On the odd radius-one cube at $M = 64$, this literal source has exactly two zero modes and a complementary absolute-energy gap at least 62. The normalized kernel frame $W$ has scalar local density $W_x^\dagger W_x = \rho_x I_2$, with $\sum_x \rho_x = 1$. All 28 retained winding matrices have exact spectral enclosures with every multiplicity and source-matrix rounding error controlled. For every retained odd-degree cube, inversion and complete spectral exhaustion prove exactly two zero modes, including weak-mass cases. Degree-three six-level windows hold two zeros and four nonzero levels, which

is no count of generations. A complete-Fock contact theorem admits 34 of 84 supplied contact cases, with a ground pair and unique neighboring charged states.

A radial mass profile also needs a total energy law. In the selected ordered-spin rotation orbit, each admissible charged shell costs at least $\pi J\kappa|q|$ with $\kappa > 0$. A hedgehog retaining nonzero degree on arbitrarily many disjoint shells therefore has infinite excess energy in that ungauged identification, and a fermion bound-mode gap omits this background cost. Screening, compact opposite-charge textures or another specified finite-energy structure are substantive dynamical choices. The compact pair below provides the second kind.

### 5.2 One local carrier for defect states, currents and mass coefficients

The radial defect operator admits a direct realization of the native matter currents and a prescribed family mass operator on the same localized modes. The construction is a choice of simulator carrier and coherent pairing service. Its finite operator identity is exact; the choice is not claimed to follow uniquely from overlap consistency.

Let $h$ be the radial operator on $\ell^2(\Lambda) \otimes \mathbb{C}^2$, with its normalized kernel frame $W : \mathbb{C}^2 \to \ker h$. Use the strong texture domain, in which the kernel has dimension two and the complementary spectrum has absolute energy at least $d > 0$. With

$$J = i\sigma_2, \qquad D_x = (-1)^{x_1+x_2+x_3}, \qquad C = DJ,$$

the source identities are

$$C^T = -C, \quad C^\dagger C = I, \quad hC = -C\overline{h}, \qquad hW = 0, \quad W^\dagger W = I_2, \quad C\overline{W} = WJ.$$

The last identity fixes the quaternion frame through evaluation at the origin and its scalar normalization Gram matrix.

Choose the canonical anticommutation relation algebra on $\ell^2(\Lambda) \otimes \mathbb{C}^2 \otimes M_{15} \otimes \mathbb{C}^3$. Here $M_{15}$ is the native matter representation, transported by its actual source intertwiner; the three family copies are declared inputs. For a symmetric all-left Higgs kernel $K \in M_{45}(\mathbb{C})$, define

$$\Delta = C \otimes \overline{K}, \qquad B = \begin{pmatrix} h \otimes I & \Delta \\ \Delta^\dagger & -\overline{h} \otimes I \end{pmatrix}, \qquad E = \operatorname{diag}(W \otimes I, \overline{W} \otimes I).$$

The antisymmetry of $\Delta$ makes it a valid local creation pairing. Multiplication gives the exact isometric intertwiner

$$BE = EB_0, \qquad B_0 = \begin{pmatrix} 0 & J \otimes \overline{K} \\ (J \otimes \overline{K})^\dagger & 0 \end{pmatrix}. \tag{11}$$

Every singular value of $K$ therefore appears with both signs and twice the multiplicity in the localized Nambu operator, including zero values. Nambu coordinates duplicate annihilation and creation operators; they introduce no physical mirror carrier.

For every actual anti-Hermitian native current $G$, the map $W \otimes I$ intertwines $I \otimes G$ with $I_2 \otimes G$. In Nambu coordinates the generator is $\operatorname{diag}(G, \overline{G})$; Hermitian charges use $\operatorname{diag}(Q, -\overline{Q})$. The Higgs background transforms with matter under the uniform internal gauge action. A fixed broken vacuum does not make every gauge generator commute with its mass operator. The resulting map of canonical anticommutation relation algebras intertwines quadratic Heisenberg evolution. On even states, tensoring with the normalized trace on the orthogonal complement gives an exact state inclusion with partial-trace recovery. This complement state is not a vacuum.

Use the positive family matrix $M(P)$ supplied by the memory construction, whose eigenvalues are the retained charged-lepton coefficients $m_i(P)/v(P)$. Choosing $H = (0, v(P)/\sqrt{2})$ and $Y_e =$

$\sqrt{2}M(P)$ realizes those coefficients in $K/v(P)$ itself. The coherent pairing service implements the stationary-mean memory coefficient times its declared gain. Equation (11) does not identify the unread finite memory channel with this Hamiltonian. The calibration provenance of the retained family law is unchanged.

An explicit record service realizes the same Hamiltonian in a controlled weak-step limit. Apply the native current probes to fresh auxiliary legs in their invariant preparations, so their complete twelve-state memory transition is independent of the spatial field. For the updated memory port $j$ choose the field pulse generated by

$$H_r = H_0 + \mathbf{1}_C(r)\, gL(P_j),$$

where $H_0$ contains the radial kinetic term and the chosen quark pairing, $L$ is the local lepton pairing map, and $g$ is the source gain. Every $H_r$ has the same core intertwiner. Linearity gives stationary mean $\overline{H} = H_0 + L(gZ_C)$, precisely the Hamiltonian above. If $\|H_r\| \le M$ on the fixed finite Fock space, the reset contraction of the complete memory law and a second-order unitary expansion give

$$\left\|\rho_N - e^{-iT\overline{H}}\rho_0 e^{iT\overline{H}}\right\|_1 \le \frac{4MT}{N(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{M^2T^2}{N}.$$

The bound includes arbitrary finite untouched references and initial memory correlations; stationary product memory removes the first term. For the displayed cube, $M = 2430\cdot 66 + 1215(1+g)$ suffices. This chosen coherent service uses every record. Its finite unread channel need not be unitary, and its bound is not uniform in volume.

For a complete finite witness, choose the retained quark design $Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1)$ and $Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2})$, and set the three minimal neutrino coefficients to zero. Their exponents are design inputs, not topological index conclusions. On the radius-one cube there are 27 sites, 2430 physical annihilator modes and 4860 Nambu rows. The localized sector has 90 annihilators and 180 Nambu rows: 84 positive, 84 negative and 12 zero rows. These numbers count operator degrees of freedom, not observed species. In particular the zero rows do not establish a relativistic massless neutrino sector.

The full complement is retained. Indeed, with $T = \mathrm{diag}(I, C^\dagger \otimes I)$,

$$T^\dagger B T = \begin{pmatrix} h \otimes I & I \otimes \overline{K} \\ I \otimes K & h \otimes I \end{pmatrix}.$$

Its energies are the sums $\lambda(h) \pm s_j(K)$, not a relativistic square-root dispersion. If $\|K\| \le M$ and $2M < d$, the localized band lies in $[-M, M]$ and the complement has absolute energy at least $d - M$. This is single-quasiparticle isolation, not isolation of all many-body energy sums. In the displayed design, choosing $v(P)$ as energy unit, radial hopping one and texture coefficient 64 admits $d = 62$ and $M = 1$.

The common spatial density follows from $W_x^\dagger W_x = \rho_x I_2$; inversion gives mean position zero. A Fourier transform supplies a momentum distribution, without making the localized state a momentum eigenstate. Occupation of these modes does not create the supplied texture winding. Original site-resolved repair can leak out of the kernel, and interactions can mix the two factors. Thus the exact common finite operator supplies neither defect formation, physical rest masses nor an interacting relativistic continuum.

The certificate retains all twelve native current operators, and its complete finite controls include an unstaggered-pairing counterexample.

### 5.3 Compact opposite-degree textures and their isolated modes

A finite-energy texture can carry opposite local winding numbers while approaching a constant exterior. On the two-spin cubic hopping carrier, choose an integer $a \geq 1$, write $R = a + 1$, and define

$$f(x, y, z) = \begin{cases} (x, y, z^2 - a^2), & \max(|x|, |y|, |z|) \leq R, \\ (0, 0, 1), & \text{otherwise.} \end{cases}$$

Set $n = f/|f|$ away from $c_\pm = (0, 0, \pm a)$ and $n(c_\pm) = 0$. The Hamiltonian is $h = T + Mn \cdot \sigma$, where the nearest-neighbor hopping has $\|T\| \leq \sqrt{3} < 2$ and $M \geq 8$. The texture and its amplitude zeros are explicit design inputs. The complete unit cubic shells about $c_+$ and $c_-$ have degrees $+1$ and $-1$ under their normalized barycentric interpolation; the exterior shell is constant. Exact rational regular-value counts check every oriented triangle. This degree classifies the interpolated texture and is not a fermionic superselection charge.

For the selected ordered-spin reference, fill the two core directions by the exterior direction. Each rotated-ground source bond has excess energy in $[0, J]$. With $s = 2a + 3$, precisely $3s^2(s + 1)$ bonds touch the support cube, so

$$0 \leq \Delta E_{\text{spin}} \leq 3s^2(s + 1)J.$$

The two zero fermion masses and the spin-core fill are distinct declared parts of the construction. The second-quantized Hamiltonian differs from that of the uniform texture by an operator of norm at most $2M(s^3 - 1)$. Its vacuum-energy difference has the same bound, independently of exterior volume. These are total texture-energy bounds; they do not assign a constituent rest mass to a localized mode.

Projection onto the two core sites gives $h = \left(\begin{smallmatrix} 0 & B^\dagger \\ B & D \end{smallmatrix}\right)$. The exterior block has no spectrum in $(-M + 2, M - 2)$, and $\|B\|^2 = b^2$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. There are exactly four low eigenvalues. Their Feshbach equations give

$$|E| \leq r := \frac{b^2}{M - 4}, \qquad \frac{\|\psi_{\text{exterior}}\|^2}{\|\psi_{\text{cores}}\|^2} \leq \frac{b^2}{(M - 2 - r)^2}.$$

These bounds are uniform in volume. For a cube of radius $L \geq R + 1$, resolvent path expansion and monotone Feshbach eigenvalue crossings give

$$|E_j(L) - E_j(\infty)| \leq \frac{2b^2}{M - r - 2} \left(\frac{2}{M - r}\right)^{2(L+1-R)}.$$

Thus the infinite lattice also has precisely four isolated modes in $[-2, 2]$. At $a = 1$, $M = 64$, $L = 3$, complete rational interval certification of all 686 eigenvalues excludes zero from all four low levels and from their infinite-volume limits. The numerical levels are approximately $\pm 0.003905899328$ and $\pm 0.003906437151$. The finite-volume filled-sea energy difference is approximately 127.9700 in the same hopping units.

Let $W$ span this four-dimensional band, $A = W^\dagger h W$, and $C = D_{\text{stagger}} i\sigma_2$. Particle-hole symmetry gives $C\overline{W} = WS$ with $S = W^\dagger C \overline{W}$ antisymmetric and unitary. The actual twelve native current operators intertwine through $W \otimes I_{45}$. The same local pairing $\Delta = C \otimes \overline{K}$ has the exact core

$$\begin{pmatrix} A \otimes I & S \otimes \overline{K} \\ (S \otimes \overline{K})^\dagger & -\overline{A} \otimes I \end{pmatrix}.$$

Its energies are those of $A$ plus or minus the singular values of $K$. Consequently the compact texture shifts the retained family coefficients; dropping $A$ gives an incorrect operator inclusion. A

local Pauli perturbation at the two cores preserves every shell datum while changing these energies. Winding alone therefore does not pin a zero mode for this carrier. The construction supplies a compact background and controlled localized spectrum, without autonomous formation, stability, a relativistic mass shell or physical particle identification.

### 5.4 Protected texture records and controlled band transport

A declared local repair law protects the compact pair's two winding charges. Let $f_x$ be its unnormalized reference texture and fix the zero cores and constant exterior. At every noncore site retain a rational vector $v_x$ with $\|v_x - f_x\| \le \eta\|f_x\|$, $0 < \eta < 1/4$, using $v_x/\|v_x\|$ in the actual Pauli potential. A one-vertex proposal is accepted only if it satisfies this budget and has nonnegative scalar product with the old vector and every other vertex of every incident shell triangle. Shared-face data are checked on both shells. Rejected proposals remain recorded identity operations.

**Proposition 5.1** (Guarded winding and uniform isolated band)**.** *Every admitted finite history preserves the two shell degrees. Its spatial Hamiltonian has exactly four isolated modes with* $|E| \le r = b^2/(M-4)$ *and complementary spectrum outside* $(-g, g)$, $g = M - 2$, *where* $b^2 = 7/4$ *for adjacent unit shells and* $b^2 = 3/2$ *for separated shells. For two admitted band projectors* $\Pi_v, \Pi_w$,

$$\|\Pi_w - \Pi_v\| \le \beta = \frac{4M\eta c}{d^2}, \qquad c = \frac{g+r}{2}, \quad d = \frac{g-r}{2}.$$

*The statements are uniform in the exterior volume.*

*Proof.* During an accepted replacement, every interpolated shell vector is a convex combination of nonzero vectors with pairwise nonnegative scalar products. Its squared norm is at least $\sum_i t_i^2\|v_i\|^2 > 0$, so normalization gives a shell homotopy. Every noncore onsite eigenvalue stays $\pm M$ and the four core coordinates stay zero; the compact-source Schur proof therefore gives the same band count and localization bounds. Normalization gives $\|H_w - H_v\| \le 4M\eta$. On $|z| = c$ both resolvents have norm at most $1/d$. Their identity and the contour formula for the two projectors give the displayed bound. □

For $\beta < 1$, define

$$S = \Pi_w\Pi_v + (1-\Pi_w)(1-\Pi_v), \qquad U = S(S^*S)^{-1/2}, \qquad S^*S = 1 - (\Pi_w - \Pi_v)^2.$$

Since $S\Pi_v = \Pi_w S$ and the positive factor commutes with $\Pi_v$, $U$ is unitary and $U\Pi_v U^* = \Pi_w$. The literal particle-hole relation $H_v C = -C\overline{H_v}$ implies $\Pi_v C = C\overline{\Pi_v}$, and the polar formula gives $UC = C\overline{U}$, hence $UCU^T = C$. Thus $U \otimes I_{45}$ transports all twelve uniform native currents and the same pairing tensor $C \otimes \overline{K}$. For $W' = UW$, the pairing matrix $W'^* C\overline{W'}$ is unchanged, while $A_w = W'^* H_w W'$ retains the actual changed spatial energies. Second quantization also transports $N_v = d\Gamma(\Pi_v \otimes I_{45})$ and $\lambda(N_v - 90)^2$.

The explicit source uses $M = 64$, $\eta = 1/512$ and four equiprobable proposals at each record: two opposite transverse changes at the shared-face center, one forbidden zero and one forbidden core change. Its three texture states have transition rows $(1/2, 1/4, 1/4)$, $(0, 3/4, 1/4)$ and $(0, 1/4, 3/4)$. All twelve state/proposal records are retained. Each branch applies its texture transition and $\sqrt{q_r}\,\Gamma(U_r)$, with identity control for a repeated texture, so the complete instrument preserves trace. The band is transported exactly at record endpoints. Omitting $U_r$ instead gives leakage probability at most $\min(1, n^2\beta^2)$ after $n$ quenches and intervening source evolution; the numerical controls exhibit nonzero leakage.

A nearest-neighbor snake through the $N = 2(2L+1)^3$ spatial-spin coordinates compiles each $U$ by adjacent Givens rotations: at most $N(N-1)/2$ two-mode gates and $N$ onsite phases. The complete $N = 686$ controls replay 234,955 two-mode gates per nontrivial texture pair. Addressed quadratic pulses, their timing and drift cancellation are supplied services; this is not a compilation into unchanged C/E repair. Intermediate pulses need not preserve the band, and the controller supplies the endpoint energy change, bounded by $2r$ on the one-body band. Erasing the guards permits zero-degree textures with the same four-mode count. The result establishes controlled protection and transport on this law, without autonomous core motion, local Gauss dressing, physical species identification or relativistic dispersion.

For this same compact pair, the two core sites have equal bipartite parity. Odd Schur hopping terms therefore cancel. At texture amplitude 64 the static four-mode band obeys $|E| \le 1/256 + \eta$, with $\eta = 2365097/60283957504 < 1/25000$. Combining it with the zero-phase charged coefficient $m_{\min} = (15 - 10\sqrt{2})/216$ gives a paired static gap greater than $1/40000$. Its complete positive charged response lies in four separated interval groups of multiplicities $8, 8, 4, 4$; each core probe on the paired Gaussian reference has combined band weight greater than $99/100$. The exact band operator retains $A = W^\dagger h W$. Its paired energies are $E_j \pm \sigma(K)$, not relativistic mass quadrature. Neither the four interval groups nor their multiplicities count physical species. The fixed texture breaks translations, and its two-component spinor space admits no nonzero scalar mass matrix anticommuting with all three kinetic Pauli matrices. Moving defects and a source-selected paired reference require independent constructions (`model/particle_candidate_atlas/DEFECT_FIELD_JOIN.md`).

### 5.5 Gauge-covariant paired compact bands

The compact source extends to supplied classical unitary internal links on its 343-site radius-three lattice. The gauge paper [5] proves that with onsite texture amplitude 64 and pairing of norm at most one, the complete Bogoliubov–de Gennes operator keeps a separated band whose particle grading defines 180 canonical annihilation modes and an integer occupation $N_r$. The quartic $\lambda(N_r - 90)^2$ of Section 5.16 is therefore well-defined on every link background and equals the source compact interaction at trivial links. Transforming link and pairing together preserves band and interaction covariantly, and declared polar controllers transport the grading between records.

The same paper gives the separate native quantum-link design: a 60-state link on the five matter blocks with exact local non-Abelian Gauss constraints on the full Fock space, and a conditional rotor completion that leaves five link-energy differences unselected. A literal native two-site raising-link example has a noninteger naive occupation even on a Gauss singlet, so the canonical band above does not transfer to operator-valued links by substitution. A dynamical quantum-link realization of the interacting compact band is not established here.

### 5.6 Native record response

The fixed biased binary source also supplies an observable response without an attached field action. Give ports $0, 1$ weight 25 and the other ten weight 8, select each of the thirty seams uniformly, and tag current and contrast repairs with equal probability. Let $P$ be its column-stochastic occupation transition and $\pi(x) \propto \prod_a \theta_a^{n_a(x)}$ its stationary law. On every complete record define $Y_e = 60\,\mathbf{1}[\text{contrast law, seam } e](n_a - n_b)$ for $e = (a, b)$, with zero score on reserved addresses. This uses the reported incoming contrast. If $B_e$ is the transition weighted by that score, then

$$\mathbf{1}^T B_e = d_e^T, \qquad B_e \pi = \operatorname{diag}(\pi) g_e, \qquad g_e = \frac{\theta_a - \theta_b}{\theta_a + \theta_b} \mathbf{1}[n_a + n_b = 1].$$

The second identity includes the measuring repair's backaction. With rows $d_e^T$ and $g_e^T$ collected in $D, G$, and $m = D\pi$, the ordered stationary covariance and its discrete-attempt resolvent are

$$C(k) = DP^{k-1}A \quad (k \geq 1), \qquad A = \operatorname{diag}(\pi)G^T - \pi m^T,$$
$$\sum_{k\geq 1} z^{-k}C(k) = D(zI - P)^{-1}A.$$

All 22 equal-weight seams have zero columns in $C(k)$, despite nonzero same-time shot noise. A passive occupation correlator omits this backaction. Exact ranks 66 for $[\mathbf{1}^T; D; \ldots; DP^{12}]$ and 65 for $[A, PA, \ldots, P^{10}A]$, together with the squarefree characteristic polynomial, show that all 65 nonstationary population relaxation poles appear. They are positive real decay factors, without a particle-mass interpretation. The biased occupation means require pair correlations: adjoining one-step images raises their span from 12 to 19.

The calculation retains all 721 interpreter slots and arbitrary untouched references. Complete record probabilities observe only diagonal source blocks, so the rank statement is population observability, not quantum-state tomography. After 1600 unread preparation attempts, the reduced source/reference half trace distance is at most $\varepsilon = 542159/10^{12}$; any fixed future $r$-fold score moment has error at most $2\,60^r\varepsilon$. The preparation archive can remain correlated. Exact population observability alone supplies no sampling or stable pole-inversion bound (`model/native_record_response`, [1]).

**Positive transfer from complete records.** Write $\Pi = \operatorname{diag}(\pi)$. Detailed balance gives the symmetric transfer $S = \Pi^{-1/2}P\Pi^{1/2}$, and the actual minimum diagonal $\min_x P_{xx} = 791/990$ gives

$$\frac{296}{495}I \leq S \leq I, \qquad H_{\mathrm{tr}} = -\log S \geq 0.$$

For each raw flag the native reversal satisfies $T_{\bar{h}} = \Pi T_h^T \Pi^{-1}$. It reverses word order and exchanges incoming and outgoing contrast using the retained current. If $F$ is any scalar future-record cylinder and $f_F$ its conditional expectation on the initial configuration, its reflected form across $n$ unread attempts is $\langle f_F, (P^T)^n f_F\rangle_\pi \geq 0$. The preceding observability rank makes the reflection quotient exactly 66-dimensional. Centered contrasts therefore have the positive reflected spectral representation

$$DP^n\Pi D^T - mm^T = \sum_{j=1}^{65} \lambda_j^n v_j v_j^\dagger, \qquad v_j \neq 0.$$

The ordinary oriented covariance $C(1)$ is different: the centered score $Y_{(0,5)} - Y_{(0,1)}$ has the naive unreflected quadratic form $-83481968/1858725825$. Reflection must use the actual record dictionary.

Complete single-record conditional probabilities span dimension 42. Two records suffice: the constant, eleven contrast functions and fifty-four ordered contrast products give an invertible feature matrix $X$. With $G = X^T\Pi X$ and $B_h = X^T\Pi T_h^T X$, correctly reflected moments of at most five attempts determine

$$A_h = G^{-1}B_h = X^{-1}T_h^T X, \qquad G \succeq \frac{55756800}{2129251819009}I.$$

The first feature is constant, so every finite chronological word has law $e_0^T G A_{h_1} \cdots A_{h_m} e_0$. All 721 raw labels, including zero slots, enter this reconstruction. The finite realization and the depth-two rank certificate supply its domain; a dimension bound or known source specification is required to

infer arbitrary histories from bounded-depth data. The Gram bound controls inverse perturbations. Uniform native mixing followed by each probe gives conditional score-bias bounds without independent resets, provided the preparation archive is inaccessible to later controls. Martingale concentration then gives finite-error intervals; the conservative sample budget exceeds the interpreter's admitted horizon and is unexecuted. This finite population result does not determine quantum coherences or a physical clock (`model/native_record_response/SHORT_RECORD_RECONSTRUCTION.md`, [1]).

This is a finite Euclidean reconstruction from scalar records. Spectral calculus defines a unitary continuation but supplies no native protocol executing it. Its pure ground direction $\sqrt{\pi}$ is not the native faithful working state, and $H_{\mathrm{tr}}$ differs from that state's modular generator $K = -\log \Pi$. For $x = \{0,1\}$ and $y = \{1,5\}$ the exact element $[S,K]_{yx} = \sqrt{2}\log(8/25)/99$ is nonzero; hence $[H_{\mathrm{tr}}, K] \neq 0$. The two clocks cannot be scalar rescalings and shifts of one another. Physical time, spatial locality, operator-valued history positivity and particle masses require additional maps (`model/native_record_response/TRANSFER.md`, [1]).

**Operational clock comparison.** The distinction has a finite operational witness. For the accessible $Y$ probe on seam $(1,7)$, the working state $\Pi$ has zero mean under every modular-clock time. The proposed identity identification with $e^{-itH_{\mathrm{tr}}}$ instead gives

$$f(t) = \mathrm{Tr}(Y e^{-itH_{\mathrm{tr}}} \Pi e^{itH_{\mathrm{tr}}}), \qquad f'(0) > \frac{7}{1000}, \qquad f(1/10) > \frac{1}{2000}.$$

A rational degree-ten logarithm enclosure proves the derivative bound. Reality makes $f$ odd, and $\|H_{\mathrm{tr}}\| < 103/200$ controls the cubic remainder. The native preparation and complete analyzer bounds leave an output half-trace separation greater than $1/5000$ if both proposed clock services have half-diamond error at most $10^{-5}$. This rejects their identification on the same working reference, including reference-preserving unitary changes of basis. It neither executes the transfer unitary nor excludes a different state/readout map. Bare repair records, with the final carrier traced, are insensitive to any interleaved diagonal phases; the declared coherent analyzer supplies the phase-sensitive readout (`model/native_clock_comparison`, [1]).

**Complete records for disjoint repairs.** For two fixed native commands with disjoint endpoints, every pair of actual raw Kraus operators commutes on the full 66- or 78-state carrier. Both executions must finish. The comparison permutes command-instance slots with their exported and retained records, and relabels the known program identifiers; their literal digests differ. The resulting complete classical-quantum channels are identical for every initial state and untouched reference. All $25^2$ raw-slot pairs, including holds and zero addresses, are retained. Adjacent swaps extend the identity to fixed schedules related by such disjoint interchanges. The literal chronological tapes need not agree, and the shared queue's implementation dependencies are unchanged. Overlapping endpoints, selectors using earlier records and memory timeouts give explicit failures. This is a finite equivalence of observation protocols; it supplies no physical event quotient, spatial dimension or light cone (`model/native_record_causality`, [1]).

**Canonical source restriction.** The fair connected federation has $V_\ell = 240\,4^\ell$ ports and a declared global occupation $m$. Its canonical state $\rho_{V,m} = \Pi_m / \binom{V}{m}$, where $\Pi_m$ projects onto total occupation $m$, is preserved by each fixed fair native seam's unread source channel. Tracing to any $v$ ports gives the full number mixture

$$\rho_W = \sum_{k=0}^{v} h(k)\rho_{v,k}, \qquad h(k) = \frac{\binom{v}{k}\binom{V-v}{m-k}}{\binom{V}{m}},$$

with impossible binomial coefficients zero. These restrictions compose exactly for nested regions. Orthogonal number sectors give the half trace distance $D(\rho_W, \rho_{v,q}) = 1 - h(q)$; imposing a sharp local number is a different state. For $p = m/V$ and $\beta_p = \mathrm{diag}(1-p, p)$, sampling with and without replacement gives

$$D(\rho_W, \beta_p^{\otimes v}) \le \min\{1, v(v-1)/(2V)\}, \qquad D(\rho_W, |0\rangle\langle 0|^{\otimes v}) \le \min\{1, vm/V\}.$$

A common complete local instrument contracts these bounds, including its records; a correlated external reference needs a joint input bound. The number law requires $O(v)$ rational operations. All thirteen outcomes of a twelve-port restriction at $\ell = 200$ are evaluated exactly with the 408-bit population integer, without allocating the global state. These are static state comparisons. They select no physical capacity map, occupation trajectory $m(V)$, finite preparation, clock or dynamical restriction. The fair federation is distinct from the biased local member used above (`model/native_capacity_restriction`, [1]).

The existing independent-cell grammar supplies a distinct dynamic extension. Append unused 66- or 78-state factors to any admitted native repair/hold program, preserving its parameters, active cell addresses and record controller. Every raw branch is $K_h \otimes I_B$. Consequently partial trace over $B$ intertwines the complete adaptive instrument for arbitrary correlated inputs and untouched references. The comparison retains chronological records, FIFO exports and matched stopping outcomes, and checks the exact lifted program identifier before relabeling it. These restrictions also compose. Declared allocation and specification queries can distinguish the programs. The host evaluator stops before executing products above 64 cells, so equality of executed histories requires a common admitted and evaluated domain. Compiled coherent source-word services have a separate grammar and are not lifted by this theorem. This proves a spectator-factor extension without connected geometry or interactions. A direct single-attempt comparison with the fair federation fails: the same coin record has probabilities 25/33 and 1/2, giving total variation at least 17/66 (`model/native_capacity_restriction/PRODUCT.md`, [1]).

**Connected weighted references.** The same endpoint rule has a consistent weighted extension on the retained connected federation graph. For positive vertex weights $w_i$, write $e_m(w)$ for the elementary symmetric polynomial and set $\pi_m(S) = \prod_{i\in S} w_i/e_m(w)$ on $|S| = m$. Every seam preserves this reference by conditional heat-bath detailed balance. Its full restriction to $A$ has number weights

$$h_A(k) = \frac{e_k(w_A)e_{m-k}(w_{A^c})}{e_m(w)}.$$

These restrictions compose. Repeating the local weight word gives the H66 reference conditional on two loads in a cell; cross-cell repair changes that local number. The faithful product family with occupations $p_i = zw_i/(1+zw_i)$ also restricts exactly for fixed fugacity $z$ and inherited weights. The modular transfer $j\log(w_a/w_b)$ is independent of total occupation and graph size. It is not a particle mass.

For a fixed independent seam selector, replacing crossing repairs by a complete hold control changes the $t$-attempt channel by at most $1-(1-\beta)^t$ in half diamond norm, where $\beta$ is its boundary-selection probability. Each cell has thirty internal and three crossing seams. Under the uniform seam selector, through the $k$th internal repair, the no-crossing probability is $(10/11)^k$ at every federation level; increasing exterior capacity does not establish vanishing error in that local clock. Symmetrizing the same population law gives local positive terms

$$h_{ab} = \frac{1}{w_a + w_b}\begin{pmatrix} w_b & -\sqrt{w_a w_b} \\ -\sqrt{w_a w_b} & w_a \end{pmatrix}$$

on $|10\rangle, |01\rangle$, zero on $|00\rangle, |11\rangle$. Their weighted sum annihilates $\sqrt{\pi_m}$. This is a relaxation representation. The weight assignment and fugacity are declared, and the native program admission rule rejects weighted connected federations; no source selection, prepared coherent ground or relativistic limit is inferred (`model/weighted_source_family`, [1]).

**Local weighted state limits.** The repeated native weight word admits a controlled local limit without enumerating its global carrier. Let $B$ be the number of twelve-port cells, declare total occupation $rB$ with $r \in \{2, 4\}$, and choose the positive fugacity solving

$$1000z^2 + 32z - 1 = 0 \quad (r = 2), \qquad 800z^2 - z - 2 = 0 \quad (r = 4).$$

The product reference $Q_B = Q_1^{\otimes B}$ then has mean $rB$, and its conditioning on that total is exactly the canonical reference $P_B$. A binomial-mode bound gives $Q_B(S = rB) \geq 1/(240B)$. Cell exchangeability and the finite entropy chain rule distribute this conditioning cost: for every set of $k$ cells, including $k$ that do not divide $B$,

$$D(P_k \| Q_1^{\otimes k}) \leq \frac{k \log(240B)}{B}, \qquad \tfrac{1}{2}\|P_k - Q_1^{\otimes k}\|_1 \leq \min\left\{1, \sqrt{\frac{k \log(240B)}{2B}}\right\}.$$

Here $D$ is classical relative entropy, the second inequality is finite Pinsker, and every local number sector is retained. A common trace-preserving local instrument contracts the bound; a numerical map with bounded gain requires that gain in the comparison. The instrument must act on every retained local number sector: a two-load coherent service does not automatically extend to this mixture. Correlated references require a joint input-state bound. For $B = 20\,4^{200}$ the one-cell bound is below $2\,10^{-60}$, computed with integer and rational enclosures. Matching mean occupation two does not confine a cell to its two-load sector. The theorem selects neither density nor physical capacity and supplies no preparation time or connected dynamical limit (`model/weighted_local_limit`, [1]).

**Local records and finite-volume response.** For a fixed repair schedule, a local observation consists of a final port region and selected complete local records at their original event positions. An untouched reference is retained; remote record values and the final exterior are traced. Reverse scanning the schedule adds the endpoints of each retained read and each unread command touching the accumulated support. Every disjoint unread complete channel cancels under partial trace. Thus the complete observed instrument factors exactly through the initial state on this dependency cone and its reference, including initial coherences and correlations. State-independent random selectors can be averaged afterward. Record-adaptive selectors and intrinsic queue halts require separate analysis; the full global archive is a different observable.

Compare with a finite port region whose crossing and exterior commands act as holds, retaining the original selector and clock. Coupling the selector tapes bounds half the diamond distance of the two local output channels by the probability that the dependency cone exits the region. An initial joint state error adds to this bound. For independent seam choices, root set $S$, maximum degree $d$, maximum seam probability $q_*$ and graph-distance margin $r$, a union over ordered paths gives the sufficient bound

$$p_{\mathrm{exit}} \leq \min\{1, |S| \tbinom{T}{r+1} (dq_*)^{r+1}\}.$$

The roots include both endpoints of every retained state-dependent read. This conservative cone can exceed the support obtained from the precise record times.

For independent uniform seam selection and repair probability one, on the actual fair connected source at levels zero through two, each cell has thirty internal seams and three distinct exterior neighbors, each with five seams leaving the one-neighbor halo. For a whole-cell root with retained internal-cell records, this reduces cone exit to four transient states $n = 0, 1, 2, 3$, with $n$ counting the halo ports reached. With $M = 630\,4^{\ell}$ seams, the only nonzero entries are

$$Q_{nn} = 1 - \frac{3+4n}{M}, \qquad Q_{n+1,n} = \frac{3-n}{M}, \qquad p_{\mathrm{exit}} = 1 - \mathbf{1}^{\top} Q^{T} e_0 \le \min\{1, 15\tbinom{T}{2}/M^2\}.$$

Here $Q^T$ denotes the $T$th power. At level zero and $T = 30$ this probability is below 0.015. The combinatorial attachment below proves the halo geometry for every returned source routing. Stopping immediately after the $k$th internal repair instead gives

$$R = 30(D - B_0)^{-1}, \qquad D_{nn} = 33 + 4n, \quad (B_0)_{n+1,n} = 3 - n, \qquad p_{\mathrm{exit}} = 1 - \mathbf{1}^{\top} R^{k} e_0.$$

The elapsed global time is retained without conditioning. The exit probability is independent of $M$, equals $677/50061$ at $k = 1$, and is at most $\min\{1, k(k+1)/120\}$. For bounded scores $|f| \le 1$ and an analysis parameter $0 \le q < 1$, the normalized Abel response error is at most

$$2\left[1 - (1-q)\mathbf{1}^{\top}(I - qR)^{-1} e_0\right] \le \min\{2, q/[30(1-q)^2]\}.$$

At $q = 1/2$ the sharper value is $151/3432$. These are complete local-record response bounds, with no pole-energy interpretation. The weighted connected extension has the same conditional locality construction but no native admission; its reference is not assigned to the fair source. The proof and bounded interpreter checks supply neither a physical clock nor a relativistic limit (`model/native_local_response`, [1]).

**Occupation-sector readout.** On any connected binary source with $V$ ports, the conserved total $m$ and complete records of $V - 1$ fixed-seam repairs along a spanning tree determine the initial configuration. If $s_i$ is the accumulated recorded displacement, the incoming contrast on $(a, b)$ gives the initial difference $d - s_a + s_b$. Propagate these differences to offsets $o_i$; the common offset is $(m - \sum_i o_i)/V$. Replaying the currents also determines the terminal configuration. Every nonzero complete word therefore has the map $\sqrt{w_h}|y_h\rangle\langle x_h|$, including arbitrary untouched references, and coarse graining by $x_h$ gives the configuration projectors. Without a supplied total, the empty and full configurations share one rank-two branch that preserves their coherence. All other supported words decode a unique configuration. This ambiguity cannot be removed by record-adaptive repair selectors.

The actual connected members at levels zero through two admit trees with 239, 959 and 3839 commands. An exact lazy query of a complete raw word uses the original continuation kernel and retains its archive; it does not turn the expanded evaluator's resource stop into a completed execution. A final tree read permits reverse-current reconstruction of a preceding complete repair history. Its terminal basis state is outcome-labelled. For any controller of at most $T$ repairs, positive placement probabilities retain a no-motion event from every basis input with probability at least $(1 - r + r p_{\min})^{T}$, where $r$ is the repair probability. On a nontrivial fixed-number sector, this obstructs exact deterministic reset of every input to one target by bounded repair-only control. It permits approximate or heralded preparation and the separately declared coherent services (`model/native_sector_readout`, [1]).

**Native correlation sectors.** For the fair connected source with $M$ seams, the population transfer is $P_q = I - \sum_{ab}(I - S_{ab})/(2M)$ on the $q$-occupation sector. Let $U$ add a site to a subset, $D = U^*$ and $W_r = \ker D_r$. The standard harmonic decomposition of the Boolean slice [27] follows from $DU - UD = (V - 2r)I$ on degree $r$. Its source-preserved degree-$r$ subspace has dimension $\binom{V}{r} - \binom{V}{r-1}$, for $r \le \min(q, V-q)$, and is unitarily equivalent to $P_r|_{W_r}$. This is a decomposition of the unread transfer, with no assertion that each recorded quantum branch preserves these subspaces.

For two disjoint native seams $(a, b), (c, d)$, put $h = e_{ac} - e_{ad} - e_{bc} + e_{bd}$ and $\phi = (n_a - n_b)(n_c - n_d)$. Then $D_2 h = 0$, and in the uniform fixed-number reference

$$v_q = \mathbb{E}\phi^2 = 4\frac{\binom{V-4}{q-2}}{\binom{V}{q}}, \qquad \frac{\mathbb{E}[\phi(X_0)\phi(X_t)]}{v_q} = \frac{h^* P_2^t h}{4}, \qquad 2 \le q \le V - 2.$$

An actual four-read experiment realizes this curve: read both seams, score their outgoing contrasts $(d_1 - 2j_1)(d_2 - 2j_2)$, run $t$ uniform attempts, then read both seams and score the incoming contrasts $d_3 d_4$. The complete fair instruments are unital; the first scored output is $\phi\tau_q$ and the last scored effect is $\phi$. Thus their joint score equals the displayed covariance with measurement backaction included. Every raw flag is retained. Using incoming contrasts at both ends changes the experiment and fails this identity. A preparation error $\epsilon$ in half trace distance gives normalized error at most $2\epsilon/v_q$; $v_q$ decreases as $V^{-2}$ at fixed $q$. The normalized poles are independent of occupation on a fixed graph, not of graph size or clock. Different correlation degrees may share poles. Neither degree nor pole is assigned a particle identity (`model/native_correlation_sectors`, [1]).

**Preparing native correlation records.** The stationary response can be prepared at the level of its complete signed score from an arbitrary quantum input in the same fixed-number sector. For possibly different past and future seam contrasts $\phi_i, \phi_j$, let $H_e = (I + S_e)/2$ on occupation functions. Pulling the entire four-read experiment back to its initial cut gives the diagonal effect

$$A_{ij,t} = H_{e_i} H_{f_i}[\phi_i P_q^t \phi_j], \qquad \|A_{ij,t} - \mathbb{E}_{\tau_q} A_{ij,t}\|_2 \le \sqrt{v_q}.$$

The two past seam averages include measurement backaction. This effect has squarefree degree at most four. Put $s = \min(4, q, V - q)$ and $N_s = \binom{V}{s} - 1$. The mean-zero polynomial space is invariant under all site permutations, even when the graph is not vertex-transitive. Its orthogonal projector therefore has constant diagonal $N_s$ in the uniform $L^2$ normalization. Cauchy–Schwarz and the exclusion gap $\gamma = \lambda_1(L)/(2M)$ give, after $n$ independent fair warmup repairs,

$$\frac{\|Z_R - C_{ij}(t)\rho_R\|_1}{v_q} \le \sqrt{N_s/v_q}\,(1 - \gamma)^n.$$

Here $Z_R$ is the terminal reference operator weighted by the complete score, and $R$ is any untouched reference initially correlated with the source. The diagonal input blocks on $R$ are positive and have total trace one, which proves this extension. Warmup records are summed without conditioning; their full joint law is not asserted to be stationary.

Integer square roots and the bound $(1 - \gamma_0)^{\lceil 1/\gamma_0 \rceil} \le 1/2$ give sufficient attempt budgets from a rational gap lower bound. On the fair 66-state source, 352 warmup repairs followed by four reads and two lag attempts give normalized error at most $33/65536$; exact propagation of all input columns gives a tighter rational enclosure. At fixed density the prefactor grows polynomially rather than with the full sector dimension. The gap and sample cost retain their capacity dependence. All raw records and host stops are retained; assigning score zero to stopped runs adds $p_{\mathrm{stop}}/v_q$ to the bound. Single-program budgets retain the original $2^{32}$-attempt limit. The same argument applies

to any separately implemented degree-four scored effect using its exact centered norm. It supplies neither full-density preparation nor a physical vacuum (`model/native_correlation_preparation`, [1]).

**Native spatial motion and clocks.** Midpoint subdivision preserves a simple closed triangular complex; each cell has three distinct neighbors. The source assigns their crossing edges to distinct ports, so each of the three halo ports has five seams leaving the halo and none joining another halo port. This proves the four-state halo premise at every combinatorial level for every returned routing. It does not assert successful floating-point construction or host execution at arbitrary levels.

In the existing auxiliary affine cell-centroid dictionary, normalized to unit circumsphere, all ports of a cell share one coordinate. Writing $n = 2^\ell$, there are $30(n^2 - n)$ within-face and $30n$ original-face hinge crossings. Their exact total squared length is

$$E_\ell = 20 - 4\sqrt{5} + \frac{-40 + 16\sqrt{5}}{3\,2^\ell} < 12.$$

For a stationary uniform single-unit source, $P = I - L/(2M)$ with $M = 630\,4^\ell$ seams and $V = 240\,4^\ell$ ports. Positivity gives $I - P^T \le T(I - P)$ and therefore

$$\mathbb{E}\|x(X_T) - x(X_0)\|^2 \le \frac{TE_\ell}{MV} \le \frac{T}{12600\,16^\ell}.$$

Markov's inequality gives a displacement-probability bound by dividing by the squared threshold. At $T = \lfloor 1260\,8^\ell t \rfloor$, this mean square is at most $t/(10\,2^\ell)$. It is a finite stationary path bound in comparison coordinates, with no physical light-speed assignment.

For a fixed set of $d$ tick seams, a completed tick consists of a geometric number of remote repairs followed by one incident repair. Both conditional transfers preserve the uniform state. The squared triangle inequality and stationarity bound displacement after $k$ completed ticks by $k^2/(10d\,4^\ell)$. Substituting the mean elapsed time into the deterministic-time bound would not justify this result. A finite attempt cap retains timeout as a separate outcome. Along a shortest path of $D$ edges to a receiver, one positive native history of weight $(2M)^{-D}$ arrives at its first incident tick; that tick has no strict port-distance support bound uniform in capacity. Global attempt locality is unchanged. A final tree detector and the body currents reconstruct the recorded path; the prepared law, auxiliary coordinate dictionary and complete archive access are explicit inputs (`model/native_spatial_scaling`, [1]).

**Spatial pair responses.** Use one declared $2 \le q \le V - 2$ and its reference $\tau_q$, the same fair graph and the same attempt clock for a menu of disjoint-seam contrast pairs. The complete reflected probes give

$$C_{ij}(T) = \frac{v_q}{4} h_i^* P_2^T h_j, \qquad B_{ij}(T) = C_{ij}(T)/v_q.$$

The matrix $B(T)$ is positive semidefinite; $B(0)$ is the probe Gram matrix, not necessarily the identity. The two-subset coefficients implement the exact intertwiner without changing the working occupation to two. For a radius-$r$ local comparison containing every probe endpoint, maximum degree $\Delta$ gives normalized response error at most $2\min\{1, |S|\binom{T}{r+1}(\Delta/M)^{r+1}\}/v_q$. The compared window retains all local number sectors, and remote raw values are traced at this observation boundary. A global archive-dependent controller is outside that locality bound.

The source also fixes a finite exclusion correction. Compressing the independent two-walker generator to distinct symmetric pairs gives $G_{\mathrm{ind}} = G_2 + 2A$, where $A$ projects onto adjacent pairs.

Its collision-retaining control has surviving transfer $R = P_2 - A/M$ and an absorbing collision flag. The identity

$$P_2^T - R^T = \sum_{k=0}^{T-1} P_2^{T-1-k}(A/M)R^k$$

is entrywise nonnegative. Its column sum is the complete collision probability, bounding the normalized probe-response difference under the initial coefficient law $|h_j|/4$. No survivor renormalization occurs. On a four-port path, the one-attempt discrepancy is $1/12$; matching the one-body transfer therefore does not determine the pair response.

For bounded macroscopic profiles, subtracting the constant and one-body components of a product of native density fields gives an exact primitive degree-two field on the same $q$ slice. With $V_\ell = 240\,4^\ell$, $q_\ell = V_\ell/3$ and $\lfloor 1260\,16^\ell \tau \rfloor$ attempts, along the existing native density subsequence, these fields converge to $W_\tau(f,g) = {:}Y_\tau(f)Y_\tau(g){:}$, with

$$\mathrm{Cov}(W_\tau(f,g), W_u(h,k)) = K_{\tau,u}(f,h)K_{\tau,u}(g,k) + K_{\tau,u}(f,k)K_{\tau,u}(g,h).$$

Here $Y$ is the inherited Gaussian density field and $K$ its heat covariance. Canonical exponential-moment bounds give uniform eighth moments and justify the product and covariance limits. A final native spanning-tree detector and reverse-current readback access these occupation snapshots; this global observation is distinct from the local four-read menu. The result is the quadratic sector of a Gaussian diffusive field. The finite exclusion correction does not by itself establish an interacting relativistic continuum (`model/native_pair_response`, [1]).

**Native attempt time and the local exchange action.** The same source fixes $G_q = \sum_e (I - S_e)$ and $P_q = I - G_q/(2M)$. Its bound $10I/21 \le P_q \le I$ gives, by scalar spectral calculus,

$$\left\| P_q^n - e^{-nG_q/(2M)} \right\| \le \frac{3}{5n}, \qquad n \ge 1.$$

Indeed, for $p \ge 10/21$ and $\delta = 1 - p$, $0 \le -\log p - \delta \le 21\delta^2/20$; maximizing $n\delta^2 e^{-n\delta}$ proves the bound. It holds on every occupation sector, independent of its dimension, and bounds each entry of the normalized stationary response matrix. The rescaled time is the exact count $t_n = n/(2M)$, without a replacement Poisson scheduler or a physical time unit.

Generator convergence has a narrower domain. On correlation degrees at most $r$, $\|G_q\| \le 2r\Delta$ and, for $M > r\Delta$,

$$0 \le -2M \log P_q - G_q \le \frac{r^2\Delta^2}{M - r\Delta} I.$$

At half occupation the full-sector generator difference is at least $M/16$, although its endpoint semigroups obey the dimension-independent bound above. Exact sparse moments give rational enclosures of both response functions. Preparation and local-window errors add with their stated normalizations. The Poisson-count representation of this exponential has a different complete history-length law; the endpoint estimate identifies neither that law nor a native unitary implementation. These results fix a finite local exchange response, with no chiral matter, Lorentzian action or physical mass assignment (`model/native_exchange_clock`, [1]).

### 5.7 An observable native contact mode

The fair twelve-port action at working reference $I_{66}/66$ has an observable correlated mode. On unordered distinct pairs write $G_2 = \sum_e (I - S_e)$ and let $W_2 = \ker D$, where $D$ sums pair coefficients

at each port. Its dimension is 54. If $Q$ projects onto $W_2$ and $A$ onto adjacent pairs, compression of the independent one-body sum gives $G_2 + 2A$. On $W_2$ this yields

$$H = G_2|_{W_2}, \qquad H_0 = H + 2QAQ \geq 2(5 - \sqrt{5})I.$$

The coefficient two follows from the literal collision loss; it is not fitted. For the adjacent-coordinate insertion $E_A$, put $U = QE_A$ and $R_0(z) = (H_0 - z)^{-1}$. Direct multiplication proves

$$(H - z)^{-1} = R_0 + 2R_0U(I - 2U^*R_0U)^{-1}U^*R_0.$$

Below the stated independent-pair threshold, the middle kernel detects eigenvalues with multiplicity. This is a finite comparison threshold, without an infinite-volume scattering interpretation.

Exact integer characteristic polynomials give a fivefold native eigenvalue $E$ at the lowest root of

$$E^3 - 30E^2 + 280E - 776 = 0, \qquad 5 < E < 51/10.$$

It lies more than $2/5$ below the comparison threshold. The actual seam contrast $(n_0 - n_1)(n_2 - n_4)$ has a twelve-dimensional cyclic primitive space. Its exact scalar resolvent has coprime numerator and denominator, so all twelve distinct primitive poles have positive weight. At this lowest line the normalized weight obeys $1/100 < w_E < 1/50$; decimals are $E \simeq 5.01827276928$ and $w_E \simeq 0.01943798293$. Complete reflected four-E records have raw weight $(2/33)w_E$ and attempt transfer factor $1 - E/60$. All raw branches and measurement disturbance are retained.

The isolated-cell mode does not extend unchanged to the connected source. For any normalized vector in the fifteen-dimensional sum of the cubic eigenspaces, every three-port boundary attachment satisfies

$$\|(I - P_{\text{cell}})G_{\text{federation}}J_{\text{cell}}f\|^2 > 1/100.$$

Exact integer principal minors verify this for all 220 possible boundary triples. Thus its unchanged embedding has residual greater than $1/10$, independent of capacity. A dressed candidate needs a separate residual, spectral isolation, localization or propagation, and accessible-residue bound. At fixed total occupation two the raw probe variance is $8/[V(V-1)]$, so normalized spectral weight alone does not ensure feasible sampling. These are native correlation poles, with no physical particle or mass assignment (`model/native_contact_response`, [1]).

### 5.8 Connected contact response

The connected comparison has a different threshold from the isolated cell. Let $V = 240\,4^\ell$, $M = 21V/8$, and let $Q_2$ project pair coefficients onto the primitive degree-two space. The actual comparison is $H_0 = Q_2(G_2 + 2A_{\text{adj}})Q_2$ on that space, where $A_{\text{adj}}$ marks adjacent occupied ports. Give every port in a cell the same auxiliary affine triangle-centroid coordinate $f$. Exact icosahedral symmetries give $\sum_i f_i = 0$, $|f_i| \leq 1$, $V/5 < A = \sum_i f_i^2 \leq V/3$, and $\sum_{ij\in E}(f_i - f_j)^2 < 4$. These are comparison coordinates. The primitive trial vector is

$$w_{ij} = f_if_j - c_i - c_j, \qquad c_i = \frac{-f_i^2 + A/[2(V-1)]}{V - 2}.$$

Its squared norm is at least $V^2/75$, its exchange energy at most $4V/3$, and its contact energy less than $6V$. Rayleigh's principle therefore gives

$$0 < \lambda_{\min}(H_0) \leq \frac{550}{V} = \frac{55}{24\,4^\ell}.$$

The bound uses the exact cell adjacency and the source routing postconditions; it does not assert that the floating constructor returns at every level. The isolated pole $E > 5$ lies above this connected comparison bottom. Its isolated threshold is therefore insufficient to certify global binding. Bound bands at fixed momentum and resonances require separate tests.

The source also supplies a controlled exterior correction. Retain the 66 pair configurations wholly inside the chosen cell and write $G_2 = \left(\begin{smallmatrix} A & B^* \\ B & D \end{smallmatrix}\right)$. The literal boundary gives $A = G_{\mathrm{cell}} + N_\partial$, $B^*B = N_\partial \le 2I$ and $0 \le D \le 24I$. For $s > 0$ set

$$F(s) = A + sI - B^*(D + sI)^{-1}B, \quad F_J(s) = A + sI - \sum_{k=0}^{J} \frac{B^*(24I - D)^k B}{(24 + s)^{k+1}}.$$

The core resolvent is $F(s)^{-1}$. Positive inverse ordering and the geometric remainder give

$$0 \le F^{-1} - F_J^{-1}, \qquad \|F^{-1} - F_J^{-1}\| \le \frac{2}{s^3}\left(\frac{24}{24 + s}\right)^{J+1}.$$

Only finite exterior paths enter each coefficient. No projection back to the cell's primitive space is inserted during an excursion. For the normalized four-E probe $u = h/2$, $s = 24$ and $J = 2$, the connected response $s\langle u, (G_2 + sI)^{-1}u\rangle$ is enclosed between 0.660081931825 and 0.660515959604 on the evaluated cell-zero windows at levels zero through two. The isolated value exceeds the connected enclosure by more than $1/120$.

For the canonical total-two reference, this response is accessible through complete native lag experiments: with $q_s = 2M/(2M + s)$,

$$s\langle u, (G_2 + sI)^{-1}u\rangle = (1 - q_s)\sum_{n \ge 0} q_s^n C(n)/v_V, \qquad v_V = \frac{8}{V(V - 1)}.$$

Truncation at $N$ costs at most $q_s^{N+1}$. Preparation errors and independent sample costs have explicit allowances; host stops remain distinct outputs. The discount is an analysis parameter. It adds no native random scheduler or physical clock. The negative-spectral-parameter resolvent bound does not continue a pole through the exterior spectrum at positive energy (`model/native_connected_contact`, [1]).

### 5.9 Positive-energy correlations from open records

On the finite configuration transfer space, the constant unit vector $\Omega$ is the unique ground of $H_M = -2M \log P_q$. It differs from the native working density $I/\binom{V}{q}$. For normalized primitive pair fields, the complete reflected record statistics determine, using the isometric primitive two-occupation representative of $H_M$,

$$W_{ij}(t) = \langle h_i, e^{-itH_M} h_j\rangle/4.$$

This matrix has a positive spectral measure on nonnegative energies. It is a reconstructed ground correlation, without a claimed unitary implementation on the working carrier.

The continuation has a controlled capacity dependence. On primitive degrees at most $r$, choose $b = \lfloor M/(2r\Delta)\rfloor \ge 1$, $Q = P^b \ge I/2$ and $u = 2Mt/b$. With $c_0 = 1$, $c_{k+1} = c_k(k - iu)/(k + 1)$ and $N = \max(1, \lceil |u| \rceil)$,

$$V_m(Q) = \sum_{k=0}^{m} c_k(I - Q)^k = \sum_{j=0}^{m} a_j Q^j, \quad \|V_m(Q) - Q^{iu}\| \le 2^N - \sum_{k=0}^{m} \binom{N + k - 1}{k} 2^{-k}.$$

Substituting the actual lag-$bj$ four-E statistics gives an explicit signed complex reconstruction. Its systematic error adds $A_m\eta$, where $A_m = \sum_j(|\Re a_j| + |\Im a_j|)$ and $\eta$ bounds each normalized prepared score. For $m+1$ settings, variance-bounded independent repetitions give sampling error at most $s$ with failure probability $p$ when each setting has at least $(m+1)A_m^2/(pv_q^2s^2)$ shots. This bound can be impractical; preparation, sampling and host-stop costs are not removed by analytic reconstruction. The displayed $t = 1/64$, order-12 example requires 116424316 attempts at level zero under its sufficient warmup bound. The corresponding levels one and two exceed the unchanged $2^{32}$ attempt ceiling, and the conservative 99% sampling certificate requires more than $10^{17}$ total repetitions. These are sufficient budgets, without an optimality claim.

The reference distinction has an operational witness. Bare native record effects are diagonal, so an ideal diagonal phase on the working carrier changes no complete record law, even with arbitrary initial reference correlations. In the transfer ground the same field instead has susceptibility $-2\Im W_{ii}(t)$, with a strictly positive finite-kick margin certified by native moments and a Taylor remainder. Identifying these interventions and readouts without an additional map would assign two different responses to one experiment. A physical dictionary must preserve the intervention statistics as well as the reconstructed correlations (`model/native_vacuum_response`, [1]).

**Complete finite energy windows.** The same blocked transfer defines positive effects $B_k = \binom{n}{k}Q^k(I-Q)^{n-k}$, $0 \le k \le n$, whose sum is the identity. Their probabilities are exact linear combinations of the complete native lag scores at $0, b, \dots, nb$. Assign $k > 0$ the counted-energy label $-(2M/b)\log(k/n)$ and retain $k = 0$ as overflow. At a transfer eigenvalue $x \ge 1/2$, the output is binomial, so

$$\Pr(|k/n - x| \ge \delta) \le \frac{1}{4n\delta^2}, \qquad |E_k - H(x)| \le \frac{2M\delta}{b(1/2-\delta)}$$

outside that exceptional event, for $0 < \delta < 1/2$. The total coefficient amplification of the measured, nonzero-lag moments is $A = 3^n - 1$. Normalized moment error $\eta$ gives histogram half-$\ell^1$ error at most $A\eta/2$. Sampling error $s$ with failure probability $p$ has the sufficient budget $nA^2/(4pv_q^2s^2)$ independent repetitions per setting. Native local preparation on the fair 66-state source makes the $n = 16$ full-vector preparation error less than $4 \times 10^{-11}$, with at most 3871 native attempts per setting. The sufficient one-percent sampling budget exceeds $3.2 \times 10^{25}$ total repetitions. This is a reconstructed spectral statistic; the effects are not thereby an executed energy instrument. Physical mass, a laboratory energy unit and an autonomous relativistic action require separate identifications (`model/native_energy_identification`, [1]).

**A complete controlled energy instrument.** The fair 66-state carrier also admits a destructive energy measurement using its declared coherent services. An exact rational eigenframe certificate gives a nearby generator $\tilde{H}$ with $\|H - \tilde{H}\| \le 27 \times 10^{-8}$ in the counted convention $H = -60\log P$, without assuming separated eigenvalues. For each certified normalized ray $u$, let $C$ be the existing source encoding and $p$ a padding address. The phase word on $z = (Cu + |p\rangle)/\sqrt{2}$ implements $U = I - 2|z\rangle\langle z|$, whose source-success block is $I - |u\rangle\langle u|$ and padding row is $-\langle u|$. Thus a padding flag detects that ray destructively; the surviving complement continues to the next filter. The sixty-six energy effects and final residual sum to the identity by telescoping. This removes the second phase word of the general weak filter, which acts trivially on this surviving complement.

All sixty-six words are separately admitted, totaling 3,853,872 pulses. Exact primitive certificates and the eigenframe residual bound give complete half-diamond error below 1/12000, including every processor failure, with stored-map gain below 1001/1000. The ideal instrument is normalized; stored outputs have that bounded gain. This is a sequential composition of declared source

services, with no monolithic repair-program admission or generic-host expansion asserted. Every stored success prefix is evaluated with a retained reference control.

The record-prepared noncentral total-two reference described below feeds this same instrument, with exact acceptance 7771/65536 and explicit preparation rejection. Exact rational matrix products give the entire accepted energy law, including the residual. No successful branch is renormalized in that full output. Repair attempts and service pulses have distinct counters; physical duration and a rest-mass identification are separate maps (`model/native_energy_measurement`, [1]).

The existing source phase experiment supplies a second ideal reference $\rho = P/56 \geq I/84$ on the same configurations. Its exact modular generator is $-\log \rho = H/60 + \log(56)I$, identifying the controlled operator families under a known factor of sixty. The preparation retains its selector timeout and numerical-density admission scope; unread repair is a different evolution. This reference and the diagonal tree reference are operationally distinct: their norm-one occupation-exchange expectations differ by $1/168$, so their half trace distance is at least $1/336$. Their separate regional-modular and energy-clock properties cannot be combined as properties of one state.

**A fair ground, local energy and complete readout.** For the connected fair binary source put $G = \sum_e (I - S_e)$, $P = I - G/(2M)$ and $H = -2M \log P$. A triangle-cover argument gives $P \geq \alpha I$, $\alpha = 10/21$, uniformly on the growing federation. The fixed-number uniform vector $\Omega$ is the unique ground on each connected occupation sector. On the sixty-six-state source a declared coherent preparation, following eleven native tree attempts, has complete half-diamond error below 1/4,000,000. A local occupation phase creates an excitation with $\langle G \rangle = 50/33$ and $H \geq G$. The matched preparation, control and complete energy detector use sixty-eight words and 3,877,451 service pulses. Scoring all failures as zero gives an unnormalized mean-score contrast greater than 3/50; the excitation is detected without a successful-branch normalization. This is a finite excitation-energy observation, with no stable-particle or rest-mass assignment (`model/native_fair_ground_readout`, [1]).

The same source admits an explicit comparison between its logarithmic clock and local exchange dynamics. Write $\tau = I/D$, $\rho = P/\operatorname{Tr} P$, $c = \operatorname{Tr}(\tau P)$, $\mu = \operatorname{Tr}(\rho G)$ and $a = (1 - \mu/(2M))^{-1}$. For the weighted Hilbert–Schmidt norm,

$$\|[H, X] - a[G, X]\|_{2,\rho} \leq \frac{\|[G, X]\| \sqrt{\operatorname{Var}_\tau(G)/c} + \frac{1}{2}\|[G, [G, X]]\|}{2M\alpha^2}.$$

Exact occupation moments bound the variance by $O(M)$ on the bounded-degree graph. Fixed local observables, times and complete classical intervention histories therefore have error $O(M^{-1/2})$. All failures and zero slots are retained. This is a state-weighted comparison, not an arbitrary-input channel-norm claim. On $A\Omega$, positivity instead gives $\|(e^{-itH} - e^{-itG})A\Omega\| \leq |t|\|[G, [G, A]]\|/(4M\alpha)$, with a corresponding fixed-word bound. A growing ground preparation is an additional premise.

Writing $h_e = I - S_e$ gives antisymmetric local currents $J_{e \to f} = i[h_e, h_f]$, supported on adjacent seams, and exact conservation of $G$. Native unread repair is a different operation: on the faithful sixty-six-state reference it increases the mean $G$ by 37/840. For its unital channel the mean increase in $H$ is at least as large. The complete energy instrument resolves a score increase greater than 0.034 for the ideal reference, with separately counted preparation error. Source-plus-apparatus conservation requires an apparatus-energy observable. None of these graph currents supplies a symmetric Lorentz stress tensor (`model/native_local_energy_clock`, [1]).

The dilute ground imposes a separate constraint on its observable limit. For a bounded number-

preserving operator $A$ on $s$ ports,

$$\mathrm{Var}_{\Omega_{V,q}}(A) \le 4\|A\|^2 \left(1 - \frac{\binom{V-s}{q}}{\binom{V}{q}}\right) \le 4\|A\|^2 \frac{sq}{V}.$$

Thus fixed-support bounded connected correlations vanish when $q/V \to 0$, uniformly in controlled time. For a fixed sequence of complete local instruments on supports $s_j$ whose fine Kraus operators each preserve local occupation, the terminal source and archive have half trace distance at most $\min\{1, 2\sum_j \sqrt{s_j q/V}\}$ from the scalar empty-sector protocol. The comparison retains every branch. A clock rescaling cannot restore a vanishing pole residue. Finite density, growing supports, rescaled fields or charged operators require separate limits; this bound does not exclude them (`model/native_dilute_ground_limit`, [1]).

### 5.10 Controlled transfer state and time

The weighted 66-state source admits a distinct operational construction using its declared coherent services. Its scalar record transfer has

$$S = \Pi^{-1/2} P \Pi^{1/2}, \quad \frac{296}{495} I \le S \le I, \quad \Omega_x = \sqrt{\pi_x}, \quad H = -\log S.$$

An eleven-E configuration detector has fine maps $\sqrt{w_h}|y_h\rangle\langle x_h|$. Retained-record routing followed by the native amplitude-tree word with rational weights $\pi_x$ changes these to $\sqrt{w_h}|\Omega\rangle\langle x_h|$. Thus the ideal prepared source factors from the retained archive and any untouched reference, even for an initially correlated input. The archive can remain correlated with that reference.

Transfer evolution has a finite residual certificate. Rational proposed columns $V$ and diagonal $(296/495)I \le \Lambda \le I$ satisfy $\|V^*V - I\| \le g = 10^{-9}$ and $\|SV - V\Lambda\| \le r = 10^{-9}$, checked by exact rational arithmetic and directed square-root intervals. The polar comparison $W = V(V^*V)^{-1/2}$ is unitary and obeys $\|W - V\| \le g$. Consequently

$$\|S - W\Lambda W^*\| \le r + 2g, \qquad \|\log S - W \log \Lambda W^*\| \le \frac{r + 2g}{296/495}.$$

The logarithm bound follows by integrating the resolvent identity. The actual compiler uses normalized rational columns of $V$ and their existing rank-one phase words; it does not execute the polar map. Telescoping the 66 phases bounds their ideal unitary error by $|t|(r + 2g)/(296/495) + 528g$. Scalar logarithm intervals and each stored primitive's certificate supply the additional numerical allowance.

One complete experiment prepares $\Omega$, phases configuration $a = (0, 1)$ by $i$, applies the controlled transfer evolution at $t = 1/10$, and decodes the initial configuration of a second eleven-E detector. The unphased control has probability $p = \pi_a = 625/7505$. Writing $u(t) = \langle a|e^{-itH}|a\rangle$, the intervention has

$$p_a(t) = p\,|1 + (i - 1)u(t)|^2.$$

Here $H_{aa} \ge 32/495$ and $\|H\| < 103/200$. A spectral Taylor remainder proves $p_a(1/10) - p > 1/1000$. The complete protocol has 68 separately admitted coherent words, at most 3,877,451 pulses and 22 native E commands. Every first processor failure and padding abort is retained and scored zero. Passing primitive gates bound complete half-diamond error below $1/11000$ per experiment and gain below $1001/1000$. Charging the full trace-norm error for both rounded, potentially unequal-trace outputs leaves a stored-map score increment above $1/2000$. These rounded scores are approximations to the ideal probability law.

This supplies a controlled finite preparation, intervention and readout for the source-derived transfer Hamiltonian. The nine-bit direct compiler and lazy detector retain the generic evaluator's separate resource stops. The parameter $t$ is dimensionless transfer time, not elapsed repair time or laboratory seconds. Autonomous unitary repair, a growing coherent embedding, a physical vacuum and a relativistic particle spectrum do not follow from this controlled experiment (`model/native_transfer_control`, [1]).

### 5.11 Local preparation and readout on one attempt clock

The fair federation admits a controlled preparation on each twelve-port cell, including every local occupation sector and an arbitrary untouched reference. Run its eleven-edge E tree, then $0 \le b \le 68$ repetitions of two ordered sweeps through the thirty internal seams. Each sixty-attempt population matrix is exactly $A_q/2^{60}$ on sector $q$. Integer enumeration for all $0 \le q \le 12$ proves double stochasticity and $\sum_y \min_x A_q(y,x)/2^{60} \ge 1/2$. Minorization therefore contracts half-$\ell^1$ distance by at least a factor $1/2$ per block.

This population argument has a complete quantum attachment. Each nonconstant tree record identifies a rank-one branch $2^{-11/2}|y\rangle\langle x|$; constant branches preserve the entire empty/full corner $P_c$. Retain the tree records and trace only the subsequent mixing-record values in the comparison. The resulting channel is within half-diamond distance $2^{-b}$ of the tree followed by

$$\mathcal{R}(\rho_{AR}) = P_c \rho_{AR} P_c + \sum_{q=1}^{11} \tau_q \otimes \sum_{|x|=q} \langle x|\rho_{AR}|x\rangle.$$

Here $\tau_q$ is the uniform local sector state. Its weights are inherited from the input. All physical records remain in the archive; replacing the source by $\tau_q$ while retaining every mixing record has error $1 - 1/\binom{12}{q}$ on a known basis input, regardless of mixing time. Thus this preparation does not erase archive correlations or select a global density or pure vacuum (`model/native_fair_local_preparation`, [1]).

Complete occupation readout uses the same native records. For a connected $b_0$-port window, track all incident currents, including boundary repairs. If $s_i$ is accumulated displacement, the first occurrence of an internal tree edge yields its initial contrast $d - s_a + s_b$. Tree integration fixes every nonconstant initial bit pattern. Global occupation $q$ is public source specification: empty and full windows are feasible precisely when $q \le V - b_0$ and $q \ge b_0$. A constant pattern is therefore resolved whenever only one is feasible. In particular, for $0 < q < 12$ a fresh eleven-E intracell tree gives exact occupation readout across every feasible local sector, with its terminal disturbance and arbitrary reference.

For a general window and $0 < q < V$, $T$ uniform E attempts cover its tree except with probability $p_{\mathrm{cov}} \le (b_0 - 1)2^{-\lfloor T/M \rfloor}$. Keep coverage failures and unresolved constants as erasures. The full labelled output has half-trace distance exactly $p_{\mathrm{cov}} + (1 - p_{\mathrm{cov}})\eta_{\mathrm{amb}}$ from an initial occupation measurement followed by the same native instrument. The unresolved constant weight is $\eta_{\mathrm{amb}}$; for the canonical fixed-$q$ marginal it is bounded by $(1 - q/V)^{b_0} + (q/V)^{b_0}$ when both constants are feasible, and vanishes otherwise. This oracle is a comparison channel, not an extra implemented service. Worst-case input error need not be small (`model/native_fair_local_readout`, [1]).

Preparation and readout share the actual attempt counter. With $b = 64$, preparation followed by a fresh intracell tree takes 3862 attempts and literal nodes; the sparse-sector comparison error is at most $2^{-64}$. This two-tree program permits $b \le 67$ under the fixed node ceiling. For a 348-port certified window at level six and canonical $q = V/3$, preparation followed by $64M$ uniform attempts uses 3852 nodes and 165154571 attempts, with combined error below $10^{-16}$. These are admitted

finite instruments, with expanded host stops retained. In a separate short-time comparison with binned Poisson clocks, reserving $L$ fixed native operations costs at most $KL/M$ for $K$ window seams, in addition to the existing cone, selector and joint-state errors. The long readout estimate is not identified with that short-time limit (`model/native_fair_local_clock`, [1]).

The occupation chart also binds an explicitly supplied proper-code load/export service to the actual twelve local source bits and exterior, intertwining every native C/E raw branch. After a fresh E tree, its decoded nonconstant branch selects an occupied/unoccupied seam. A beam, optional phase, inverse beam and complete compiled E detector use 305 pulses. Their ideal unconditional score increment equals the actual nonconstant branch weight $a$; certified stored primitives give score increment at least $a\,9999999/10000000$, retaining constant outcomes, all first failures and rounded number leakage. The rounded register stays in the apparatus. The supplied quantum coupling is an operational premise; pulses and native attempts have distinct counters without a derived physical time conversion. Sector-uniform mixed states alone are invariant under these number-preserving controls. No autonomous field action or particle mass follows from this conditional experiment.

**A faithful noncentral reference from the same records.** One sixty-E mixing block after the tree has exact integer transition counts $A_q[x, y]$ with denominator $2^{60}$. Every column is faithful and nonuniform on each recognized local sector $1 \le q \le 11$. Choose the explicit observer protocol $a_i = 1 + \mathrm{depth}(i)$ on the native tree and $W(x) = \prod_{i \in x} a_i$. Define $L_q = \min_{x,y} \lfloor A_q[x, y]/W(x) \rfloor$. Accepting the first $L_q W(x)$ lexicographic coin paths to endpoint $x$ prepares $\sigma_q = \mathrm{diag}(W)/Z_q$ exactly, independently of the recognized tree terminal state $y$. Exact backward path counts compute the flag from the retained tape. Conditional acceptance is at least $1/12$, and $\sigma_q \ge I/8192$. Rejects and coherent constant-code outcomes remain explicit; unconditional acceptance can vanish on a constant input. The complete source word has seventy-one attempts. The reference observation retains the tree and flag and traces only the sixty mixing values. Since $-\log \sigma_q = (\log Z_q)I - \sum_i (\log a_i) n_i$, natural number-preserving regional modular actions agree on overlaps. This observer-selected protocol supplies no unique vacuum or geometric modular clock (`model/native_noncentral_reference`, [1]).

**Occupation current and complete local histories.** For a rational profile $\phi$ on the retained ports, extend $\phi$ by zero outside and set $A_\phi = \sum_v \phi_v n_v$. Each actual seam record gives $n'_a = n_a - j$, $n'_b = n_b + j$. Thus its weighted occupation increment is $(\phi_b - \phi_a)j$, including crossing seams. A fine history is a scaled partial permutation $K_h$; if $\Delta_h$ is the summed increment, then

$$K_h^* A_\phi K_h = K_h^* K_h A_\phi + \Delta_h K_h^* K_h.$$

This proves the discrete continuity identity for arbitrary input and untouched reference. The decoded initial mask and retained currents determine every local snapshot on the same attempt clock. Keeping that initial label preserves the complete instrument comparison, so its error has no factor for the number of snapshots. Erasures receive no invented absolute trajectory. In the 3862-attempt preparation/readout join this history bound concerns the final eleven detector attempts, with preparation mixing values traced as specified. Occupation conservation supplies no physical gauge charge or stress-energy tensor by itself (`model/native_local_charge_history`, [1]).

## 5.12 Complete local records at growing capacity

The capacity comparison also admits complete local quantum instruments. Fix observed seams and final ports with endpoints in a finite set $S$, and a radius-$r$ neighbourhood $B$ with $v$ ports and

$K$ internal seams. Retain every observed native tag, coin, raw value and current, its order, and the final quantum state with an untouched reference. Remote records and exact global tick addresses are traced by this observation, while remaining available in the native archive. Choose a common partition $0 = t_0 < \cdots < t_b = t$ with $Mt_j$ integral and run $T = Mt$ native attempts. Each observed event is assigned its partition bin; its order within the bin is retained. Here $t$ is per-seam attempt time, twice the exchange time above, with no physical calibration.

The complete instrument differs from its restriction to $B$ by at most

$$a_N = \min\left\{1, |S|\binom{T}{r+1}(\Delta/M)^{r+1}\right\}$$

in half-diamond norm. Conditional on the input-independent selector word, the backward cone seeded at terminal $S$ contains every observed record root. Leaving $B$ requires a time-ordered path of $r+1$ seams, which proves the bound for arbitrary coherent inputs and references.

Compare the restricted process to independent rate-one Poisson attempt clocks on its $K$ seams, applying the same native instrument at each event. In one interval of length $1/M$, the native window selector is empty with probability $1-\lambda$, or has each seam with probability $1/M$, where $\lambda = K/M$. The corresponding Poisson probabilities are $e^{-\lambda}$ and $e^{-\lambda}/M$, with every multiple-event word retained. Their exact total variation is $\lambda(1-e^{-\lambda}) \le \lambda^2$. Coupling all $T$ intervals and applying the identical conditional quantum maps therefore gives

$$\tfrac{1}{2}\|\Phi_B - \Phi_{\mathrm{Pois},B}\|_\diamond \le d_N := \min\{1, tK^2/M\}.$$

Aligned boundaries give the same bin label on matched events. Exact continuous timestamps have a singular comparison with native grid times, and the full global record-count laws differ, so neither observation can replace the declared local binned one.

If initial joint window/reference states differ by half-trace distance $\epsilon$, their complete observed outputs differ by at most $\min\{1, \epsilon + a_N + d_N\}$. For a canonical fixed-$q$ source and an independent common reference, comparison to a declared Bernoulli density $p$ gives

$$\epsilon \le \min\{1, v(v-1)/(2V) + v|q/V - p|\}.$$

Matching rooted windows thus give an explicit comparison between source capacities. If all fixed-radius windows stabilize and their state restrictions converge, the additional Poisson cone bound $|S|(\Delta t)^{r+1}/(r+1)!$ constructs a limiting local instrument. The evaluated source levels zero through two have identical cell-zero radius-one windows with 15 ports and 33 seams, with a verified orientation-preserving record map. This establishes neither a unique all-radius routed refinement nor the physical density or spatial dimension. The proof and exact rational controls retain every overflow and distinguish kernel analysis from bounded native execution (`model/native_local_history_limit`, [1]).

### 5.13 Rooted source refinement

The exact oriented midpoint stencil supplies a constructive rooted cell topology. Choose the cell that follows child zero at every subdivision around one original vertex. Inserting an additional leading child zero identifies its radius-$r$ dual ball, including incident edges, whenever $2^\ell > r+2$. In each of the five incident original faces, a dual step increases the triangular grid's radial coordinate by at most one. This proves stabilization of every finite corner ball and consistency under restriction. The resulting countable graph retains the five-face corner; no assertion about an arbitrary root sequence follows.

A recursive local evaluator also computes the literal midpoint coordinates, face orientation and port scores without allocating the sphere. It reproduces coordinates bit for bit and routed neighbor rows at the independently evaluated levels zero through three. Exact binary32 score intervals certify first-maximum decisions, including stable ties. Collision repairs require a unique certified assignment. Uncertain decisions are rejected. The finite address domain extends through level fourteen, where every source edge key fits its signed integer representation. These are finite numerical attachments to the oriented topology.

The certified tables give matching windows of cell radii one, two, four and eight at specified levels through fourteen. Keeping every port of a radius-$r$ cell ball contains the required radius-$r$ port ball. Canonical names preserve source endpoint order and therefore intertwine the complete native instruments, including signed currents and zero slots. For declared density $p$, the canonical totals $q_\ell = \lfloor p\,240\,4^\ell \rfloor$ have the local state bound above; the Bernoulli comparison states restrict consistently. The radius-four, level-fourteen window has 348 ports. The observation retains records on the thirty root-internal seams and the final state on the twelve root ports; other window records are traced. At $p = 1/3$ and $t = 1/630$, its complete local-output comparison bound is below $10^{-6}$. The corresponding 268435456-attempt description is admitted. The expanded host retains its unevaluated-carrier stop; this is a finite kernel bound, without a claimed execution of the global experiment.

An infinite routed process requires a specified arithmetic continuation. Finite-width level tags and coordinate collapse in bounded local tests prevent treating the literal numerical constructor as a total such specification. Exact coordinates, arbitrary precision and combinatorial port assignment are different continuations. The construction chooses none of them and establishes no unique all-radius routed limit, physical density or Lorentzian field (`model/native_rooted_refinement`, [1]).

### 5.14 Native clock observables

The declared coherent clock services also give operational noncommuting measurements on the same biased 66-state source. On the existing seam $(1, 7)$, pair $|1, k\rangle$ with $|7, k\rangle$ for each of the ten spectators. Let $X, Y, Z$ be the direct sum of the usual Pauli matrices on these pairs, zero on the 46-dimensional complement $P_0 = I - Q$. The existing balanced cap and quadrature words give analyzers $U_A$ with $U_A^\dagger Z U_A = A$. One native contrast-reading repair after the analyzer therefore has effects

$$P_{A,\pm} = \frac{Q \pm A}{2}, \qquad P_{A,0} = P_0, \qquad A \in \{X, Y, Z\}.$$

All 25 raw addresses are retained, including 19 zero maps. The six live maps have coin factors $8/33$ and $25/33$. The zero-contrast maps preserve all coherences within $P_0$, including coherence between empty and doubly occupied endpoints. The raw update is destructive. Its retained displacement record permits an explicit correction: apply two existing balanced cap rotations when the repair moved, then undo the analyzer. Every fine corrected Kraus operator is a scalar sign times $\sqrt{p_{\mathrm{coin}}} P_{A,\delta}$ on all 66 states. This gives the three-outcome Lüders instrument, with its coin refinement and arbitrary untouched references.

For the full normalized stationary density, no odd-sector conditioning is needed:

$$\langle X \rangle = \langle Y \rangle = 0, \quad \langle Z \rangle = \frac{1649}{7505}, \quad \operatorname{Var} X = \operatorname{Var} Y = \frac{3201}{7505}, \quad \frac{|\langle [X, Y] \rangle|^2}{4} = \left(\frac{1649}{7505}\right)^2.$$

These actual observables and state supply the finite Robertson theorem's inputs. With $\Delta =$

$\log(25/8)$, the working modular clock acts by

$$U_s = \rho_*^{-is}, \qquad U_s^\dagger X U_s = \cos(s\Delta)X - \sin(s\Delta)Y.$$

The literal nine-qubit analyzer words have full-address certificates, retained export failures and a separate multiplication-rounding allowance. These bounds concern the stored numerical primitive matrices, with no hardware or unrounded analytic exponential error claim. The coherent services and their feedback composition are declared observer operations; they are outside the bare repair/hold grammar. This finite measurement menu supplies neither universal implementation independence for every complex effect nor a physical position–momentum relation (`model/native_clock_observables`, [1]).

**Configuration readout and repeatable preparation.** There is a bounded contrast-record detector on this same sector. Perform one native E repair on each edge of a fixed spanning tree of the twelve ports. Let $u_i$ be the accumulated recorded change of occupation at port $i$. An incoming contrast $d$ on $(a, b)$ gives the initial difference $n_a(0)-n_b(0) = d-u_a+u_b$; the retained current updates $u$ after that read. The eleven differences and $\sum_i n_i(0) = 2$ determine the initial configuration. Thus every nonzero fine history has a rank-one map $K_h = \sqrt{p_h}\,|x_h\rangle\langle x_0|$, and coarse graining by the decoded $x_0$ gives the full computational projective measurement. All $25^{11}$ formal raw words are retained in an exact lazy representation; $66\,2^{11} = 135168$ words have nonzero maps. One queue slot and ten exports suffice for the eleven-step record protocol. Streamed branch continuation uses the existing native transition kernel and authenticates its records. The stock expanded evaluator's resource stop after three steps is retained as a separate return; it is not equated with the lazy evaluator. The nine-bit coherent words use the direct source compiler; the generic word evaluator has a six-qubit host ceiling.

The terminal configuration is also known from the currents. Existing coherent bit routing maps its code either back to $x_0$ or to a fixed target code. The ideal corrected maps therefore give the basis Lüders instrument or the replacement channel $\rho \mapsto |x_*\rangle\langle x_*|\,\mathrm{Tr}\,\rho$, including untouched references. This supplies repeatable preparation to the existing pair-analyzer architecture: its 4291 settings span all 4356 Hermitian coordinates. Complete compiler bounds and export outcomes accompany these compositions. The rounded maps have bounded normalization error; their positive record functionals retain full rank without output renormalization. Coherent service semantics are declared, and finite tomographic coverage does not prove universal coexistent-effect realization or a growing field-observable family (`model/native_effect_coverage`, [1]).

**Rank-one measurement synthesis.** The supplied coherent services implement more than the fixed tomographic menu. For any nonzero Gaussian-rational vector $v$ on the 66 source states and rational $0 \le \lambda \le 1$, with each input numerator and denominator of at most 64 bits, consider $E = \lambda|v\rangle\langle v|/\langle v, v\rangle$. In the existing 512-address register, append the amplitude $\sqrt{1-\lambda}$ at one padding address. A binary amplitude tree and diagonal phase word prepare the resulting normalized vector from address zero. Executing the inverse word and routing zero to a genuine source code before complete export and tree readout gives $E$ as the ideal selected effect. All other source outcomes, padding aborts and processor failures are retained. No extra register or conditioning on source-code success is used, and an effect alone specifies no Lüders update.

Directed integer trigonometric enclosures select a 41-bit angle grid; exact Walsh identities compile its layers into existing native pulses. The full-register operator error from target angles is at most $11\,2^{-41}$. The emitted word uses at most 23579 pulses, below the unchanged source-word limit. The certificate checks every emitted stored six-dimensional primitive, including negative inverse

requests and fixed basis-change angles, against its full ideal matrix and rejects a failed bound. For a passing word with $N$ pulses and $g = 2\,10^{-11}$, put

$$x = Ng, \qquad e = \frac{x}{1-x} + 11\,2^{-41}, \qquad F_N = \frac{Ng^2}{(1-x)^2}.$$

Telescoping the good-code products and retaining every first processor failure gives complete half-diamond error at most $[e(2+e) + F_N]/2 < 10^{-6}$, including arbitrary untouched references. These are bounds on the stored primitive maps; floating propagation arrays are diagnostics. The direct nine-bit compiler executes the tape, while the generic six-bit word evaluator returns its resource stop. The real projector onto $(3|0\rangle + 4|65\rangle)/5$ also has an 80-pulse construction with error below $2\,10^{-9}$, despite lying outside the positive cone of the fixed balanced-analyzer menu. Fixed precision and word length supply neither arbitrary-accuracy dense effects nor universal coexistent refinement. The composition consumes the declared coherent services and Kraus probability semantics; it does not derive those premises from repair counting (`model/native_effect_synthesis`, [1]).

**Complete coexistent weak measurements.** The same source services give joint noncommuting effects with their disturbance. For a normalized rational-direction vector $u$, projector $P = |u\rangle\langle u|$ and rational $0 \le s \le 1$, couple $u$ to one existing padding address by a rank-one phase. Complete source export and a second correcting phase give the ideal surviving contraction $L = I + (s-1)P$ and abort effect $(1-s^2)P$. The abort retains its reference state and terminates; the success remains a quantum input to the next stage. For at most eight such stages, put $A_0 = I$ and $A_j = L_j A_{j-1}$. The complete effects are

$$E_j = A_{j-1}^*(1-s_j^2)P_j A_{j-1}, \qquad E_{m+1} = A_m^* A_m, \qquad \sum_{j=1}^{m+1} E_j = I.$$

The last identity telescopes, including every earlier disturbance. With $u_1 = |0\rangle$, $s_1 = 3/5$, $u_2 = (|0\rangle + |65\rangle)/\sqrt{2}$ and $s_2 = 4/5$, the $(0, 65)$ entry of $[E_1, E_2]$ is $216/3125$. Their coexistence follows from one common complete instrument.

Each stage uses two separately admitted 58392-pulse words in the existing nine-bit work register. Every first processor failure, padding abort and residual outcome is retained, with no intermediate destructive tree readout. Conditional on all exact primitive checks passing, a gain-aware complete half-diamond bound for eight stages is below $1/50000$. The generic evaluator retains its resource stop; the direct compiler and lazy complete instrument have their own stated domain. This bounded family supplies noncommuting coexistent effects but does not implement every effect or discharge the universal operational Born premises (`model/native_coexistent_effects`, [1]).

**A prepared phase contrast under fair repair.** A separate twelve-bit occupation-register experiment targets the fair source's mixed state $\rho = P/\operatorname{Tr} P$, where $P = \frac{1}{30}\sum_e Q_e$, $Q_e = (I + \mathrm{swap}_e)/2$ and $\operatorname{Tr} P = 56$. Declared coherent services target the ideal successful blocks $q\rho$ and $qD\rho D$, with $D = (-1)^{n_0}$. Their ideal signed single-occupation exchange readouts on ports $(0, 1)$ are $\pm q/168$, $\pm 5q/1008$ and $\pm q/240$ before repair, after C and after E, respectively, where $q = 1 - (368/2048)^{16}$ is the unconditional selector acceptance. The exact difference operator is multiplied by $5/6$ under C and $7/10$ under E, although the two repairs share their population transfer. A separate rational implementation checks the numerical source execution's prepared density blocks with Frobenius tolerance $2\,10^{-10}$, checks outside-carrier mass separately, and compares every outcome with its target. Source failures, selector timeout and all terminal labels are retained. These bounded

numerical checks supply no arbitrary-reference channel-error certificate. The mixed preparation differs from the transfer ground; its decaying phase contrast identifies neither a protected sector nor a particle mass. Matching a population transfer alone does not establish a common state-and-instrument dictionary (`model/native_phase_experiment`, [1]).

### 5.15 Protected charge and its native reference

An explicit composition of the biased native repair and its persistent ordering bit connects protection to a state and a modular clock. For the binary total-two source, let $\theta_0 = \theta_1 = 25$ and $\theta_i = 8$ otherwise. Its reference is $\rho_* = 7505^{-1} \sum_x \prod_i \theta_i^{x_i} |x\rangle\langle x|$. The ordered branch is a weighted partial injection with an ordering flip $X$ when transport crosses the occupied spectator. All branches commute with $Q = I_{66} \otimes X$. Their complete star algebra has commutant $\mathrm{span}\{I, Q\}$. A faithful invariant state makes this the adjoint fixed space, so the stationary densities are exactly

$$\rho_\alpha = \rho_* \otimes [\alpha P_+ + (1-\alpha) P_-], \qquad P_\pm = (I \pm X)/2, \quad 0 \le \alpha \le 1.$$

The weight $\alpha$ is conserved. With the ordering bit prepared in either sharp sector, 1600 unread attempts prepare the source/reference marginal within $10^{-6}$ of its stationary product; the complete archive is not claimed independent. The proof includes arbitrary untouched references.

For $0 < \alpha < 1$, put $K_\alpha = -\log \rho_\alpha$. A recorded current $j$ on seam $(a, b)$ obeys $[K_\alpha, K_h] = j \log(\theta_a/\theta_b) K_h$. Thus every complete branch is covariant under the same modular clock. The negative exchange sector has the actual two-occupation CAR transport sign. Its closed odd exchange returns to the initial occupation and has zero modular energy change. The two protected sectors differ in mean modular energy by $\log[\alpha/(1-\alpha)]$, set by their conserved reference weight. Protection therefore supplies no nonzero rest energy. The potential commutes with occupations; its unitary flow is not the stochastic motion. These statements concern the composed 132-state source, whose ordering memory is an additional declared service (`model/protected_reference`, [1]).

The fixed-selector raw instrument has the exact weighted reverse $\rho_\alpha^{1/2} K_h^\dagger \rho_\alpha^{-1/2} = K_{\bar{h}}$, where a label involution exchanges each moving outcome with its reverse and fixes the holding and zero slots. This proves Kubo–Martin–Schwinger detailed balance and equality of stationary fixed-length record probabilities under reversed order and labels. At $\alpha = 0, 1$ the identity uses the invariant supported corner. It concerns stationary preparation and this selector; it supplies neither a reversal theorem for arbitrary feedback nor physical time reversal (`model/protected_reference/REVERSAL.md`, [19, 1]).

**Complete gauge transport.** The same literal quantum-link matrices supply a complete locally covariant operation for each of the five matter blocks, including colored matter. For a block of dimension $d$, let $A = \sum_{ab} |a\rangle\langle b| \otimes U_{ab}$ and $B = A/\sqrt{d}$. Both the raising coordinates and their conditional rotor completion obey $(A^\dagger A)^2 = d A^\dagger A$. On the two-position edge define

$$T = |x\rangle\langle y| \otimes B, \qquad P_y = T^\dagger T, \quad P_x = T T^\dagger.$$

The endpoint intertwining equations imply that $T, P_x, P_y$ commute with the total local Gauss action. Since $P_x P_y = 0$, the complete directional instrument is $\{T, I - P_y\}$, its reverse is $\{T^\dagger, I - P_x\}$, and $W = T + T^\dagger + I - P_x - P_y$ is a Hermitian unitary. Every blocked flux state is retained. The exact identities cover all sixty link states and all twelve currents; the hard-core spectator contributes the usual CAR sign.

The covariant unitaries $U(c, s) = I + (c-1)(P_x + P_y) - is(T + T^\dagger)$, $c^2 + s^2 = 1$, expose the unselected operation parameter. On a fixed gauge-singlet preparation with a dual charge reference

at $y$, the complete detector $\{P_x, I-P_x\}$ gives probability $s^2$. The rational choices $(c,s) = (3/5, 4/5)$ and $(5/13, 12/13)$ give $16/25$ and $144/169$ on that same experiment. Covariance and completeness therefore do not determine its evolution. This finite operation uses a declared matter/link extension, without equivalence to the original C/E law or a full collision extension (`model/gauge_defect_transport`, [1]).

The protected reference and this transport admit a specified common edge process. On native seam $(0, 11)$ the move $\{5, 11\} \to \{0, 5\}$ crosses the spectator, so its joint operator is $\widetilde{T} = |\{0,5\}\rangle\langle\{5,11\}| \otimes X \otimes B$. The forward and reverse coins have probabilities $25/33$ and $8/33$. Their complete transport/hold instruments preserve $Q$ and the reference $\rho_* \otimes \sigma_\alpha \otimes I_{60d}/(60d)$, where $\sigma_\alpha = \alpha P_+ + (1-\alpha)P_-$, by exact endpoint detailed balance. They retain the other 64 configurations as holds. This full reference is gauge invariant without support on the joint Gauss kernel. A declared dual reference gives an orthonormal Gauss-singlet pair $\psi_x, \psi_y$ and stationary corner state $(25|\psi_x\rangle\langle\psi_x| + 8|\psi_y\rangle\langle\psi_y|)/33$. In a fixed memory-charge sector one unread attempt replaces every admitted corner input by this mixture, including reference-entangled inputs. Its complete records are retained before taking that marginal. The same modular reference assigns forward energy change $\log(8/25)$. This is a declared one-edge law, with a designated moving internal fiber and separate singlet preparation. It supplies no all-seam gauge network, global mixing theorem, two-particle colored identity or selected coherent Hamiltonian (`model/protected_reference/GAUGE_JOIN.md`, [1]).

The same corner supplies a finite thermodynamic instance. Summing all four outcomes gives the diagonal kernel $P(i,j) = p_j$, $p = (25/33, 8/33)$. With $E_i = -\log p_i$ and multiplier one, the Lean theorem `corner_same_kernel_four_laws` identifies this kernel with the heat bath consumed by the finite four-law core (`Lean/Thermodynamics/ProtectedGaugeCorner.lean`, [1]). It gives the first-law split, relative-entropy contraction and multiplier identifiability. The energy is nonconstant, so the replacement exchanges modular heat for some inputs. Its fixed-spectrum mathematical Gibbs family has zero low-temperature entropy; the fixed native coin supplies no implemented temperature variation, physical calibration or uniform capacity limit.

**Gauge-invariant propagation on trees.** The configured sixty-state quantum-link Hamiltonian has a complete spatial sector on a finite rooted tree $\mathcal{T}$. Retain all five native matter blocks, fix total fermion number one, and put a static dual reference for one block $r$ of dimension $d_r$ at the root. Every block has nonzero central charge and scalar current commutant. Leaf elimination therefore forces all links off the root-to-fermion path to vacuum; irreducibility and inequivalent central charges leave one contraction along that path. The entire Gauss kernel has an orthonormal basis $|v; r\rangle$, one state for every vertex, including the root.

For the owning literal raising-link hopping of strength $J$ and the electric occupation-projector cost $\epsilon \geq 0$, the exact restriction is

$$H\big|_{\ker G} = J\sqrt{d_r}\ \mathrm{Adj}(\mathcal{T}) + \epsilon\, \mathrm{diag}(\mathrm{dist}(0, v)).$$

The factor $\sqrt{d_r}$ comes from the norm of the contracted flux string; substituting normalized partial-isometry hopping would change the configured action. Position projectors commute with Gauss and form a complete detector, with every position and repeated-position outcome retained. Their readout can change hopping energy, whose exact work balance is accounted for.

On an additionally specified infinite bounded-degree tree, positive $\epsilon$ gives a self-adjoint operator with compact resolvent: its diagonal potential tends to infinity and hopping is bounded. For initial support inside radius $R_0$, compression to radius $R$ gives the complete continuous-time position-

record total-variation bound

$$\min\left\{1,\ 2\sum_{n\geq R-R_0+1}\frac{(D_{\max}|J|\sqrt{d_r}\,t)^n}{n!}\right\},$$

where $D_{\max}$ bounds degree and $t$ is the total evolution time across all readouts. The executed Cayley clock has a separate finite-step comparison allowance. These reference-bound dressed levels use a specified graph, link, coupling and clock. They supply no freely propagating asymptotic particles, relativistic dispersion or physical mass (`model/native_gauge_strings`, [1]).

**Source coupling readout and its color-action boundary.** On one link, the same Gauss-singlet source states give the exact block

$$H_r=\begin{pmatrix}0 & J\sqrt{d_r}\\ J\sqrt{d_r} & \epsilon\end{pmatrix},\qquad p_1(t)=\frac{4d_rJ^2}{\epsilon^2+4d_rJ^2}\sin^2\left(\frac{t\sqrt{\epsilon^2+4d_rJ^2}}{2}\right).$$

Here $p_1$ is an unconditional endpoint probability; the other outcome and all measurement back-action are retained. For $J\neq 0$, $p_1''(0)=2d_rJ^2$ and $\epsilon^2=-p_1''''(0)/p_1''(0)-2p_1''(0)$. Two finite-time probability intervals at $t$ and $2t$ supply rational Taylor enclosures, avoiding an assumed derivative oracle. The configured source sets $J=\epsilon=1$ in its model clock, hence determines the clock-independent ratio $\epsilon/|J|=1$. Reading these specifications is sufficient to fix that model ratio.

The admitted color currents have $\mathrm{tr}_3(T_aT_b)=\delta_{ab}/2$, whereas their selected fifteen-state matter lift has $\mathrm{Tr}_{15}(T_aT_b)=2\delta_{ab}$. These trace conventions must not be interchanged. The five nonvacuum source link sectors have $(C_3,C_2,Y^2)$ equal to

$$(0,0,1),\ (4/3,0,1/9),\ (4/3,3/4,1/36),$$
$$(0,3/4,1/4),\ (4/3,0,4/9).$$

Their configured electric energies are all one. No coefficients $a,b,c$ give this vector as $aC_3+bC_2+cY^2$: the second and fifth entries force $c=0$, contradicting the first. The zero-energy link vacuum excludes an added common identity. Thus the source electric law cannot be directly identified with a quadratic gauge Laplacian. This does not exclude an emergent color action after a controlled coarse-graining.

For a separately justified continuum action $S_E=(\kappa/2)\int \mathrm{tr}_3\,\mathcal{F}_{\mu\nu}\mathcal{F}_{\mu\nu}$, with $\mathcal{A}=g_sA$, $\alpha_s=1/(4\pi\kappa)$ in the specified normalization. The finite gauge-invariance theorem permits every positive $\kappa$. The missing source identification is the continuum action coefficient and its renormalization scheme and scale. Running a supplied coupling does not select this boundary value. The exact finite readout and coefficient comparison are in `model/source_strong_coupling` [1].

**Complete color-loop readout in the interacting Higgs source.** The separate Casimir-weighted action retains its full 438- and 1,596-state Gauss sectors, with empty colored-fermion occupation, three identity family matrices and the stated scalar cutoff. Write $H=eE-m(L+L^*)+hT+gS+sN_H$, where $S$ is the actual Higgs vertex and $T$ includes the hopping sign. Its three colored link blocks have $(d_r,c_r)=(3,4),(6,25/8),(3,3)$, with $c_r=2C_3+C_2/2+(6Y)^2/12$. The vacuum input and its actual reference-neutral addition image are labeled $q=0,1$. The complete instrument resolves scalar occupation and both link blocks; all output states and zero-probability records are retained. Summing both scalar occupations in the equal-block loop outcome gives

$$\frac{p_r(t)}{d_r^2}=m^2t^2-m^2\left(\frac{e^2c_r^2}{3}+18m^2+\frac{qh^2}{3}\right)t^4+O(t^6).$$

The exact full-source moments cancel the Higgs contribution at fourth order; a sixth-order control detects its nonzero interacting contribution. For the two dimension-three blocks, let $M$ be the quadratic coefficient divided by nine and $D$ the difference of quartic coefficients divided by nine. Then $M = m^2$, $D = -7e^2m^2/3$ and $(e/|m|)^2 = -3D/(7M^2)$. This reads a clock-invariant bare ratio when $m \neq 0, e > 0$. Six positive times $t_j = j/1024$, exact polynomial evolution through order 28 and norm-controlled remainders enclose the default value one. The full source, diagonal energies and detector labels are regenerated. The preparation, flux instrument and Hamiltonian clock are declared finite interfaces. No repair-only flux compiler, chiral colored continuum or renormalized strong coupling is inferred (`model/color_loop_readout`, [1]).

**A local symmetry-sector excitation.** On the actual open eight-site selected spin ground at $J = 1, B = 1/4$, a one-site $Z$ pulse flips the global internal $X$ parity and leaves every initial exterior expectation unchanged. Second quantization includes the minus sign on doubly occupied sites; the full centered Hamiltonian and every complete recorded C/E branch commute with this parity. The protected single-occupancy spin code has identity repair branches with fair coin records. The ideal selected-state energy excess is enclosed by $[1.825861195059, 1.825861195062]$. Complete native preparation and detection execute in 1,090 and 1,092 pulses and certify pointer contrast above 0.67. The local commutator bound and complete source/spin comparison retain contrast above $2/5$ through at most ten repairs and time $1/16$ on $c = 12^4, U = 12^8$. This lifetime tape is an analytic construction; evolution spreads the excitation. The parity is a finite symmetry charge, not a topological boundary invariant or a physical mass (`model/defect_localization_readout`, [1]).

**Continuous charge, creation work and conjugate fields.** The exact full-Fock spin generator induces $Q = \sum_x \epsilon_x X_x/2$ in the Marshall frame. It commutes with the source action and every recorded repair branch. Ground uniqueness and a charge-reversing source automorphism force $Q\Omega = 0$. The complete local instrument $A_q = (Z_0 - qi\epsilon_0 Y_0)/2$, $q = \pm 1$, obeys $[Q, A_q] = qA_q$ and $\sum_q A_q^* A_q = I$. A local $Z$ followed by one $X$ pointer prepares these coherent branches. Measuring every local $X$ and grouping the outcomes gives the charge probabilities but dephases within a sector, so it is a different preparation.

The original weights for $q = -1, +1$ are approximately 0.83547507093 and 0.16452492907. A complete native Pauli estimator and an independent controlled-clock phase measurement recover the unconditioned work $W_q = \langle A_q^*[H_B, A_q]\rangle$. The conditional mean excitation energies are approximately 1.84463098607 and 1.73054620199 at $J = 1$. Both channels have positive spectral variance. Their first return-survival value 0.99 occurs in the certified source-time intervals $[0.1726210202, 0.1726210331]$ and $[0.1044566248, 0.1044566376]$, respectively. Return survival can rebound. A fixed uniformly chosen local-$X$ detector retains original-weight charge contrast $w_q/8 > 1/50$ at every source time. Thus charge conservation, return retention and a particle decay lifetime are distinct assertions. The generator action over an autonomous interval is $TE_{\mathrm{exc}}$ while ongoing system power is zero. Scalar bare-repair effects cannot measure these different energies; the coherent controls supply the sensitive readout (`model/source_energy_effort` and `model/charge_channel_lifetime`, [1]).

For the ordered source, define the second staggered configuration

$$Q_{2,p}(F) = p^2 \sum_x \epsilon_x F(px) S_x^2.$$

Append its bounded groups to the existing $Q_{3,p}, P_{2,p}$ histories. Internal rotation symmetry supplies

its variance and energy bounds. The equal-time estimate

$$\|[Q_{2,p}(F), Q_{3,p}(G)]\| \le \frac{p^4}{2} \sum_x |F(px)G(px)| = O(p)$$

and the full-history transport bounds construct commuting configuration groups on a common further subsequence. The source U(1) acts there with charged vectors $\Phi_\pm\Omega = (Q_3 \mp iQ_2)\Omega$. An odd microscopic translation combined with the internal axis-two $\pi$ rotation induces charge conjugation, commuting with the limiting energy and spatial translations. It gives

$$\mu_{\Phi_+\Omega} = \mu_{\Phi_-\Omega} = 2\mu_{Q_3\Omega}.$$

These measures have nonzero positive gapless spectral weight. This ordered-zero-field construction supplies no second uniform canonical momentum; no Lorentz spin, sharp shell or physical antiparticle follows from this internal symmetry (`model/protected_defect_catalog`, [1]).

For a normalized source-created vector $a \in \operatorname{Dom} K$, retain its full complement: $r = (I - |a\rangle\langle a|)Ka$ and $D = K$ compressed to that complement. Then

$$\langle a, (K+z)^{-1}a\rangle = \left[z + \langle K\rangle_a - \langle r, (D+z)^{-1}r\rangle\right]^{-1}.$$

The positive memory term is nonzero in both finite charge channels. Bounded sine preparation and the source resolvent filter construct this response on the same continuum history subsequence with nonvanishing original probability. Their filtered first two moments converge through bounded spectral integrands. This retains the spectral complement without fitting a mass or assuming a one-frequency field (`model/spin_dynamical_memory`, [1]).

A bounded response test avoids energy-domain assumptions. For an actual source-preparation isometry V and $S = z(K+z)^{-1}$, $z > 0$, put $A = V^*SV$. Then $D = V^*S^2V - A^2 = R^*R$, where $R = (I - VV^*)SV$. The response reduces K exactly when D vanishes. Identifying $A = z(\omega(P)+z)^{-1}$ additionally requires the actual spatial translation action. Small D controls bounded polynomials, but time comparison also needs energy-tail control. The complete native dimer distinguishes a nonreducing preparation from a reducing two-energy span, without identifying either as a particle mass (`model/spin_dynamical_memory/RESOLVENT_CLOSURE.md`, [1]).

There is a constructive hierarchy for this test. For admitted source creators $\psi_a = A_a\Omega$, let $\mathcal{H}_r = \operatorname{span}\{S^j\psi_a : 0 \le j < r\}$ and compress S to $T_r$ there. Splitting powers across the two retained vectors proves equality of source and compressed moments through degree $2r-1$. Bernstein polynomial approximation consequently gives, for a real L-Lipschitz score f and original weight $w = \|\psi_a\|^2$,

$$|\langle\psi_a, f(S)\psi_a\rangle - \langle\psi_a, f(T_r)\psi_a\rangle| \le \frac{wL}{\sqrt{2r-1}}.$$

Coherent Gamma–Cauchy time superpositions prepare $S^\alpha A_a\Omega$ from the same local histories, with time-tail weight at most $2\alpha/(\pi zT)$. Complete failed preparations, Gram floors and window boundary errors remain explicit. The closure of the source cyclic union reduces K; adjoining actual translation orbits also reduces those translations. Finite Ritz nodes are quadrature nodes, not proved source energy levels. The exact dimer resolves energies $9/8, 5/4$ and two finite reflection labels with all failures retained, without a continuum momentum or particle assignment (`model/spin_dynamical_memory/SPECTRAL_BRANCH.md`, [1]).

An isolated pole is not required for every charged physical sector [17]. A controlled infrared edge has a separate readout criterion. For an original subprobability measure $\nu$ with edge $\theta = \inf\operatorname{supp}\nu$, assume the source estimate $\nu([\theta, \theta+\epsilon]) \ge a\epsilon^\alpha$ for every sufficiently small positive $\epsilon$. Certified CDF

bounds $\nu((-\infty, x_-]) < a\epsilon^\alpha$ and $\nu((-\infty, x_+]) > 0$ imply $x_- - \epsilon < \theta \le x_+$, by interval inclusion. An empty finite bin cannot supply the tail premise. A source-derived relativistic dictionary and inclusive or dressed asymptotic detectors are additional requirements; the theorem assigns no OPH infrared mass (`model/spin_shell_readout/THRESHOLD_READOUT.md`, [1]).

**Bounded charge and pair preparation.** The finite selected reference can be prepared along a gapped source bond ramp. An exact basis change puts its positive-field action in the Lee–Yang class, with full gap at least $B/8$; a quantitative adiabatic bound and complete native compilation give an explicit preparation error. Interpolating boundary bonds preserves that gap. Source locality then bounds the interior marginal error by a surface factor times decaying functions of boundary distance, uniformly in exterior volume at fixed B. This supplies controlled finite positive-field references, without a zero-field marginal rate or a particle mass. The source substitutions, imported gap and adiabatic theorems, and complete implementation bounds are given in [6] and `model/spin_fixed_field_locality` [1].

A distinct joint family keeps $p = n^{-2}$, $B = Jn^{-5}$, the positive-field ground and its $H_B$ clock together. Its gap controls order and bond fluctuations. Auxiliary transverse and longitudinal phases then control the mixed momentum commutator through complete histories. The field limit has two regular pairs $(Q_3, P_2)$ and $(Q_2, P_3)$, with symplectic coefficient $m \int FG$, $m > 1/10$, and a full internal U(1). Even microscopic translations give common continuous spatial translations. Fixed-scale preparation and local records have one complete error budget. This is a chosen preparation/scaling prescription, not a theorem selecting it from unread repair or identifying it with the ordered-zero-field history state (`model/spin_joint_field_limit`, [1]). The actual filtered current has norm $O(p^2)$. A sufficient source-norm route to the linear force is a specific residual $o(p^2)$, measured with its original probability and retained complements. This cancellation and a sharp spectral branch are unproved.

The two-site positive ground marginal also gives a strict correction to mean-spin factorization. With transverse bond coefficient $\kappa_B$ and staggered moment $m_B$, the source dimer variational bound implies $\liminf(\kappa_B - m_B^2) \ge 61/14400$. Exact ground double commutators then exclude the unfiltered force coefficients $12m_B, m_B$ obtained by replacing each longitudinal spin by its mean. Corrected coefficients and an infrared sharp component are allowed; the result supplies no pole (`model/spin_joint_field_limit/QUANTUM_FORCE_SLACK.md`, [1]).

The actual two-force equations admit a useful static reduction. Here P is canonical field momentum, distinct from spatial translation momentum. For $q = Q\Omega$, $a = P\Omega$, put $h_Q = \|K^{1/2}q\|^2$, $\chi_Q = \|K^{-1/2}q\|^2$ and similarly for P, and $c = \mathrm{Im}\langle q, a\rangle$. With positive susceptibilities,

$$u = c/\chi_P, \quad b = c/\chi_Q, \qquad \delta_Q = h_Q - c^2/\chi_P \ge 0, \quad \delta_P = h_P - c^2/\chi_Q \ge 0.$$

These are exactly the squared $K^{-1/2}$-form norms of $iKq - ua$ and $iKa + bq$. On a positive energy window $[\ell, M]$ their ordinary norms are at most $\sqrt{M\delta_Q}$ and $\sqrt{M\delta_P}$. For $c \ne 0$, substitution bounds the original Q weight outside an $\epsilon$ tube about $\omega = \sqrt{ub}$ by $M(M\sqrt{\delta_Q} + |u|\sqrt{\delta_P})^2/(\omega^2\epsilon^2)$, divided by the actual preparation normalization squared. The filtered cross moment must use the same filter; it is not the unfiltered order commutator. Growing preparation normalization can suppress original success even for a nonzero field vector. Uniform source saturation remains unproved. The actual joint dimer has a zero-defect pair and a positive-defect pair on the same sharp finite energy line: polarization mismatch alone can make this sufficient whole-pair test fail (`model/spin_joint_field_limit/WARD_DYNAMICS.md`, [1]). Its actual conserved charge projectors resolve the latter into two zero-defect branches, each with original Q/P preparation weight greater than 1/25. Their filtered cross moments have opposite signs. This is a complete finite-source

resolution, without a continuum species assignment. For the uniform pair on $K_0 = H_B - E_B$, the source identity gives $\delta_P = 0$ and $\delta_Q = M_{-3}\,\mathrm{Var}_\pi(E^2)$, with $M_j = \int E^j\,d\nu$ for its configuration energy measure and $d\pi = E^{-3}d\nu/M_{-3}$. With N sites and $m = N^{-1}\langle\sum_x \epsilon_x S_x^1\rangle$, the joint clock obeys $h_L/\chi_L \le R = 6/(mn) + n^{-6}$ and $\mathrm{Var}_\pi(E^2) \le R^2/4$, with energy rescaled to L. Finite boxes with $m > 1/20$ are constructible. The normalized uniform preparation has original weight at least $m/(8N)$ below $\sqrt{2R}$; this supplies no uniform residue or nonzero-momentum shell. Fixed cube spectral lines have positive variance and do not identify the growing limit (`model/spin_joint_field_limit/UNIFORM_MODE.md`, [1]).

Circle projection of $\arctan(X_\theta)/\pi$ gives bounded charges on this joint family. Real even tests in $\delta|k_0|$ caps about $\pm k_0$, $0 < \delta \le 1/4$, give each charge and direction original probability $1/774400$ in energy $[9|k_0|/774400, 580800|k_0|]$. Inversion pairs the caps using the linear test's support; seven records retain nonlinear complements. A six-site periodic control has thirteen translation/energy records and positive directed variance. Narrow spatial phases give no sharp time phase (`model/spin_joint_field_limit/DIRECTED_CHARGED_BANDS.md`, [1]).

The ordered-zero-field prescription has a separate bound. The rational infrared integral and biased trial $(I + 10M)G$ certify $m > 1/10$. For a unit annular test, $x = \sqrt{r}Q_3(F)$, $y = \sqrt{r}Q_2(F)$, take $B_+$ as the first circle coefficient of $(x\cos\theta + y\sin\theta)/[1+(x\cos\theta+y\sin\theta)^2]$ and $B_- = B_+^*$. Their norms are at most $1/2$; the positive square-root complement completes the instrument. Canonical and energy moments give each charge original probability $p_* = 1/1757600$ in $r/2 \le |P| \le r$, $3p_*r \le L \le r/p_*$. Source quadratures converge on the same ordered history subsequence; no spectral pole follows (`model/spin_charge_dispersion`, [1]).

Since translated configuration fields commute, finite spatial averaging of $A = B_+^* B_+$ gives arbitrarily far rational displacements with $\|\alpha_a(B_+)B_+\Omega\|^2 \ge p_*^2/4$. This uses positivity and translation invariance without clustering. For every fixed $n$, Jensen's inequality applied to the spatial average of $A$ gives separated preparations of every charge-sign pattern, with original weight at least $p_*^n/4$ and all $3^n$ records retained. An explicit cube contains suitable profile centres; selecting them effectively also requires the source-state readout bounds. Charge conjugation gives the negative-charge pair. The source energy-form bound gives $\langle L\rangle_{\text{unnormalized}} \le 27r/4$ for each pair; an upper cutoff $54r/p_*^2$ retains at least $p_*^2/8$ original probability, possibly including zero energy. On the finite cube the pair columns span each entire 28-dimensional charge-$\pm 2$ sector, with a complete detector and connected retarded response. Asymptotic factorization, a comparison dynamics and a controlled long-time limit are additional requirements for particle scattering (`model/source_charge_scattering`, [1]).

**Local creation, positive transfer and the scattering comparison.** The actual canonical translations shift the first configuration while fixing the second. Their strong continuity, the configuration rotation symmetry and Fubini's theorem imply absolute continuity of every finite independent joint configuration law. No second canonical momentum is assumed. Nonzero polynomials in distinct bounded charge creators have strictly positive vacuum norm; their finite Gram matrices are nondegenerate. This does not supply a tensor-product particle inner product.

Cutting off an annular test in physical space gives compactly supported bounded charge creators. A direct canonical-commutator estimate retains original joint-band probability $9/200000000$ for each charge, with $r/2 \le |P| \le r$ and $ar \le L \le br$, where $d = 3m_0/1000$, $a = 3d^2/2$, $b = 2/d^2$ and $m_0 = 1/10$. Time filtering with the same source clock gives $C_+, C_-$ of norm at most $1/2$ and positive energy transfer in $[ar/2, 2br]$. Analytic Fourier norm bounds fix a scale-independent normalization $M$; the inner-band probability is at least $p = 9/(200000000M^2)$. The positive square-root complement completes the three-record instrument. Both operators raise energy and are exchanged by charge conjugation; they are not each other's adjoints.

The inherited causal cone bounds their separated commutators and time derivatives. Applying Jensen's inequality to the spatially averaged positive effect, and controlling the noncommuting product by those bounds, gives equal-sign $n$-charge preparations of original probability at least $p^n/4$. All $3^n$ records remain. The selected n-charge branch has its entire energy in $[nar/2, 2nbr]$; the complementary records need not lie in that interval. Every nonzero integer internal charge therefore has positive spectrum approaching zero. This statement does not identify each charge with an elementary massless species.

For a proposed comparison dispersion $w(k)$, the actual single-packet derivative norm squared is

$$\int |E - w(k)|^2 |\widehat{f}(k)|^2 \, d\mu_{C\Omega}(E, k).$$

It is independent of time. A fixed broad energy band alone cannot make this defect integrable on an infinite time tail. The exact product derivative estimate separates this single-packet term from the controlled inter-packet commutators. Minimizing over measurable $w$ leaves the conditional energy variance at fixed momentum. Narrowing a filter onto a zero-weight graph loses its original success probability; a preparation with positive variance can nevertheless contain a distinct sharp subbranch. Source spectral branches, comparison-space normalization and long-time estimates are additional scattering inputs (`model/spin_multipacket_comparison`, [1]).

**Spatial quantum numbers and a common matter clock.** The actual proper cubic site rotations preserve the selected source and act on its charged fields by scalar test-function pullback. The binary-cubic central two-pi element consequently acts trivially. This does not determine a continuum spin: the harmonic polynomial $x_1^4 + x_2^4 + x_3^4 - 3|x|^4/5$ is a cubic singlet in spin four. The local spin creators are even full-source fermion bilinears and commute on disjoint sites. A physical spin-statistics assignment requires their continuous spacetime representation and spectral-sector attachment.

The centered Hubbard ground has interacting local CAR histories. Finite projectors admit algebraic preparation; pressure secants certify infinite-ground scores. At $U/c = 8$ its symmetric one-site density is $\mathrm{diag}(d, 1/2 - d, 1/2 - d, d)$, $1/90000 < d < 233/1600$. Use its proper-cubic average and actual magnetic translations. The composite fields $b = 2(n_{\mathrm{opposite}} - 1/2)a$ give $L_{\mathrm{ret}} V_0 = V_0 h + (U/2) V_1$. Three local ground correlations determine

$$C(k) = \kappa h(k), \quad G(k) = [g_0 + 2g_2 \sum_j \cos(2k_j)] I, \quad R^* R(k) = G(k) - \kappa^2 h(k)^2,$$

Here $h = c \sum_j \cos k_j \Gamma_j$ and $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij} I$. The same algebra gives $R^* R \geq 18c^2/125$ on $S = \sum_j \cos^2 k_j \leq 3/250$. Finite-range moments and their uniform exponential bounds define a unique spectral measure at every momentum, weakly continuous in local Bloch charts. At $k_* = (\pi/2)^3$, magnetic and proper-cubic symmetries, inherited site inversion and particle-hole symmetry give one even scalar measure on all sixteen components. All eight transported valley probes share it. Its zero residue is $m_0/(\alpha^2 + m_0)$, with $m_0 I = R^* 1_{\{0\}}(L_\perp) R$ and $\alpha = U/2$. Write $Q_{xy} = S_x \cdot S_y + \eta_x \cdot \eta_y$ and $r = 6^{-1/2} \sum_{|y|_1 = 1} a_y$. The full-CAR stars $12 Q_{0r} \pm \sum_{j, \epsilon = \pm 1} Q_{0, 2\epsilon e_j}$ have minima $(-13 - \sqrt{577})/4$ and $(-11 - \sqrt{673})/4$. Thus $G(k) \leq \Gamma c^2 I$, $\Gamma = (31 + \sqrt{577})/4$. At $U/c = 8$, the nodal atom and every packet's ballistic fixed-window limsup are at most $\Gamma/(16 + \Gamma)$. The same source has exact local sixth and eighth moment observables, reduced by its state symmetries to 493 and 27965 Hermitian Majorana words. In units $c = 1$, write $A = 16$, $H = A + s$, $s = \|R\|^2$, $t = \|L_\perp R\|^2$ and $u = \|L_\perp^2 R\|^2$, their moments are $M_6 = A(H^2 + t)$ and $M_8 = A(H^3 + 2Ht + u)$. For $u > 0$ the atom is at most $(su - t^2)/(Hu - t^2)$. The selected-ground moment values are undetermined (`model/source_family_compatibility/NODAL_NEXT_MOMENTS.md`, [1]).

On this fixed normalized ground under $c = v/a$, positive nodal atom is equivalent to positive original weight in a bounded physical-energy window for packets with $\|k - k_*\|_1 \le aK$. The window includes zero; no dispersion follows. For $U/c \to \infty$ with $c = v/a$, original fixed-energy weight vanishes (`model/source_family_compatibility`, [1]).

With this fixed reference, $F_a(p, B) = \nu_{k_*+ap}(aB/v)$ obeys $F_a(\lambda p, \lambda B) = F_{\lambda a}(p, B)$. Set $d\mu_a = dp\,\mathrm{tr}(dF_a)/16$. A full joint limit has $\mu(\lambda A) = \lambda^3\mu(A)$. Local finiteness excludes positive-weight nonzero mass shells in that limit. Whole geometric-refinement convergence only forces discrete mass dilates. Resonances and different source trajectories require separate analysis.

The default cube has $3I_{16}/80 < R^*R < 61I_{16}/1000$, without a species assignment. On the separate weak-contact path $c = v/a$, $0 \le U = o(c^{2/7})$, eight kinetic nodes yield free Weyl fields with winding $-(-1)^{|b|}$ and opposite conjugate helicities. Period-two masses require supplied coefficients (`model/source_family_compatibility/KINETIC_NODE_MAP.md`, [1]). Complete interference instruments read Lorentzian scores with explicit errors. A native 8194-record dimer execution retains original spectral weight above 0.595 in $[0, 1]$; infinite-ground pressure certificates require a strict ground margin [6]. The cube's 256 clock records resolve profiles $f_{k,s}(x) = (-1)^{k\cdot x}/\sqrt{8}$; Table 3 gives original addition weights. Conjugation maps removal to addition at $k \oplus (1, 1, 1)$. All complements remain; profile labels and conditional means do not identify momentum or rest mass (`model/source_family_compatibility/CUBE_SPECTRUM.md`, [1]).

| $k_3$ | Original addition weight | Conditional mean $E$ | Window weight |
|---|---|---|---|
| 0 | $[0.583370, 0.584440]$ | $[0.479760, 0.482820]$ | $\ge 0.466330$ |
| 1 | $[0.415560, 0.416630]$ | $[0.522980, 0.527380]$ | $\ge 0.335250$ |

Table 3: Cube source energies $E = K - E_0$ at $c = 1/8, U = 1$. The last column bounds original weight in $[0.125, 0.935390]$, including finite clock resolution. Conditional variances are at least 0.009860 and 0.008300. Coherent services are supplied; the full tape is not executed. Phase records are not sharp energy projectors; no physical masses are assigned.

The signed action also has an invariant aligned-hole Fock sector with bands $\epsilon_\pm(k) = U/2 \pm c\sqrt{\sum_j \cos^2 k_j}$, with the actual $\pi$-flux translations and complete vacancy position detectors. A prescribed band-minimum limit gives $cP^2/(2\sqrt{3})$ and the operational position–momentum uncertainty relation. This free nonrelativistic sector does not describe the full interacting ground or identify a physical species (`model/source_family_compatibility/ALIGNED_HOLE_DISPERSION.md`, [1]).

The internal export matches fifteen source weights, including color conjugation, while eighteen comparison channels retain distinct family, defect and mass fields. Shared family blocks and neutrino flavour labels do not establish mass eigenstates. Physical assignments and mass predictions are empty (`model/standard_model_source_map`, [1]).

Particle identification requires preparation, controlled charge transport and complete energy, momentum and interaction readouts on one dynamics. The memory charge, compact texture and gauge–Higgs response have distinct carriers and actions; their separate constructions do not establish that map.

### 5.16 A quartic interaction on the complete compact band

The spatial source is the cubic operator $h = T + Mn \cdot \sigma$, $\|T\| < 2$, $M \ge 8$. Its direction is the normalization of $(x, y, z^2 - a^2)$ inside the cube of radius $a + 1$ and a constant outside, with zero amplitude at the two cores $(0, 0, \pm a)$. The four-band bound is $r = b^2/(M - 4)$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. The spectral and particle–hole identities hold for this source

by the compact-pair construction. Let $W$ contain the four localized modes and set $A = W^\dagger h W$. The isometry $E = W \otimes I_{45}$ embeds 180 annihilators into the complete finite field. The current and pairing intertwiners hold for every complete memory record, with $A$ retained in the quadratic core. Write $P = EE^\dagger$ and choose

$$V = \lambda(N_W - 90)^2, \qquad N_W = d\Gamma(P), \qquad \lambda \geq 0. \tag{12}$$

This is an explicit controller design. The particle–hole identity $C\overline{W} = WS$ ensures that the pairing has no localized/complementary cross term. All terms are even, so $H_r^{\mathrm{full}} = H_{r,W} \otimes I + I \otimes H_{r,\perp}$ on every record. The interaction is parity even and commutes with all twelve native internal currents. In the localized occupation basis it contains the quartic term $2\lambda \sum_{i<j} n_i n_j$. The pairing Hamiltonian does not conserve $N_W$, and its commutator with $V$ need not vanish. Every record preserves the same localized algebra because $V$ acts entirely on that factor.

Hold the texture fixed throughout the memory averaging interval. The complete record limit has a bound independent of exterior volume. Writing the compact energies as $\{-e_2, -e_1, e_1, e_2\}$ with $e_i \leq r$, the kinetic norm is $45(e_1 + e_2) \leq 90r$. If $g$ is the source feedback gain, the pairing bound and $0 \leq N_W \leq 180$ give

$$\|H_{r,W}\| \leq M_W = 90r + 90(1 + g) + 8100\lambda.$$

The radial two-column zero-mode construction is a separate special case with ninety annihilators, centre 45 and bound $45(1+g)+2025\lambda$. The actual memory transition has the reset decomposition $T = (1-p)R + pS$, and contracts zero-sum operator-valued memory blocks by at most $p$. At step size $t/n$, its reduced core state consequently satisfies

$$\|\rho_{W,n}(t) - e^{-it\bar{H}_W} \rho_W(0) e^{it\bar{H}_W}\|_1 \leq \frac{4M_W t}{n(1-p)} + \left(\frac{8}{1-p} + 4\right) \frac{M_W^2 t^2}{n}. \tag{13}$$

The estimate includes arbitrary initial classical–quantum memory correlations and a finite untouched reference. The first term vanishes for stationary product memory. The mean Hamiltonian contains the same memory-derived mass matrix and the interaction (12). Uniformity concerns this fixed localized algebra under its source isometries; it gives no uniform continuum estimate for all spatial observables.

Equation (13) compares the unread core. There is a separate estimate for the full native history register. For a fixed initial memory label, purify the core/reference state. Write $p_\omega$ for the field-independent history probability, $\psi_\omega = U_\omega \psi$, and $\bar{\psi} = e^{-it\bar{H}_W}\psi$. If $\delta$ is the unread trace-norm bound, then $\sum_\omega p_\omega(1 - |\langle\bar{\psi}|\psi_\omega\rangle|^2) \leq \delta/2$. The pure-state distance identity and Jensen's inequality bound the full history/core trace distance from $\sum_\omega p_\omega |\omega\rangle\langle\omega| \otimes |\bar{\psi}\rangle\langle\bar{\psi}|$ by $\sqrt{2\delta}$. Partial trace covers mixed inputs. For initial classical memory/core correlations, condition on the initial memory label and retain that tag in the ideal history law; convexity preserves the same bound. The complete final record/core rate is therefore $O(n^{-1/2})$. Opposite qubit phase records saturate this square-root relation. No uniform estimate for rare conditioned histories or intervening field measurements follows.

The interaction is defined through a spectral projector and need not be strictly onsite. For the radial special case, the resolvent estimate at texture coefficient 64 and $W_x^\dagger W_x = \rho_x I_2$ gives, for the projection $A_L$ onto $|x|_\infty < L$,

$$\|(I - A_L)E\|^2 \leq \epsilon_L = \frac{3}{2 \cdot 62^2}\, 32^{-2(L-1)}.$$

Thus $\|P - A_L P A_L\| \le 2\sqrt{\epsilon_L}$, and finite rank gives $\|N_W - d\Gamma(A_L P A_L)\| \le 360\sqrt{\epsilon_L}$. The radial interaction $\lambda(N_W - 45)^2$ is exponentially localized about the prescribed origin, without being a strictly onsite interaction.

On this radial subspace there is an exact obstruction for the onsite density alternative. On the two-particle sector let $D_g = \sum_x g_x \bigwedge^2 P_x$ and $E_2 = \bigwedge^2 E$. Direct compression gives

$$E_2^\dagger D_g (I - E_2 E_2^\dagger) D_g E_2 = \left[ \sum_x g_x^2 \rho_x^2 - \left( \sum_x g_x \rho_x^2 \right)^2 \right] I. \tag{14}$$

Writing $c = \sum_x \rho_x^2 < 1$, Cauchy–Schwarz bounds the bracket below by $(1-c)\sum_x g_x^2 \rho_x^2$. Every nonzero member of this onsite density class leaks from the source subspace on which it has support. This statement does not exclude a dressed low-energy subspace or other local interaction tensors.

**Connected dynamics and spectral shifts.** The compact mean pairing operator has 84 charged two-mode blocks and twelve unpaired neutrino modes. Select the two spatial pairs of one charged-lepton singular value $m$. Fix one occupation in each of the other 82 blocks and fill six neutral modes. Pair creation and annihilation kill these spectators, whose number is 88. Subtracting their fixed energy gives the exact reducing four-mode Hamiltonian

$$H_4 = e_1(n_1 - n_2) + e_2(n_3 - n_4) + m(a_1^\dagger a_2^\dagger + a_2 a_1 + a_3^\dagger a_4^\dagger + a_4 a_3) + \lambda(N_4 - 2)^2.$$

This is an invariant sector of the mean Hamiltonian, not its asserted vacuum or an invariant sector of every finite memory pulse. Pair parities split its sixteen levels into the four even/even values $0, 4\lambda, 2\lambda \pm 2\sqrt{\lambda^2 + m^2}$, four odd/odd values $\pm e_1 \pm e_2$, and eight mixed values $\lambda \pm e_i \pm m$. Consequently

$$E_0 = \min\{2\lambda - 2\sqrt{\lambda^2 + m^2}, -e_1 - e_2, \lambda - m - \max(e_1, e_2)\}.$$

This enumeration is a complete diagonalization by pair parity. For the unique odd/odd ground, a positive-energy orbital has two addition-plus-removal poles $e_1 + \lambda \pm m$, each of residue $1/2$. At $M = 64$, half-separation one and $\lambda = 1/128$, rational coefficient and exterior-volume enclosures place all three inherited charged-lepton slices in this ground sector:

| Quadratic coefficient $m$ | Lower pole | Upper pole |
|---:|---:|---:|
| 0.000002072030404 | 0.01171632730 | 0.01172047136 |
| 0.0004284302055 | 0.01128996912 | 0.01214682953 |
| 0.007205207119 | 0.004513192209 | 0.01892360645 |

The unit is the declared source pulse energy. These are slice poles, without a physical mass assignment or a determination of the full interacting vacuum. The compact exterior error $\delta_L$ bounds each displayed pole error by $\delta_L$ in this certified sector; no complementary mode is discarded.

The full 180-mode ground also has an explicit variational bracket. List the charged singular values, including colour multiplicities, as $m_j$, $j = 1, \dots, 21$. Each occurs in two pairing blocks at each spatial energy $e_i$, with quadratic levels $\{\pm m_j, \pm e_i\}$. The twelve neutral modes contribute $-3(e_1 + e_2)$, giving

$$E_{\mathrm{quad}} = -2 \sum_{j=1}^{21} [\max(m_j, e_1) + \max(m_j, e_2)] - 3(e_1 + e_2).$$

Positivity of $V$ makes this a lower bound. A quadratic product ground has mean occupation 90 and number variance one per even pairing block, zero per odd block. If $b$ blocks are even, its trial

energy is $E_{\mathrm{quad}} + \lambda b$. A completely blocked trial state has exact occupation 90, zero interaction energy and kinetic energy $-45(e_1 + e_2)$. Thus, retaining every localized mode,

$$E_{\mathrm{quad}} \le E_{\mathrm{full}} \le \min\{E_{\mathrm{quad}} + \lambda b, -45(e_1 + e_2)\}.$$

The specified coefficient witness has $b = 28$. These bounds constrain the full vacuum energy without identifying its excitation spectrum.

The radial control sets $e_1 = e_2 = 0$. Its ground energy is $2\lambda - 2\sqrt{\lambda^2 + m^2}$, and its single-mode spectral poles and weights are

$$\omega_\pm = 2\sqrt{\lambda^2 + m^2} - \lambda \pm m, \qquad w_\pm = \frac{1 \mp m/\sqrt{\lambda^2 + m^2}}{2}.$$

In this radial control the cross-pair density cumulant is $-\lambda/(4\sqrt{\lambda^2 + m^2})$, while its normal and anomalous cross-pair two-point contractions vanish. Hence positive $\lambda$ gives a non-Gaussian connected interaction. These spectral shifts preclude identifying the quadratic mass coefficients with the interacting excitation energies without further calculation.

**A common finite energy and modular response.** For the same full localized Hamiltonian choose $\rho_\beta = Z^{-1} e^{-\beta \bar{H}_W}$, $\beta > 0$. This state is faithful. A global spectral reset with Kraus operators $\sqrt{p_i}|i\rangle\langle j|$ prepares it and preserves total probability. That reset is a declared service, not an implementation by the original site repair. For $H(s) = \bar{H}_W + sX$, Duhamel differentiation gives

$$\rho'_\beta(0) = -\beta \int_0^1 \rho_\beta^u (X - \langle X\rangle) \rho_\beta^{1-u}\, du.$$

The positive logarithmic-mean kernel yields $F' = \langle X\rangle$, $F'' = -\beta C_{\rho_\beta}(X, X) \le 0$, and $S' = \beta(E' - \langle X\rangle)$, where $C_\rho$ is the Bogoliubov–Kubo–Mori covariance. The noncommuting choice $X = (N_4 - 2)^2$ is checked against an independent matrix-exponential derivative. These are finite identities for one nonquadratic Hamiltonian. They supply no geometric modular flow, local relativistic stress, area normalization or geometric entropy stationarity, and therefore imply no Einstein dynamics. Exact characteristic polynomials, complete finite spectra, record controls and the onsite leakage calculation accompany the proof.

**Intervening observations.** The bounds above extend to adaptive experiments. On the compact 180-mode core at $\lambda = 1/128$, every record's localized Fock generator obeys $\|H_{r,W}\| \le M_W \le 7869/32$. Between $m$ evolution intervals, apply arbitrary complete quantum instruments to the field and any auxiliary observer register, chosen from every retained native and detector record, with all failure outcomes kept. Replacing actual interval channels by ideal mean-dynamics channels one at a time bounds the complete-history distance by $\sum_j \min\{2, \sqrt{2\delta_j}\}$, where $\delta_j$ is the unread bound of interval $j$. The native memory continues under its actual transition law, since conditioning does not justify resetting it to a stationary product. Exact resource schedules and a complete finite execution check the correspondence. The bounds concern a declared temporal refinement, and adapting at every microscopic step needs its own budget.

## 5.17 Work on the moving compact interacting band

A deformation must act on the same Hamiltonian used to identify the matter states. On the compact opposite-degree texture, vary the actual hopping by $h_s = sT + MV$, with $1/2 \le s \le 3/2$,

$M \geq 8$ and $\|T\| < 2$. The texture, memory coefficients and quartic coupling are fixed. The four-dimensional spectral band has energies in $[-3, 3]$ and separation at least $M - 6$ from its complete complement. Let $P_s$ be its projector, $\mathsf{C}$ the source particle-hole matrix, and $K_\omega$ the internal pairing kernel on a complete memory record $\omega$. The same source law is

$$N_s = \mathrm{d}\Gamma(P_s \otimes I_{45}), \qquad H_{\omega,s} = \mathrm{d}\Gamma(h_s \otimes I_{45}) + \mathrm{pair}(\mathsf{C} \otimes \overline{K_\omega}) + \lambda(N_s - 90)^2.$$

Its exact force conjugate to the hopping control is

$$\partial_s H_{\omega,s} = \mathrm{d}\Gamma(T \otimes I_{45}) + \lambda\{N_s - 90, \mathrm{d}\Gamma(P_s' \otimes I_{45})\}. \tag{15}$$

The second term follows by differentiating the noncommuting square. Freezing the localization projector omits this contribution to source work.

**Proposition 5.2** (Localized force with the moving band retained)**.** *The band admits a unitary comparison frame $W_s$ in which its pairing form is constant and its intrinsic force is*

$$J_s = \mathrm{d}\Gamma((W_s^* T W_s) \otimes I_{45}), \qquad \|J_s\| \leq 360, \qquad \|P_s'\| \leq \frac{6}{M-6}.$$

*For two even localized states with the same even complement reference, their contrast of* (15) *equals their contrast of $J_s$. The bounds are independent of the containing finite cube.*

*Proof.* For band vectors $p$ and complementary eigenvectors $q$, $\langle q, P_s' p\rangle = \langle q, Tp\rangle/(E_p - E_q)$ and the diagonal blocks of $P_s'$ vanish. Hence $\|P_s'\|^2 \leq \|P_s'\|_{\mathrm{HS}}^2 \leq 8\|T\|^2/(M-6)^2 < 36/(M-6)^2$. Set $L_s = [P_s', P_s]$ and solve $U_s' = L_s U_s$. Then $P_s = U_s P_1 U_s^*$ and $W_s = U_s W_1$. Particle-hole covariance gives $\mathsf{C}\overline{U_s} = U_s \mathsf{C}$, so the pairing form is fixed. Differentiating $A_s = W_s^* h_s W_s$ gives $A_s' = W_s^*(T + [h_s, L_s])W_s = W_s^* T W_s$: $h_s$ is block diagonal and $L_s$ off diagonal. Number and pairing therefore have no derivative in this transported frame. The 180-mode second-quantization norm is at most $180\|T\| < 360$. In an even product state the cross-band terms have zero expectation; subtracting the same complement reference cancels its remaining energy. □

At fixed $s$, a complete-memory trace-norm error $\delta$ consequently gives a force-readout error at most $360\delta$. The comparison transporter is an explicit mathematical identification; Hamiltonian evolution with changing $s$ does not implement it automatically. For the supplied Gibbs reference of the same mean Hamiltonian at fixed $\beta > 0$, Duhamel differentiation gives

$$F_{\mathrm{thermal}}' = \mathrm{Tr}(\rho_s J_s), \qquad S' = \beta(E' - \mathrm{Tr}(\rho_s J_s)).$$

The complete compact source and a reducing sixteen-state interacting slice check these derivatives, retaining the spatial splitting and work performed by the hopping control. The hopping amplitude is a declared control. Its work derivative supplies no identification with a metric variation or conserved Lorentz stress, and this finite Gibbs identity supplies no regional geometric modular flow, area normalization or fixed-volume generalized-entropy stationarity.

### 5.18 A finite gauge–Higgs response with complete detector records

A separate two-vertex control joins leptonic fermion hopping, two oppositely oriented native 60-state gauge links and the charged-lepton Higgs tensor. There are three families and two spin components. The scalar at vertex zero has vacuum or one conjugate-doublet quantum; colored fermions have zero occupation and the up/down vertices are disabled. Colored link sectors are retained. The conserved quantity is $Q_* = N_F - 2N_H$, and Gauss constraints imply $N_L = N_H$,

$N_e = Q_* + N_H$ and bulk hypercharge $6Y = 6Q_*$. A reference of charge $-6Q_*$ makes the complete state neutral. The scalar cap defines this finite model, rather than an invariant truncation of an unrestricted bosonic field.

For $F$ families the complete physical dimensions are

$$d_0 = 6 + 48F^2, \qquad d_1 = 16F + 64F^3 - 20F^2.$$

At $F = 3$ these are 438 and 1596. Exact Gauss kernels, independent representation counts and the full operator images verify completeness and zero complement leakage. The charged scalar-vacuum subspace has dimension 48, with its entire 1548-dimensional occupied complement retained. Writing $A = PHP$, $C = QHP$ and $D = QHQ$ gives

$$P(z - H)^{-1}P = [z - A - \Sigma(z)]^{-1}, \qquad \Sigma(z) = C^\dagger(z - D)^{-1}C.$$

For $\operatorname{Im} z > 0$, $-\operatorname{Im}\Sigma(z) \succeq 0$. In the declared unit-strength control, $\|\Sigma(1+0.5i)\| \simeq 1.793879153064$ and the hopping/Higgs commutator is nonzero. This verifies interaction in the finite system.

Let $H_0\Omega = E_0\Omega$ be its prepared neutral ground and $b_j = c_j^\dagger d_{\mathrm{ref}}^\dagger$ the six family/spin creation maps. The odd reference makes $b_j$ even. Each is a partial isometry, and the full signed response is

$$C_{jk}(t) = \langle\Omega|b_j^\dagger e^{-i(H_1 - E_0)t}b_k|\Omega\rangle = \sum_n e^{-i(E_{1n} - E_0)t}\langle b_j\Omega|n\rangle\langle n|b_k\Omega\rangle.$$

Every spectral row and zero residue is retained. For the specified family matrices, unit dynamical strengths and scalar energy, and zero fermion/reference energies, the first three distinct numerical addition frequencies are $2.58004293348 \times 10^{-8}$, 0.001006653510723 and 0.452014700447579. Exact-input 50/80-digit residual refinement supports the tiny positive first difference; it supplies no interval-certified enclosure or ordering.

These frequencies enter a complete measurement, not a postselected creation curve. On the direct sum of the two sectors set

$$G_j = \begin{pmatrix} 0 & b_j^\dagger \\ b_j & 0 \end{pmatrix}, \qquad W_{s,j} = I - G_j^2 - isG_j, \qquad s = \pm 1.$$

Since $G_j^3 = G_j$, $W_{s,j}$ is unitary. With $U(t) = e^{-it(H_0 \oplus H_1)}$, a detector ancilla compares the forward paths $W_{+,j}U(t)$ and $U(t)W_{s,k}$. Their overlap is $F_{s,jk} = A_{jk} + sC_{jk}$, where $A_{jk} = \langle a_j|e^{-i(H_0 - E_0)t}|a_k\rangle$ and $a_j = (I - b_j^\dagger b_j)\Omega$ retains blocked creation. The two detector bases give

$$p_\pm^X = \tfrac{1}{2}(1 \pm \operatorname{Re} F_{s,jk}), \qquad p_\pm^Y = \tfrac{1}{2}(1 \pm \operatorname{Im} F_{s,jk}), \qquad C_{jk} = \tfrac{1}{2}(F_{+,jk} - F_{-,jk}).$$

The maps $(W_{+,j}U \pm UW_{s,k})/2$ and $(W_{+,j}U \mp iUW_{s,k})/2$ retain both outcomes and their back-action on arbitrary inputs, including purifications. The controlled ancilla, quarter-turn probes and ground preparation are supplied services.

For independent prepared shots at times $t_i$, compare $H_1$ with $H_1 - \delta\Pi$, where $\Pi$ is a spectral projector. The two pre-readout pure states have overlap $z = 1 + (e^{i\delta t} - 1)w/2$, $w = \|\Pi b_k\Omega\|^2 \le 1$, hence $1 - |z|^2 \le \delta^2 t^2/4$. Multiplication of fidelities and contraction under the complete measurement give the record-only bound

$$\mathrm{TV} \le \min\left\{1, \frac{|\delta|}{2}\sqrt{\sum_i t_i^2}\right\}.$$

The supplied schedule has $24 \times 4096$ shots at each $t = i/10$, $0 \le i \le 128$: 12,681,216 retained bits. For $|\delta| \le 3 \times 10^{-8}$ the bound is strictly below $1/2500$. An observer who reads the simulator

specification can have further information excluding that comparison Hamiltonian. Applying this bound to an exactly vanishing smallest line also requires an enclosure absent from the numerical precision ladder.

Finally, $\epsilon_F N_F + \epsilon_H N_H = \epsilon_F Q_* + (2\epsilon_F + \epsilon_H) N_H$. Changing $(\epsilon_F, \epsilon_H)$ by $(\delta, -2\delta)$ shifts addition frequencies if the reference energy is fixed. A common gauge-potential shift changes the compensating reference by $-\delta$ and cancels exactly in the complete Hamiltonian. This is a reference-calibration distinction, not freedom to fit intrinsic masses. The graph, scalar cap, couplings, preparation and physical clock are not selected by these finite identities. Full certificates and record instruments are in the repository package `model/interacting_extension` [1].

**Native state and carrier bounds.** The 66-dimensional occupation factor $\rho_*$ of Section 5.15 has eigenvalues 625/7505, 200/7505 and 64/7505, with multiplicities $1, 20, 45$. Its ideal modular commutator, divided by $\log(25/8)$, obeys $D(D^2 - I)(D^2 - 4I) = 0$. Any injective target-algebra map into this source that intertwines the two derivations must preserve that polynomial identity. This is a condition on an exact common clock, without a numerical exclusion of the Higgs response lines. The full $\rho_\alpha$ also carries its reference-dependent memory gap.

An isometric image of $\rho_*$ retains its 66 nonzero eigenvalues, so it has trace distance at least $1 - 625/7505 = 1376/1501$ from every pure target. A pure field subsystem is possible if other retained quantum memory carries the mixedness; its preparation and records must then be supplied. Separately, if an entire quantum channel crosses a single 66-dimensional carrier, arbitrary classical records cannot raise its entanglement fidelity above $66/D$ on a $D$-dimensional input. With no additional quantum path or entanglement assistance, its half-diamond distance from the identity is therefore at least 62/73 for $D = 438$, or 255/266 for $D = 1596$. The bound concerns all inputs and references, rather than one prepared ground-state experiment. The named observer's coherent work register is an additional quantum resource. These exact tests rule out direct identifications while leaving explicit active preparations and larger encodings to their own comparison theorems (`model/source_state_bridge`, [1]).

### 5.19 Compensated probe uncertainty and collision records

The six creation maps above also supply an exact uncertainty relation on the complete finite carrier. For any one map $b$, set

$$P_0 = b^\dagger b, \quad P_1 = bb^\dagger, \quad P = \mathrm{diag}(P_0, P_1), \quad D = \mathrm{diag}(P_0, -P_1),$$

$$X = \begin{pmatrix} 0 & b^\dagger \\ b & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -ib^\dagger \\ ib & 0 \end{pmatrix}.$$

Each normalized $b$ is a signed partial permutation of rank 384. Consequently $X^2 = Y^2 = D^2 = P$, $[X, Y] = 2iD$ and $\{X, Y\} = 0$. For either $A = X$ or $Y$, the spectral projectors $(P - A)/2$, $I - P$ and $(P + A)/2$ have ranks 384, 1266 and 384. The zero outcome retains 54 neutral blocked states and 1212 charged states outside the range. These identities are checked in exact rational complex arithmetic for all six source maps.

For any density supported on the neutral sector, including a reduced density entangled with an arbitrary reference, let $w = \mathrm{Tr}(\rho P_0) = 1 - \langle n_j \rangle$. Both means vanish, both variances equal $w$, and

$$(p_{-1}, p_0, p_{+1}) = (w/2, 1 - w, w/2), \qquad \Delta X \, \Delta Y = w = \tfrac{1}{2} |\langle [X, Y] \rangle|.$$

Thus the supplied finite density saturates Robertson's inequality [18]. The tracial neutral state gives $(32/73, 9/73, 32/73)$. The complete maps $\rho \mapsto E_a \rho E_a$ preserve the untouched reference marginal

when outcomes are summed. These are exact projective instruments, distinct from the two-outcome Ramsey readout; a native pulse compiler for them is not asserted. The density and complex measurement semantics are those of the specified finite Higgs source.

These even charge-compensated quadratures do not identify physical position and momentum. Finite matrices satisfy $\mathrm{Tr}[Q,P]=0$, hence $\|[Q,P]-i\hbar I\|\geq|\hbar|$ for nonzero $\hbar$. The source's separate fluctuation and ordered-spin constructions obtain canonical field limits on their stated states and domains; the number of independently prepared fluctuation copies is not the physical memory capacity. The ordered field's Gram form constructs a position conjugate to its existing spatial translation generator, on an explicit field-vector domain. A complete extension includes a no-position outcome. Declared coherent source services implement complete finite detectors, and finite Gram compressions give ordered convergence on fixed states and references. This gives no effective source-size modulus or moment-tail bound. A larger response space, generated by all time-evolved field vectors, admits a covariant position measure and a time-invariant dark complement. It need not be a particle sector. Protected total-load coordinates are scalar and cannot supply that position. Finite uncertainty alone selects no Planck scale or particle interpretation [6]. The proof, full instruments and source bindings are in the `model/particle_identification` package [1].

**Capacity and complete local experiments.** The native interpreter has an exact record-capacity result. Under its `export_oldest` policy, capacities one through eight give the same complete experimental channel after reversible resegmentation of archive and queue, with carrier, commands, parameters and evaluator limits fixed. Each adaptive selector reads only the last retained record; export preserves it. Induction over raw branches therefore gives zero channel distance, including arbitrary references and resource-stop outcomes. The archive is part of the output; readable capacity specifications and resource costs differ. With the `halt` policy, two holds at capacities one and two instead have disjoint timeout/completion statuses. This theorem concerns record slots, not physical global capacity or a capacity-dependent particle action; its proof and exact controls are in `model/particle_identification/NATIVE_CAPACITY.md` [1].

For the specified full-Fock Weyl/contact action at fixed spacing, nested local buffers give a stronger comparison than agreement of a few frequencies. Let $\epsilon_0$ bound the initial trace distance of the joint largest-buffer, observer and reference state. At stage $i$, let $\delta_i$ be the complete operator locality error, $\eta_i$ the unitary splitting error, and $\zeta_i$ the half-diamond error of the complete instrument, uniform over incoming states, references and previous records. For target $(t_i,c,U)$ and simulator $(s_i,c',U')$, let $E_i,V_i$ count buffer edges and sites. Then

$$a_i=E_i|t_ic-s_ic'|+V_i|t_iU-s_iU'|,\qquad D_{\mathrm{final}}\leq\min\left\{1,\epsilon_0+\sum_i(\delta_i/2+\eta_i+a_i+\zeta_i)\right\}.$$

Here $D_{\mathrm{final}}$ is trace distance on the entire retained local quantum state and classical record. Each edge has two hopping terms of norm $c/2$; each site has a contact projector. Duhamel's formula bounds the local action change by $a_i$, and complete-channel contraction telescopes all stages, including adaptive instruments and failure records. Durations are fixed by stage. The bound has no ambient-volume factor, but comparison across physical capacities requires a common local identification and controlled couplings, clock, joint input and instruments. No such capacity-to-action map follows from volume independence. Nor does convergence at fixed time imply control of arbitrarily small spectral gaps.

**Spatial scattering and physical identification.** The contact source has complete two-body lattice wave operators and a unitary scattering matrix on its absolutely continuous subspace. Crit-

ical tuning of the moving-contact family gives a nontrivial two-body continuum limit in three dimensions, with all eight edge valleys retained. This limit is nonrelativistic. Its separated bound pairs prevent identifying the empty additive Fock state as a ground vacuum. The accompanying packet construction controls incoming and outgoing time tails, packet truncation, preparation, finite volume, splitting and native readout, and separates a retained interaction event from its matched free law by more than 3/10. That fine-spacing finite schedule is constructive; the executed five-qubit control is an interaction echo, not an executed asymptotic collision. The respective packages are `model/contact_scattering`, `model/moving_contact_scattering` and `model/scattering_packet_readout` [1].

A collider identification requires one source-selected state, spatial action, clock and detector family to join these ingredients. Separated incoming packets, their interacting propagation and complete outgoing bins must share quantitative errors, including soft and failed events. The effective theory must control its relevant relativistic symmetries, regulator sectors and resolution window. An infinite ultraviolet limit is not required merely to compare a finite-resolution experiment, but a free dispersion relation cannot replace the interacting record law. Stable neutral shells, charged inclusive thresholds, resonance line shapes and scheme-dependent confined parameters then require their respective physical definitions, as described below.

# 6 Comparison with the measured spectrum

## 6.1 Observables and conventions

At finite volume the spectral measure of a prepared operator $O$ is

$$d\mu_O(\omega) = \sum_n |\langle n|O|\Omega\rangle|^2 \delta(\omega - (E_n - E_\Omega))\, d\omega.$$

A visible eigenvalue requires nonzero residue in the actual detector channel, and the physical dictionary must transport $O$, the state, the Hamiltonian, current charges and the clock together. Localized wavefunctions are not momentum eigenstates, and a Fourier transform of a pinned mode is no moving-defect dispersion experiment.

In a justified relativistic regime a stable massive excitation has a common rest-energy and propagation relation. An unstable particle requires a resonance convention and width [15]. Charged sectors coupled to a massless photon require an infrared treatment rather than an assumed isolated sharp pole [17, 20], and color confinement prevents reading a free color-mode frequency as an observable quark or gluon mass. Running charged masses and mixing data must share scale and scheme [13, 14]. The finite source spectra of Section 3 have no certified running-to-pole, confinement, or laboratory-clock map, and they enter no row of the table below.

## 6.2 The twelve comparison values

The comparison branch is specified by a boundary packet: pixel coordinate $P = 1.630971465307864$, unified coupling $\alpha_U = 0.04112426184$, external unit $E_* = 1.22089 \times 10^{19}\,\mathrm{GeV}$, vacuum expectation value $v = 246.60010\,\mathrm{GeV}$ and scale $Q_0 = 91.58958701\,\mathrm{GeV}$. Its $W$ and $Z$ couplings use one-loop supersymmetric running, its Higgs and top values use imposed ultraviolet criticality, and its quark Yukawa couplings are the six-power texture

$$Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1), \qquad Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2}).$$

No proof here selects these values from finite repair. Multiloop outputs come from the pinned Standard Model in Dimensional Regularization code [12], with two corrections to its electroweak

decoupling routine, the threshold-function operator and the placement of the strong-coupling assignment. A replay reproduces the five light quarks and the $W$, $Z$, Higgs and top to maximum relative discrepancy $5.85 \times 10^{-11}$. Table 4 gives the result against the 2026 Particle Data Group tables [16].

Table 4: The twelve nonzero comparison values, in GeV. All twelve lie outside their printed intervals. No covariance-aware joint significance or total theory-error bound is asserted.

| | Output | Reference | Residual | Convention |
|---|---|---|---|---|
| $e$ | 0.000510962911 | 0.00051099895069 | −70.53 ppm | Pole premise |
| $\mu$ | 0.105650931342 | 0.1056583755 | −70.45 ppm | Pole premise |
| $\tau$ | 1.77680479525 | 1.77693 | −70.46 ppm | Pole premise |
| $u$ | 0.00668144352 | 0.00216 | +209.33% | $\overline{\text{MS}}$, 2 GeV |
| $d$ | 0.00665065487 | 0.0047 | +41.50% | $\overline{\text{MS}}$, 2 GeV |
| $s$ | 0.239423082 | 0.0929 | +157.72% | $\overline{\text{MS}}$, 2 GeV |
| $c$ | 1.56844907 | 1.2729 | +23.22% | $\overline{\text{MS}}$ self-scale |
| $b$ | 6.59077908 | 4.186 | +57.45% | $\overline{\text{MS}}$ self-scale |
| $t$ | 170.611817 | 172.1 | −0.865% | Pole, cross-section comparator |
| $W$ | 79.8563905 | 80.3625 | −0.630% | Breit–Wigner |
| $Z$ | 90.5845168 | 91.1879 | −0.662% | Breit–Wigner |
| $H$ | 124.842549 | 125.13 | −0.230% | Pole |

For positive outputs $x_i$ and intervals $[l_i, u_i]$, one common factor is feasible exactly when $\max_i l_i/x_i \le \min_i u_i/x_i$. The intersection is empty for all twelve values, for the charged triple alone and for $W, Z$ alone. This is an exact statement about the printed rectangle.

### 6.3 Charged leptons

The lepton circulant is chosen near the measured pole-mass Koide coordinates. Its response constants $50, 31, 310, 512, 77, 21, 27, 135$, its determinant exponent and the pole reading are target-informed: with the measured electron or muon mass the law returns $m_\tau = 1776.93\,\text{MeV}$, within $2 \times 10^{-3}$ of the world-average uncertainty, and this agreement carries no evidential weight. Absolute values use the external unit $E_*$. Summing the three cosines and their squares proves $Q = \sum_j m_j/(\sum_j \sqrt{m_j})^2 = (1 + e^{-2\chi})/3$, and the packet has $Q = 0.66666446$, $\rho = \sqrt{2}e^{-\chi} = 1.41420889$ and $\delta = 2/9 + \zeta = 0.22222476$, against the same coordinates of the measured pole triple to nine digits.

Reading the outputs as pole masses is an adopted modeling premise. Under it the discrepancy is almost entirely a common scale of about −70.5 parts per million. The muon/electron ratio residual is +0.0725 parts per million and the tau/electron ratio residual is +0.0662 parts per million. A common normalization cannot remove the first, and the electron and muon intervals admit no shared factor. The exact Koide point $\rho = \sqrt{2}$, $\delta = 2/9$ gives $m_\mu/m_e$ off by +9.83 parts per million and $m_\tau/m_e$ off by +30.9 parts per million; its tau ratio lies in the printed rectangle and its muon ratio does not. Assigning the same outputs running masses at $Q_0$ and applying the retained one-loop matching relation instead shifts the pole comparisons by $(4.854, 2.870, 1.820)$ percent. Proximity to pole coordinates selects neither scheme.

### 6.4 Quarks

The declared texture has the transported light ratios $m_s/m_d \simeq 35.99993$ and $m_u/m_d \simeq 1.00463$, in the modified minimal subtraction scheme with quantum chromodynamics and electrodynamics, four active quarks and three charged leptons at 2 GeV. The rectangle of the two quoted strange and down intervals is approximately $[19.33, 20.22]$, and the directly quoted $m_u/m_d$ interval is $[0.442, 0.482]$. A common multiplier cancels in each ratio, so no energy unit can repair the light-quark shape, and this texture is rejected. Precise identification with the reference quantum-chromodynamic masses also needs an electromagnetic scheme conversion.

For a coefficient $m = A\,6^{-n}$ and ratio $r =$ output/reference, holding the transport factor fixed gives $n_{\text{required}} = n + \log_6 r$. The $u, d, s, c, b$ coordinates are approximately 6.63023, 6.19375, 4.52837, 3.11653 and 2.25334. Self-scale masses and threshold dependence require a coupled rerun for an actual changed boundary, so these are diagnostic coordinates and no selected replacement texture. Matched five-flavor running gives $\alpha_s(91.1879\,\text{GeV}) = 0.11944066$ for the stipulated branch; a reference-ratio estimate of 0.11945742 is a different operation.

### 6.5 Weak bosons, Higgs and top

The $W, Z, H, t$ residuals depend on the renormalization convention and attachment scale of the supplied vacuum expectation value. The source value agrees with the reference model's $v(172.4\,\text{GeV}) = 246.60030\,\text{GeV}$ within one part per million, while the branch attaches it at $Q_0$, where the reference value is 248.38564 GeV. Table 5 substitutes reference inputs at $Q_0$ to diagnose this sensitivity. A substitution derives no attachment, and the numerical equality supplies no scale-selection theorem. The controls omit the stored finite-light shifts of 1.7 and 13 keV.

Table 5: Percent residuals under reference substitutions at $Q_0$.

| Inputs replaced by reference values | $W$ | $Z$ | $H$ | $t$ |
|---|---|---|---|---|
| None | −0.630 | −0.662 | −0.230 | −0.865 |
| Gauge couplings | −0.717 | −0.698 | −0.241 | −0.899 |
| $v$ | +0.076 | +0.047 | +0.436 | −0.200 |
| $v$ and gauge couplings | −0.012 | +0.011 | +0.425 | −0.233 |
| $v$, $y_t$, $\lambda$ | +0.063 | +0.029 | +0.110 | +0.245 |
| All running inputs | −0.025 | −0.008 | +0.099 | +0.210 |

The weak vectors use the stated Breit–Wigner convention, and the top comparison uses the $172.1 \pm 0.6$ GeV cross-section pole extraction, distinct from a direct event-generator mass. Loop-order truncation, reference-input uncertainties and physical model error are distinct from numerical replay precision, and criticality itself is sensitive to loop order: at two loops the replayed boundary has $\lambda = 2.3 \times 10^{-8}$, and at three loops it has $\lambda = 9.7 \times 10^{-4}$.

### 6.6 Massless and neutral sectors

The photon and gluon zeros are conditional action coefficients. The minimal neutrino action gives three zero masses and zero oscillation gaps, contrary to the nonzero measured gaps, and individual absolute neutrino masses are unmeasured. The specified unshifted amplitude-to-mass attachment is excluded by Proposition 7.3. No particle abundance or early-Universe population is computed by this branch.

These are conditional comparisons, including target-informed choices. None of these constructions is a derivation of the measured spectrum. The finite gauge and matter results select no continuous mass spectrum, mixing matrices or population history. A joint physical test requires one source law, its state and complete record map, compatible mass conventions and declared experimental and theoretical uncertainties. Exact equality to measured central values is not an experimental theorem; equality of observable laws is the stronger mathematical target.

# 7 What records and symmetries exclude

Each statement below is exact on its stated domain and removes a tempting route to a spectrum. None excludes OPH as a whole; together they locate the missing selection.

## 7.1 Coherent phases and family averaging

The native event rates $p = 1/12$, $q = 1/6$ satisfy $2\sqrt{p/q} = \sqrt{2}$ and $p + q - 2pq = 2/9$, the base coordinates of the charged circulant.

**Proposition 7.1** (Classical records select no coherent phase)**.** *For a finite instrument with Kraus operators $K_{ra}$ and arbitrary real $\theta_{ra}$, replacing $K_{ra}$ by $e^{i\theta_{ra}}K_{ra}$ leaves every outcome map, every repeated composition and every record-dependent feedback experiment unchanged, on the full matrix space and with arbitrary entangled references. A well-defined function of the instrument maps therefore cannot depend on these phases.*

A coherent sum $\sum_{ra} c_{ra}K_{ra}$ thus does not follow from the instrument alone, and setting a probability equal to an angle in radians is an additional rule. An explicit same-rate counterfamily, separate from the native 128-record instrument, has identical classical instruments and different coherent spectra. A source-selected Stinespring isometry with an accessible coherent reference can carry the missing information, and such a reference is outside this exclusion.

**Proposition 7.2** (Real cyclic averaging forces a repeated eigenvalue)**.** *For a real regular three-cycle $R$ and $P = (I + R + R^2)/3$,*

$$\tfrac{1}{3}\sum_{j=0}^{2} R^j X (R^T)^j = aP + b(I - P), \qquad X = X^T \in M_3(\mathbb{R}).$$

The two-dimensional rotation plane forces the repeat, so cyclic averaging of a real symmetric kernel cannot select three distinct family values. An equivariant construction from real inputs has real outputs, and a nonreal chiral response needs an input or selection rule supplying the oriented complex structure.

## 7.2 The passive family metric

For the native core-biased reference and nonnegative seam weights, stationarity gives $\sigma^T(I-T)\sigma = (157/90000)W_{\mathrm{cross}}$, which forces all eight core-to-outer weights to vanish. Every retained branch conserves a nonzero core-charge coordinate. The harmonic extension $U$ of the native passive Schur calculation obeys $B^TU = GI$ and

$$Z = U^TU/G = I + (U - B)^T(U - B)/G \geq I.$$

The Schur mass matrix has the conserved family direction as a null vector, and the maximal 22-seam mixture attains rank two. Positive normalization is an invertible congruence and preserves

rank, so it cannot generate the third charged family coefficient. A different reference or active recovery changes the premises of this obstruction.

### 7.3 Neutral hierarchy

**Proposition 7.3** (Neutral squared-gap bound)**.** *Put $z = e^{-ih}$, $U = I+(z-1)P_2$, $V = I+(z-1)P_3$ on the bright plane of the two native rank-one family projectors, which have $\mathrm{tr}(P_2P_3) = 1/5$, and let $F = \left(\begin{smallmatrix} (1-p)U & U \\ pV & 0 \end{smallmatrix}\right)$ be the recurrent transfer. For $0 \le p \le 1/12$, $0 < h \le 1/2$, its two roots of modulus above $1/2$ have phase coefficients $\omega, 1-\omega$ with*

$$0 \le \omega \le a = \tfrac{1}{2}\left(1 - \tfrac{95}{96}\sqrt{\tfrac{653}{848}}\right).$$

*Under the common-scale attachment $(m_1, m_2, m_3) = M(0, \omega, 1-\omega)$, the adjacent squared-gap ratio is at most $a^2/(1-2a) < 0.004988$.*

The determinant of $F$ reduces to a four-dimensional calculation, with the other ten memory inputs retained as a transient block of induced norm at most $p$. Substituting $\lambda = e^{-ih/2}x$ and $y = x - p/x$ reduces the quartic to $y = (1-p)\cos(h/2) \pm i\sqrt{C_0}\sin(h/2)$, with $C_0 = (1-p)^2 + 4p/5$, and elementary bounds on the slow root give the estimate. The measured normal-ordering ratio is 0.03108, with printed rectangle $[0.03011, 0.03208]$, which excludes this attachment. Arbitrary neutral actions, offsets and other mass readouts are outside the proposition, and it fixes no absolute neutrino mass.

### 7.4 Residues, references and mass readouts

The complete instrument determines which spectral values an observer can read. Let

$$T = \begin{pmatrix} 1-p & 1 \\ p & 0 \end{pmatrix}, \qquad \pi = \frac{(1,p)}{1+p}, \qquad U_j = \exp(-ihX \otimes P_j),$$

with $0 \le p \le 1/12$. On the bright plane $B$, choose its antisymmetric area form $J$ and put $A = |+\rangle\langle -| \otimes J$, $O = A + A^*$, $P_B = I_2 \otimes B$. Then $U_j A U_j^* = e^{-ih}A$ and $O^2 = P_B$. The three projectors $(P_B + O)/2$, $(P_B - O)/2$ and $I - P_B$ give a complete readout with values $+1, -1, 0$. Starting from $\pi$ and the stationary bright, dark or tracial response state $P_B/4$, $(I - P_B)/2$ or $I_6/6$ gives, respectively,

$$C_B(n) = \cos(nh), \qquad C_D(n) = 0, \qquad C_{\mathrm{tr}}(n) = \tfrac{2}{3}\cos(nh).$$

The same transfer frequency therefore has a nonzero or a zero measured residue depending on the supplied state and detector, with the first measurement operation included.

Density records also cannot fix absolute field frequencies. Adjoin a reference line and extend each branch by $V_j(\delta) = 1 \oplus e^{-i\delta h}U_j$. All complete record probabilities and conditional density matrices confined to the original response sector are independent of $\delta$, while coherences with the added reference acquire that phase, $F_\delta = e^{-i\delta h}F_0$.

A coherent family potential need not be a scalar mass. For the supplied Dirac symbol $\alpha_i = Z \otimes \sigma_i \otimes I$, a potential $I_4 \otimes A$ has zero projection onto the spin-invariant terms anticommuting with all $\alpha_i$. For Hermitian $A, M$ and $H_h(k) = \alpha \cdot k + X \otimes I_2 \otimes M + hI_4 \otimes A$,

$$\mathrm{Tr}[(H_h(k)^2 - |k|^2 I)^2] - \mathrm{Tr}[H_h(0)^4] = 16h^2|k|^2\,\mathrm{Tr}(A^2),$$

so a nonzero potential in this truncated Hamiltonian is no collection of constant relativistic masses. Finally, cyclic trace gives $\mathrm{Tr}([K,P]H) = \mathrm{Tr}(K[P,H])$. If $[P,H] = 0$, $[K,P] = iH$ and $H$ is Hermitian, then $\mathrm{Tr}(H^2) = 0$ and $H = 0$. Exact global finite translation and boost identities therefore require a different domain from a nonzero relativistic Hamiltonian; controlled sectors and infinite limits carry their own error bounds.

### 7.5 Comparing capacities

Finite tests can compare whole record laws without enumerating histories.

**Proposition 7.4** (Sufficient criterion for equal record laws)**.** *Let $A_y, B_y$ be nonnegative outcome matrices of two finite classical instruments with column-stochastic sums, and $C$ a nonnegative column-stochastic map. If $CA_y = B_yC$ for every record $y$ and $Cp = q$, then all finite joint record laws coincide. With $\epsilon_0 = \|Cp - q\|_1$ and $\delta = \max_j \sum_{y,i} |(CA_y - B_yC)_{ij}|$, the record total-variation distance after $n$ steps is at most* $\min(1, (\epsilon_0 + n\delta)/2)$.

Induction gives the exact case, and a telescoping sum of $\ell_1$ contractions gives the approximate one. Equality of unread channels is insufficient: the one-state instruments $A_0 = 1, A_1 = 0$ and $B_0 = 0, B_1 = 1$ have the same unread channel and disjoint records. Determinism supplies neither the intertwiner nor a compatible initialization.

## 8 The catalogue and what a source-selected spectrum must supply

### 8.1 What the displayed source assigns

The common finite construction specifies three distinct charged-lepton coefficients, six quark coefficients with color multiplicities, and three zero neutrino coefficients in a 45-component internal kernel. The gauge and Higgs directions belong to the separate conditional bosonic action.

| Channel | Displayed source input | Established finite meaning |
|---|---|---|
| Charged leptons | $Y_e = \sqrt{2}\, gZ_C$ | Three memory-derived singular coefficients on the native charged-lepton slots |
| Up and down flavors | Diagonal powers of 1/6 | Six declared coefficients on native color slots; this displayed choice has identity quark mixing |
| Minimal neutrino slots | Zero | Three zero internal coefficients, without a massless relativistic particle proof |
| Gauge and scalar fields | Canonical one-doublet action, $g_i$, $\lambda_H$, $v$ | Conditional symmetry breaking and quadratic masses; no bosonic defect-state inclusion in the CAR construction |

This is a complete accounting of the displayed model's assignments, not a complete catalogue of nature. Two mathematical distinctions prevent a false physical spectrum. First, the compact four-mode band has a nonzero operator $A = W^\dagger hW$, whose energies enter the exact common operator and cannot be removed to retain the zero-mode mass list. Second, the positive quartic

coupling changes the spectral poles. The pixel feedback calculation uses quadratic coefficients, so the compact interacting operator and its exact slice readouts require a recalculated electromagnetic response before a common interacting feedback fixed point can be claimed.

### 8.2 The required attachments

**A constructed lattice sector and its refinement boundary.** For the declared staggered-field spin source, an onsite frame gives $\widetilde{H}_B = (J/4)\sum_{xy}(X_xX_y - Y_xY_y - Z_xZ_y - I) - (B/2)\sum_x Z_x$. Its normalized bond perturbation has grouped norm at most $3J/B$. The simple onsite excitation is separated from all multiparticle sums, so sufficiently small positive $J/B$ gives an isolated analytic lattice particle band [28]. Exact full-source perturbation, including ground subtraction, gives

$$E(k) = B + 3J + \frac{J^2}{2B}\left[3 - \left(\sum_{j=1}^{3}\cos k_j\right)^2\right] + O(J^3/B^2).$$

The local cluster construction gives a uniform momentum remainder for sufficiently small coupling; no numerical admission threshold or remainder constant is asserted. Opposite sublattices carry conjugate internal charges with equal energies. The complete local creator and its failure branch preserve original fermion grading, so this is no single-fermion assignment. On an eight-site cube at $J = 1, B = 400$, the product reference has Rayleigh residual squared 3 and excited-spectrum separation at least 394, hence ground half-trace error at most $1/197$. A complete 167-pulse native preparation and charge instrument retains all 608 first-failure slots, with full-output error below $1/100$ from the exact-ground instrument. This finite certificate does not evaluate the infinite-volume smallness threshold or isolate its particle-band weight. The source calculations and their complete instrument are given in `model/high_field_particle` [1].

The same ground generator has gap at least $B/8$ and a field-independent Lieb–Robinson speed at most $72J$. The isolated-shell velocity bound [29] therefore implies, at physical spacing $a$ and common positive energy conversion $s$,

$$sE_{\min} \le M_{\max} \quad \Longrightarrow \quad v_{\text{physical}} \le 72asJ \le 576a(J/B)M_{\max}.$$

For bounded $J/B$, the physical dispersion has a vanishing Lipschitz constant as $a \to 0$. A nonconstant relativistic finite-rest-energy limit with local observable weight is excluded on this clock and spatial map. This result does not exclude a fixed-cutoff measurement or the distinct ordered family with $B/J \to 0$. Its pole concentration and physical labels require their own source results.

A sufficient concentration test on that positive-field source uses a packet $\psi$ away from zero spatial momentum and the joint spectral ratio $X = K_B/[J\sqrt{2\sum_j(1-\cos P_j)}]$. Write $w = \|\psi\|^2$, $h = \langle\psi, X\psi\rangle$ and $\chi = \langle\psi, X^{-1}\psi\rangle < \infty$. For $c, \epsilon > 0$,

$$D_c = h - 2cw + c^2\chi = \int \frac{(x-c)^2}{x}\, d\nu_\psi(x) \ge 0, \qquad \nu_\psi(|X-c| \ge \epsilon) \le \frac{c+\epsilon}{\epsilon^2} D_c.$$

This needs no upper energy cutoff. A complete preparation $\psi/\Lambda$ retains the original probability factor $1/\Lambda^2$. Exact source commutators determine $h$ from static bond and order correlations, while equal-time correlations determine $w$. The remaining dynamical statistic $\chi$ has bounded joint-score enclosures using the positive-field gap. Its required saturation on the growing source is unproved; the finite certificate does not identify a physical speed or particle.

A comparison of one sector has fewer dependencies than the complete spectrum. Fix one admitted source, prepared ground, clock and accessible operator menu. For two neutral channels the vacuum-relative Euclidean correlations have the form

$$C_i(t,p) = \langle O_i(p)\Omega, e^{-t(H-E_0)}O_i(p)\Omega\rangle = \int_0^\infty e^{-tE}\, d\mu_{i,p}(E).$$

The complete source readouts determine the correlation data; a persistent isolated contribution with nonzero original weight must be established with volume and cutoff errors. If its dispersion is relativistic with common speed $v$, its rest energy $\Delta_i = m_i v^2$ gives $m_1/m_2 = \Delta_1/\Delta_2$. A common positive clock rescaling cancels from this ratio. A finite box level or a decay threshold alone supplies no stable particle identification. Matching the physical symmetry labels and detector response is a separate obligation, and target masses cannot select the source parameters. A finite-resolution comparison needs controlled errors over its measurement window; neither the full Standard Model catalogue nor Einstein dynamics is a prerequisite.

This reduction reuses the exact bound-pair poles, complete local charged responses and controlled one-dimensional massive dispersion. Their respective reference states and scaling laws cannot be exchanged. In particular, the bound-pair rest/curvature ratio differs from the relativistic value, and the one-dimensional running family admits arbitrary positive limiting mass through the same initial source point. Those exclusions restrict the interpretation; recomputing their spectra does not select physical masses.

Table 6 gives the additional attachments for the complete spectrum. Measured inputs calibrate the quantities they fix; an independent spectrum prediction requires the listed source derivation.

Table 6: Attachments between the finite stack and a physical spectrum.

| **Attachment** | **Required result** |
|---|---|
| Charged-family source law | Physical scalar Yukawa eigenvalues, phase and flavor frames selected without measured-target fitting |
| Gauge, Higgs and top parameters | Gauge content, running and matching prescription, vacuum scale and the criticality premise derived from the same source |
| Neutral masses and mixing | A nonzero-gap mechanism and complete readout outside the excluded minimal and unshifted-phase branches |
| Physical records | Matched preparation, instruments, clock and estimator, retaining all failure outcomes, with a complete error budget |
| Capacity and refinement | A full-history equivalence or controlled limit for the physical source, beyond the four-address encoding |
| Early-Universe distribution | The relevant state, finite-temperature dynamics, interactions and population observables from the same source |

These combine into one joint map,

(same source, vacuum, clocks, instruments)
$\longrightarrow$ (total defect energies, current residues, dispersion, vertices)
$\longrightarrow$ (one response kernel and its pixel fixed point).

Its first row must supply an autonomous finite-energy defect process in place of an externally fixed texture. Its second must retain all spatial regulator modes until an actual decoupling or continuum theorem controls them. The third must use the first two, including thresholds and physical normalization. These are composition requirements on one permitted design; uniqueness among all possible simulator designs is not a prerequisite.

### 8.3 Interaction, chiral continuum and responsive geometry

Two controls separate spatial propagation from electric motion of a chiral subspace. The complete free three-dimensional overlap Hamiltonian on a flat gauge fiber has $E(p) \to |p|$ and modified axial weight tending to one under refinement; all spatial momenta and internal components are retained. It is vectorlike. A flat fiber has zero Haar measure and supplies no normalizable vacuum for a dynamical gauge field. A separate four-edge compact Abelian flux ring retains all five native charges, with electric energy, matter hopping $J$ and a loop term $\kappa$. For initial flux support $|\ell| \le K_0$, cutoff $K > K_0$, total time $T$ and $r = \lceil (K - K_0 + 1)/\max(1, |q|) \rceil$, a Dyson bound for complete position records, including untouched auxiliary references, gives

$$\mathrm{TV}(\text{complete records}) \le \min \left\{ 1, 2 \sum_{n \ge r} \frac{[(2|J| + |\kappa|)T]^n}{n!} \right\}.$$

At $J = \kappa = 1$, $K = 48$, $K_0 = 0$, $T = 1/2$, this is below 0.000248607 uniformly for the five charges. It controls the stated finite-time preparation and complete observations, not an unbounded-flux ground state without a tail bound.

For the source Euclidean overlap projector $P(\theta)$ on a local admissible charge-six patch, electric differentiation must preserve its range. With $C = [P, P']$, $\nabla = \partial_\theta + C$ obeys $\nabla P = 0$; parallel transport has generator $-C$. For $VV^\dagger = P$, $V^\dagger V = I$, the compressed parent kinetic energy is

$$V^\dagger(-\partial_\theta^2)V = -(\partial_\theta + A)^2 + \Phi, \qquad A = V^\dagger V', \qquad \Phi = V'^\dagger(I - P)V' = V'^\dagger V' + A^2 \succeq 0.$$

Omitting $\Phi$ changes the parent model. A nine-point Dirichlet control keeps all 576 parent components and its 288-dimensional chiral subspace. For positive parent kinetic matrix $K$ and $H_\Lambda = K + \Lambda Q$, let $a_j$ be the eigenvalues of $PKP$. The entire complement is controlled by

$$0 \le a_j - \lambda_j(H_\Lambda) \le \frac{\|PKQ\|^2}{\Lambda - a_k}, \qquad j \le k, \quad \Lambda > a_k.$$

The differential scalar $\Phi$ is positive; the finite-grid difference from a polar-link connection-only discretization need not be. Local frame covariance supplies neither a global anomaly-compatible chiral measure nor physical axial conservation. These spatial, flux and electric constructions are separate controlled models [1].

The quartic result gives a connected interaction on a fixed localized algebra, with a bound uniform in the surrounding radial volume. Its spatial tails and fixed number of localized modes do not define an interacting local 3+1 field theory. A controlled common field limit must also establish the intended chiral measure, anomaly realization, local gauge constraints, surviving interactions and regulator census. Finite internal anomaly sums are necessary representation checks; a chiral lattice measure has additional locality and integrability requirements [21, 22, 23].

The same matter energy and stress must then enter the geometric response. The companion Einstein implication uses compatible clocks and metric, conserved stress, a geometric modular normalization and controlled fixed-proper-volume generalized-entropy variations [4]. The finite Gibbs first law proved for the interacting localized Hamiltonian is a common-operator identity, and it supplies no geometric area term, small-ball remainder, Ward/Bianchi map or universal Newton normalization.

## 9 Finite selection tests for particle readout

Several supplied field controls admit exact tests against the finite source. The following calculations retain the stated source, complete records and operator domains. They determine which coefficients

follow from those data and which require an additional selection law. No measured particle mass enters their construction. Transport to another carrier requires an identification preserving its state, currents and clock; equality of spectra alone supplies no such identification.

For the coherent three-cycle $R$, write $C = aI + rb(e^{i\theta}R + e^{-i\theta}R^2)$ and $Y = C^2/3$, where $a = \sqrt{5}/6$, $b = a/\sqrt{2}$ and $0 \le r \le 1$. We use the common nonnegative-root chamber $|\theta| \le \pi/12$. The exact source calculation gives

$$\operatorname{Tr} Y = \frac{5}{36}(1+r^2), \qquad \operatorname{Tr} Y^2 = \frac{25}{3888} + \frac{25r^2}{648} + \frac{25r^4}{2592} + \frac{25\sqrt{2}\,r^3}{1944}\cos 3\theta.$$

With a declared neutral Higgs number $h > 0$, the charged free-CAR sea energy at kinetic norm $p$ is $-2\sum_j \sqrt{p^2 + h^2 y_j^2}$, where $y_j$ are the eigenvalues of $Y$. At $p = 0$ it equals $-2h \operatorname{Tr} Y$, independent of phase. Convexity in each root proves that external energy minimization sets $r = 1$ for every positive-weighted finite momentum set. Rational bounds at $p = 1/256$, $h = 1$ prove an interior phase minimum exists; changing $p$ can reverse a phase comparison. The momentum prescription is supplied.

The actual finite quantum scalar has vacuum and two doublet states. For its gauge-invariant pair-creation vertex $W(Y)$, the conserved $N_F - 2N_H = 0$ sector has dimension $1 + 2\binom{18}{2} = 307$. The vacuum and $W(Y)$ applied to it span an invariant two-dimensional subspace; the scalar cutoff forbids another creation step. With common number energies and $d = 2\epsilon_F + \epsilon_H$, its levels are

$$E_\pm = \frac{d \pm \sqrt{d^2 + 16 \operatorname{Tr} Y^2}}{2}; \qquad 305 \text{ retained dark levels have energy } d.$$

The lower level is the whole-sector ground. Minimization selects $\theta = 0, r = 1$, whose action ratios are $1 : 1 : (34 + 24\sqrt{2})$. This sector calculation excludes spatial hopping and supplies neither the global interacting ground nor a Higgs expectation value. A dynamical phase register requires its own mixing or preparation law and common energy reference.

The finite record-budget response uses the literal reset memory, event probabilities $p_C = 1/12, p_E = 1/6$ and declared C-enabled coherent pulses. Its first-derivative coefficient $Y_n$ has trace $1/12$, rank two at one attempt and rank three at every $n \ge 2$. Exact invariant bounds give, for ordered positive eigenvalues,

$$\frac{\lambda_2(Y_n)}{\lambda_1(Y_n)} < \frac{1171058}{6633} < 177, \qquad \frac{\lambda_3(Y_n)}{\lambda_1(Y_n)} < \frac{8616}{11} < 784.$$

The complete direct archive has $24^n$ positive histories and capacity $n \log 24$ nats. Varying this budget cannot reproduce observed lepton ratios by a common linear mass conversion. These coefficients are derivatives of a noisy finite response; no interacting mass identification follows. Pixel capacity and archive capacity require a separate attachment.

An independent record-clock calculation uses all sixty seam/coin outcomes on the binary total-two carrier. Its transition $\mathsf{P}$, occupation matrix $B$ and graph Laplacian $L$ satisfy $\mathsf{P}B = B(I - L/60)$. Consequently the occupation correlations decay at exact rates $e_\lambda = -\log(1 - \lambda/60)$ for $\lambda = 5 - \sqrt{5}, 6, 5 + \sqrt{5}$. Common attempt blocking and common energy conversion cancel from their ratios. These are record-relaxation rates and reconstructed $-\log \mathsf{P}$ gaps; the charged action has no proved identification with them, and GeV units require an energy reference.

The finite band and canonical dressing give two further operator connections. Uniform sampling of the twelve incidence-defined near rotations has displacement cost $2I - \sum_g R_g/6$ on the full 36-state pair. Its strictly least positive eigenvalue is $(5 - \sqrt{5})/3$, of rank three, with both zero-cost

states retained. This identifies the declared cooling band without a dimension window; selection of that controller and its physical family attachment are extra premises.

For two CAR modes and every level of the 60-state quantum link, let $S = c_x^\dagger U c_y + c_y^\dagger U^\dagger c_x$ and $W = e^{i\pi S/4}$. The operators $b_i = W c_i W^\dagger$ satisfy exact CAR and retain all endpoint gauge transformations. Their canonical one-particle projector has rank 60, including 59 dark directions beyond the rank-one edge projector. Its sign construction satisfies the Ginsparg–Wilson identity but has 118 dark zero columns; it supplies no relativistic kinetic regulator. A local interacting chiral limit requires a selected kinetic and complement Hamiltonian, compatible quantum electric transport, uniform locality bounds and a positive physical state on the same clock.

### 9.1 Complete interacting records on the specified finite source

The finite observer construction uses that same scalar/full-CAR vertex, with $G = 4\,\mathrm{Tr}(Y^\dagger Y) > 0$ and $\Delta = \sqrt{d^2 + 4G}$. It resolves both bright eigenprojectors and the entire 305-dimensional dark projector. A phase-shifted vertex pulse, with the number term paused or refocused, prepares the neutral-sector ground by a unitary that fixes the dark inputs. Complete occupation and energy instruments retain every outcome, including outside-code outcomes when the finite sector is embedded in its larger carrier. The finite coherent-instrument compiler supplies implementation and error bounds for these declared services; execution of its full pulse tape is not claimed.

Let $J_{01}(t)$ be the original joint probability of an empty scalar record followed by an occupied record after evolution for time $t$. For any reproducible input with empty weight $a_0 > 0$,

$$J_{01}(t) = a_0 \frac{4G}{\Delta^2} \sin^2(\Delta t/2), \qquad \frac{J_{01}(2t)}{4J_{01}(t)} = \cos^2(\Delta t/2).$$

On $0 < \Delta t < \pi$ this determines the gap without ground-state preparation or renormalizing selected outcomes. The declared test coefficient $Y = \mathrm{diag}(1, 2, 3)$ and $d = 1$ gives $G = 56$, $\Delta = 15$ and levels $-7, 8, 1$. At $t = \pi/45$, the probability ratio is exactly $3/4$. These are model energies with no measured mass input.

A faithful Gibbs state of this same Hamiltonian provides a common modular clock. For the chosen $\beta = \log 2$, its per-state spectral weights are $32768/71809$, $1/71809$ and $128/71809$, the last repeated 305 times. Thus $-\log \rho = \beta H + \log Z\, I$ and $\rho^{is} A \rho^{-is} = e^{-i\beta s H} A e^{i\beta s H}$. The full field density is gauge invariant, though its support includes nonsinglet field states. Its canonical purification with a conjugate charged reference is jointly gauge invariant. This is a declared computable preparation, without an assertion that passive repair prepares a thermal vacuum.

Charged probes use the same operator $H = W + W^\dagger + dN_H + \epsilon_F Q^*$, where $Q^* = N_F - 2N_H$. The entire 36-state removal sector has energy $d - \epsilon_F$. In the 1650-state addition sector, the literal pair map $W_1$ has Gram matrix $S = GI - \gamma$, with blocks $\gamma_L = YY^\dagger$ and $\gamma_e = 2(Y^\dagger Y)^T$ in the all-left creation convention. For squared singular values $s_j$ of $Y$, its eigenvalues are $G - s_j$ with multiplicity four and $G - 2s_j$ with multiplicity two. Each produces energies

$$E_{\lambda,\pm} = \epsilon_F + \frac{d \pm \sqrt{d^2 + 4\lambda}}{2},$$

with 1614 dark states at $\epsilon_F + d$. Addition and removal operators coupled to conjugate probe references form a complete instrument by the CAR identity. Their original spectral weights are computed from the prepared neutral ground. Changing $\epsilon_F$ at fixed $d$ leaves all neutral records invariant while changing charged gaps: neutral measurements do not determine this independent coefficient. These lines have no asserted spatial dispersion or physical lepton identity.

When both addition lines of a resolved channel have nonzero original probe weight, their separation $\delta_j$ and the neutral gap give $s_j = (\Delta^2 - \delta_j^2)/(4\kappa)$, with $\kappa = 1$ for a doublet channel and $\kappa = 2$ for a singlet channel. The additive $\epsilon_F$ offset cancels, and a common clock scale cancels from the resulting coefficient ratios. Exact reconstruction gives $1 : 2 : 3$ for the declared diagonal fixture and preserves the repeated pair of the coherent zero-phase source. This is an operational measurement of source coefficients, not a charged-lepton rest-mass identification or a selection of the target-fitted phase.

On the positive-$C$ branch, the source identity $\sum_j \sqrt{3y_j} = 3a$ removes a common coefficient scale. Thus $r^2 = 3\sum_j y_j/(\sum_j \sqrt{y_j})^2 - 1$, and the two trace identities above determine $\cos 3\theta$. The unordered spectrum retains the equivalence $\theta \mapsto \pm\theta + 2\pi k/3$; at $r = 0$ the phase is unobservable. Complete spectral records therefore reconstruct the configured controller within that branch. They do not select its setting independently of a target fit.

## 9.2 The full scalar ground and its occupation records

For the declared test action $Y = \mathrm{diag}(1, 2, 3)$, $d = 1$, $\epsilon_F = 0$, a complete-carrier calculation covers all $3 \cdot 2^{18} = 786432$ inputs. Splitting the scalar-empty and occupied spaces gives $H = \left(\begin{smallmatrix} 0 & A^\dagger \\ A & I \end{smallmatrix}\right)$. Six signed three-mode cells reduce the full Gram comparison to matrices of dimension at most $3^6 = 729$. Exact positive-integer Perron witnesses and Sturm root counts give a unique largest Gram eigenvalue $\lambda$ at the largest root of

$$x^6 - 244x^5 + 23737x^4 - 1171722x^3 + 30638880x^2 - 394771968x + 1881169920.$$

The unique full ground has $E_0 = (1 - \sqrt{1 + 4\lambda})/2 \simeq -7.94630530665$ and is separated from every other state by more than 0.04690056. It has $Q^* = 6$, zero hypercharge and singlet weak-gauge and source-spin actions. A controlled preparation from a six-fermion singlet seed has certified infidelity below $2.6 \times 10^{-11}$. The closed pair action cannot reach this sector from empty $Q^* = 0$; the seed and controller work are part of the specified experiment.

For the full occupation apparatus, of ranks $2^{18}$ and $2^{19}$, put $\Delta = \sqrt{1 + 4\lambda}$, $\alpha = (1 + 1/\Delta)/2$ and $\beta_b = 1 - \alpha$. Original two-time probabilities are

$$(J_{00}, J_{01}, J_{10}, J_{11}) = (\alpha(1 - T), \alpha T, \beta_b T, \beta_b(1 - T)), \qquad T = \frac{4\lambda}{\Delta^2} \sin^2(\Delta t/2).$$

Exact intervals enclose these records at ticks $1/32$ and $1/16$. The occupation transition $\Delta \simeq 16.89261061331$ differs from the full isolation gap. These source energies are not assigned rest masses; the neutral-sector charged residues do not describe this other prepared ground. Family singular-value decomposition transports the block method to general $Y$, but the displayed root and numerical certificate belong only to the specified test action.

A complete charged calculation uses this same full ground, action and clock. Compensated CAR probes have effects $I - N_F/18$ and $N_F/18$; their sum is identity. All twelve family/multiplet/addition-or-removal channels retain their original weights and conjugate references. Exact reducing blocks, of dimensions at most 240, contain every source image of the probe. Dyadic eigenframe proposals are admitted by exact integer Gram and residual bounds, including degenerate projectors and the preparation error. First-family doublet removal and addition have lines near 0.0710 and 0.4873 in the same source-energy units, with many other retained lines. They are not one-particle mass shells.

Adding $\epsilon_F Q^*$ shifts addition/removal frequencies by $+\epsilon_F$ and $-\epsilon_F$ about this same reference, leaving their weights and conditional variances unchanged. For the first-family doublet, the

addition-minus-removal variance lies in $[-0.12243, -0.08993]$. No common offset therefore makes even those normalized spectral shapes equal. This is a statement about the specified reference, not physical CPT violation or an exclusion of every offset-dependent vacuum. The global ground must be recomputed outside its certified stability window. The two-line coefficient inverse above belongs to the separate $Q^* = 0$ preparation.

### 9.3 Gauge-coordinate cutoffs and recorded evolution

A separate declared field program uses the retained native representation on a cubic lattice and the coordinate Hilbert space $L^2(G, \mathrm{Haar})$ on each link. A bounded gauge-covariant polynomial of the Wilson kernel, full Dirac fermion occupations and positive electric Casimirs define one self-adjoint interacting Hamiltonian. Coordinate-word degree defines finite gauge-invariant truncations. If the initial word degree is $K_0$, the interaction increases degree by at most $d_c$, and its integrated norm is $X$, the complete record-law cutoff error is bounded by $2\sum_{n\geq r} X^n/n!$, with $r = \lfloor (K - K_0)/d_c \rfloor + 1$. Instruments must preserve the filtration or have their degree included in this accounting.

An executed one-cell quantum-flux/spinor control retains all flux outcomes at two times and has exact initial current six in its declared units. Gaussian-integer Taylor evolution computes unnormalized rational record weights, with an explicit operator remainder. At flux cutoff $K = 30$, the complete numerical half-$\ell^1$ error is below $1.155 \times 10^{-35}$; representation-cutoff error is bounded separately. This supplies a controlled finite interacting experiment. An interacting chiral continuum, its vacuum, physical particle dispersion and GeV matching are additional dynamical statements, independent of whether architectural choices are unique.

The same-field locality estimate uses even interactions and graded fermion commutators on the uncut coordinate field. Onsite spectral clipping treats the unbounded electric generators; its uniform bounded-observable estimate passes to the strong unitary limit. At fixed polynomial order and spacing, if the weighted local interaction sum is $L$, disjoint homogeneous operators satisfy

$$\|[\alpha_t(A_X), B_Y]_{\mathrm{gr}}\| \leq 2\|A_X\|\|B_Y\||X|\, 2^{-\operatorname{dist}(X,Y)}(e^{2L|t|} - 1).$$

Local complete instruments inherit this bound. Comparing finite boxes uses the same initial box/reference marginal and gives a volume-error modulus independent of the exterior size. The declared lowest-order control bounds a remote detector contrast at distance 32 and duration $1/64$ below $10^{-7}$. These are fixed-spacing estimates, without uniform relativistic refinement. Total-representation-degree compression is nonlocal: compressed distant shifts can have commutator norm one. Its complete-record cutoff error must therefore be added to parent locality estimates, rather than asserting exact cutoff locality.

A finite coordinate preparation uses the normalized gauge average of $\prod_e [(1 + \mathrm{Re}\,\mathrm{Tr}_{15}\, U_e/15)/2]^L$ on a periodic cubic carrier. Haar covering and positive polynomial bounds control normalization and the probability of a Wilson gap below $1/2$. The state lies in a finite representation-word code; its electric energy has a Casimir bound. A complete Householder preparation acts as identity on the entire orthogonal complement. The conservative gap planner uses large codes and gives an extremely short finite-time overlap comparison, without a uniform-volume vacuum theorem. An executed one-cell, degree-one character preparation has 1728 original electric records, including 397 zero complements, with mean electric energy $99/9100$. This smaller seed alone does not certify a useful Wilson gap. The coordinate Hilbert space is an analytic parent for these finite executable preparation and evolution codes.

A distinct large-plaquette-cost configuration admits all-time gap control. The represented plaquette admissibility theorem implies $\mathbf{1}_{\mathrm{bad}} \leq 48000S$ for $S = \sum_p (1 - \mathrm{Re}\,\mathrm{Tr}_{15}\, U_p/15)$. Energy

conservation for $H_E + K_m + \kappa S$, with $H_E \geq 0$, yields

$$\Pr_t(\text{bad}) \leq 48000 \left[ \langle S \rangle_0 + \frac{\langle H_E \rangle_0 + 2\|K_m\|}{\kappa} \right].$$

Compression preserves this positive-operator inequality on finite codes containing the preparation. The compressed effect is not a claim about the spectral gap of a compressed Wilson matrix. An explicit configuration has polynomial-versus-exact-sign complete final-record error below 0.000208 on the periodic parent at time one, with very large declared coefficient and code size. Finite-cutoff, preparation, compiler and finite-volume errors are additional. This finite energy argument does not supply a physical gauge coupling, vacuum or uniform interacting refinement.

A single prepared field experiment composes these bounds on all sixty fermion modes per cell. At spacing one and polynomial order zero, a periodic side-21 gauge preparation and filled origin induce the exact radius-five whole-box marginal, including boundary charge correlations. A finite conjugate purification prepares that marginal. With initial word degree $K_0$, cutoff $K = K_0 + 12$, the same large $\kappa$ throughout, and one fixed rational duration, the complete local occupation pointer has all 61 outcomes. The comparison retains untouched external references and traces the evolving exterior; it is not a global-state approximation. Periodic Wilson-gap control is not applied to the open-boundary Wilson matrix.

Six native hopping bonds give exactly $\|P_{59} H_K \psi\|^2 = 2880/289$ on the occupied-origin input, independent of its prepared link state. A unitary Taylor bound gives a strictly positive finite transfer probability. At the conservative duration $T \simeq 1.23896 \times 10^{-152}$, its enclosure is $[1.3779, 2.4609] \times 10^{-303}$. Parent sign, box and cutoff errors sum to less than $1.05065 \times 10^{-149}$ before additional implementation errors. This bound is too coarse to resolve that rare event in the exact-sign parent. The tiny duration and the large declared coefficient provide a finite controlled experiment, without a uniform physical observation interval, interacting chiral limit or species map. In this planner, $\kappa$ grows with the preparation's total electric energy, while the locality rate contains $192\kappa$. For side length proportional to box radius $R$, even fixed polynomial degree gives $\kappa \gtrsim R^3$; the resulting exponential-in-$\kappa T$ bound does not decay against $2^{-R}$ at fixed positive $T$. This is a limit of the estimate, not an exclusion of a relativistic phase. A local energy/admissibility or low-mode bound, with uniform interacting correlations, is needed for that refinement argument.

A distinct mixed preparation removes the global normalization cost from the local-density argument. Normalize each degree-$L$ link polynomial independently, tensor the full maximally mixed Fock density, and twirl the joint density under site gauge transformations. Gauge-invariant electric and plaquette means are unchanged by this trace-preserving twirl. A canonical conjugate reference retains the source's nonsinglet charges and makes the joint preparation invariant. This is not a projection onto a field-only singlet vacuum.

For $L = n^2$, the squared link norm is at least $(59/60)^2(1+8n)^{-450}$. An exact dyadic tail bound and a four-link plaquette estimate give volume-independent initial densities $e_E, s_0$. Translation symmetry and the same conserved action imply

$$\Pr_t(\text{rough plaquette } p) \leq 48000 \left[ s_0 + \frac{e_E + 2b}{\kappa} \right], \qquad \|K_m\| \leq bV.$$

The finite compressed roughness effects obey the same inequality, with both outcomes and the reference retained. At order 128 and unit spacing, the exact planner fixes $L = 2^{64}$ and $\kappa \simeq 2.98009 \times 10^{50}$ for every volume. Its all-time bound is $11567/17179869184 < 6.733 \times 10^{-7}$. These large declared parameters have no observed-coupling identification. They supply uniform local roughness, not a global Wilson gap or a physical mass gap.

The full Wilson decomposition also yields a local spectral estimate. On the actual $60V$ one-body space, $H_W^2 = I + P + E$ with $P \geq 0$ and $\|E\|_1 \leq 20 \sum_p \|I - R(U_p)\|_1$. For $\Pi = \mathbf{1}_{|H_W|<1/2}$ this implies $\operatorname{Tr}\Pi \leq 400\sqrt{6VS}$. Translation symmetry and Jensen's inequality give

$$\langle \operatorname{Tr}(P_X \Pi P_X)\rangle \leq 400|X|\sqrt{6\langle S\rangle/V} \leq 2\sqrt{5\eta}\,|X|$$

when the same source density obeys $\langle S\rangle/V \leq \eta/48000$. Spectral calculus then bounds the state-averaged squared local Hilbert–Schmidt error of the one-body sign approximation by $60|X|\epsilon_m^2 + (2+\epsilon_m)^2\langle \operatorname{Tr}(P_X\Pi P_X)\rangle$. Compression gives a complete two-outcome finite-code effect, without identifying it with a truncated-kernel projector. This controls local spectral weight, not low-mode spatial decay or the interacting detector response to the sign operator.

The same mixed preparation has exact compatible local marginals: tracing exterior registers leaves only the independent Haar variables incident to retained links. At fixed spacing, polynomial order and local coefficients, locality therefore makes complete local record laws Cauchy as volume grows at any fixed finite duration. A word-cutoff Dyson bound then gives finite simulators for that limit. The declared order-zero, time-one example has total volume-plus-cutoff error at most $2^{-23}$, with enormous radius and code-size bounds rather than a materialized simulation. Complete number pointers preserve the filtration; other apparatus must include its degree cost. Global canonical purifiers are not identified across volumes. Correlated untouched references require compatible joint marginals or a common local preparation operation.

This state has a nonzero local dynamical diagnostic: $\langle \dot{n}_x^2\rangle = 1440/289$ and $\partial_t^2\langle (n_x' - n_x)^2\rangle|_{t=0} = 2880/289$ for the complete two-time occupation law. These are exact native current identities, not a particle dispersion. The resulting polynomial field has a controlled fixed-spacing thermodynamic record limit. The local spectral estimate alone does not control the sign operator's dependence on the dynamical electric/link variables in a many-body Duhamel comparison. A different declared regulator supplies a direct local-interaction construction, without claiming equivalence to the discontinuous action.

### 9.4 Electric sign response and a smooth interacting action

An exact native gauge family exposes the distinction between spectral weight and electric response. For $0 < \delta < 1/2$, write $c = (1-\delta)/2$, $s = \sqrt{1-c^2}$, $\cos\theta = c$, and take weak links $U_1(u) = \exp(i(\theta+u)\sigma_1)$, $U_2(u) = \exp(i(\theta-u)\sigma_2)$, $U_3 = I$, with trivial color. On all sixty native Dirac/internal rows at $u = 0$, the squared Wilson kernel has eigenvalues and ranks

$$(\delta^2, 16), \qquad (1, 28), \qquad (3+2\delta, 16).$$

The source electric metric gives the unit tangent $X = \frac{1}{4}\partial_u$, and

$$\|X \operatorname{sign} H_W\| = \frac{1-\delta}{4\delta}, \qquad \|X \operatorname{sign} H_W\|_{\mathrm{HS}}^2 = \frac{(1-\delta)^2}{\delta^2}.$$

A near-critical packet with original weight $\delta$ can have vanishing weighted low-mode count and diverging weighted derivative cost. Each smooth packet has finite electric energy; no uniform energy bound or identification with the configured mixed preparation is asserted. This disproves an inference from low-mode weight alone. It neither excludes the exact-sign theory nor proves failure of that preparation.

A smooth alternative uses the same gauge-coordinate carrier and clock units, with declared positive smoothing $\delta$ and spacing $a$:

$$h_\delta = \frac{\beta + \sqrt{1+\delta^2}\, H_W (H_W^2 + \delta^2 I)^{-1/2}}{a}, \qquad H_\delta = H_E + d\Gamma(h_\delta) - \kappa \sum_p \frac{\operatorname{Re}\operatorname{Tr}_{15} U_p}{15}.$$

The flat-source normalization sets zero free energy at zero momentum and leading Dirac velocity one; it uses no measured mass. Since $\|H_W\| < 8$, set $D = 64+\delta^2$, $q = 64/D < 1$ and $Z = (64I - H_W^2)/D$. The binomial expansion

$$\sqrt{1+\delta^2}\, H_W (H_W^2 + \delta^2 I)^{-1/2} = \frac{\sqrt{1+\delta^2}}{\sqrt{D}} \sum_{n\geq 0} \frac{\binom{2n}{n}}{4^n} H_W Z^n$$

converges uniformly over every gauge configuration and volume. The $n$th term has range $2n+1$ and norm at most $(1+\delta)q^n$. Rooting its full second quantization at a cell costs at most the native rank sixty. Exponentially weighted interaction norms are therefore summable. The onsite unbounded electric generator is handled in the interaction picture; no derivative of the sign function is needed.

The resulting graded locality estimate controls a local Duhamel comparison between $H_\delta$ and its finite polynomials. Complete instrument histories, all occupations, and untouched references are retained. With the compatible mixed-state marginals, the same estimate gives a thermodynamic local-record limit and explicit finite-box and representation-cutoff approximations. In that volume comparison, volume-specific canonical purifiers are traced; independent untouched references or compatible references from a common local preparation may be retained. For $\delta = a = T = 1$, the declared plan allocates $2^{-20}$ to each of these three errors, giving total half-trace distance at most $3/2^{20}$. Its degree, box and cutoff are enormous certified finite resources, not a materialized simulation. Preparation, apparatus compilation and numerical errors have separate budgets. The roughness estimate is recomputed for this action rather than transferred from a different polynomial.

The free symbol has one zero corner and $h_\delta(p) = \boldsymbol{\alpha}\cdot p + a|p|^2\beta/2 + O(a^2)$ at fixed $p, \delta$. Other corners have energy at least $2/a$. This is a vectorlike free statement. The interacting physical refinement, chiral matter, particle poles and masses are not consequences of the fixed-spacing locality theorem. Sending $\delta \to 0$ is required only to reproduce the discontinuous action. The allowed $\delta \to \infty$ endpoint is the ordinary finite-range Wilson action $(\beta + H_W)/a$; it may instead be declared directly. Neither choice inherits an overlap chirality theorem.

### 9.5 Position and momentum on the referenced field

The actual periodic field has translations $T_S(b)$ moving every matter and gauge-link register. Their fermionic action is the signed exterior lift. They commute with the uniform action and coordinate-word cutoff, and normalize the local gauge group. This algebra applies to polynomials and the uniform smooth functional calculus above. On the mixed preparation's canonical purification, the admitted translation also moves the conjugate reference: $T_J(b) = T_S(b) \otimes \overline{T_S(b)}$. It preserves the joint Gauss-zero subspace; source-only translation need not do so. Character orthogonality gives complete conserved crystal-momentum projections $\Pi_k = V^{-1}\sum_b e^{-2\pi i k\cdot b/s} T_J(b)$ on $V = s^3$ cells, $s \geq 2$. The initial purification has joint crystal momentum zero. These are whole-field/reference records, without a single-particle identification. The momentum instrument is global; the local-record theorem does not assert a thermodynamic limit of these global projectors.

A complete position instrument has effects $F_x = n_x N^{-1} \mathbf{1}_{N>0}$ and $F_{\mathrm{vac}} = \mathbf{1}_{N=0}$, with square-root Kraus maps. It samples the site of a uniformly chosen occupied mode and retains vacuum. This menu is also global because $N^{-1}$ depends on the whole field; the local-record theorem instead applies to local occupation pointers. Canonical purifications are volume-specific, without an asserted cross-volume identification. For the actual maximally mixed sixty-mode-per-cell fermion marginal, define

$$A = \frac{1-2^{-60V}}{V}, \qquad C = \mathbb{E}\left[\frac{\sqrt{n_x n_y}}{N}\mathbf{1}_{N>0}\right] \quad (x \neq y).$$

At preparation time the complete sequential Born law is

$$p(x,0)=\frac{A+(V-1)C}{V},\qquad p(x,k\neq 0)=\frac{A-C}{V},\qquad p(\mathrm{vac},k)=2^{-60V}\mathbf{1}_{k=0}.$$

Here $A-C=\frac{1}{2}\mathbb{E}[(\sqrt{n_x}-\sqrt{n_y})^2/N;N>0]>0$. Position measurement thus changes conserved total-momentum records, with no selected number sector or discarded outcome. This explicit initial-state formula is not claimed for the subsequently interacting state, whose complete records use its evolved density.

Finite Fourier momentum and the position instrument's first moment obey the existing finite Robertson theorem on these actual operators. The variance of the recorded unsharp position is at least the variance of that first-moment operator. In a separate complete two-site $\mathbb{Z}_2$ control these variances are $15/64$ and $11/64$, respectively, while $\operatorname{Var}P=3\pi^2/16$ and $\langle[X,P]\rangle=i\pi/4$ for its stated phase preparation. This control illustrates the instrument algebra; it is not the native sixty-mode experiment. Finite-dimensional trace excludes $[X,P]=i\hbar I$. A universal physical uncertainty normalization and particle momentum require their own continuum identification.

### 9.6 A propagation estimate uniform in physical refinement

The physical-distance construction fixes positive smoothing and declares $0\leq\kappa(a)\leq\kappa_0/a$. The weighted interaction bound is then $W_a\leq W_*/a$. Arbitrary nonnegative onsite electric coefficients are removed in the interaction picture; their magnitude does not enter this propagation estimate. For actual supports separated by assigned distance $d=aD$, the graded commutator chain gives

$$\|[\alpha_{a,t}(A_X),B_Y]_{\mathrm{gr}}\|\leq 2\|A_X\|\|B_Y\||X|\exp\left[-\frac{\log(\lambda)d-2W_*|t|}{a}\right].$$

The weighted row sum includes every destination in $Y$, without a factor growing with the remote support. For the directly declared ordinary Wilson action, each native full-Fock bond has norm $30/a$, giving $W_*=360\lambda+48\kappa_0\lambda^2$. These constants retain all sixty native modes per cell. Neither action is identified with a chiral physical theory by this estimate.

Let $0<a\leq 1$ and $\lambda>1$ in the declared units. Let the observed supports fit inside a fixed physical cube of half-width $R$, and let $k(a)\leq Ka^{-p}$ complete local interventions occur by duration $T$. Deferred measurement and the actual cell count give a half-trace bound

$$\min\{1,Ca^{-(p+3)}e^{-g/a}\},\qquad C=2(2K+1)(2R+3)^3,\quad g=\frac{\lambda-1}{\lambda}d-2W_*T>0.$$

It retains every local record and untouched reference. The remote apparatus is private, with no record sent directly to the observer. The envelope increases with $a$ for $a\leq g/(p+3)$, so an exact dyadic planner certifies every smaller real spacing. With Wilson $\lambda=2$, $\kappa_0=1/100$, $R=1/10$, $T=1/1000$, $d=4$ and $K=2,p=0$, the bound is at most $2^{-41}$ for every $a\leq 1/128$. This is a conservative source cone, not a measured light speed. Separate complete implementation errors on both experiments add to it; choosing each at most $a$ preserves vanishing influence outside the cone. A convergent interacting state and observable family is an additional premise, not a consequence of this causal bound.

### 9.7 An interacting thermal preparation with complete flux records

The same-source thermal construction uses the unchanged charge-six control: sixteen Dirac occupations times thirteen electric fluxes, with all 208 states and $-4I\leq H\leq 22I$. Its declared

target is $\rho_b = e^{-bH}/Z$, $b = 1/64$. Put $A = b(22I - H)/2$ and $Q_M = \sum_{j=0}^{M} A^j/j!$. The positive Gaussian-rational state $\rho_M = Q_M^2/\operatorname{Tr} Q_M^2$ has a canonical purification prepared deterministically by a joint gauge- and parity-preserving Householder unitary. No success outcome is selected. For $x = 13b < M + 2$, its half-trace error to the target purification is at most

$$\epsilon_{\mathrm{prep}} = \frac{2x^{M+1}}{(M+1)!}\frac{1}{1 - x/(M+2)}.$$

At $M = 12$ this is below $3.27 \times 10^{-19}$. The source density is stationary; its purification need not be stationary under $H \otimes I$.

All 169 initial/final flux labels are retained. A rational Cayley unitary with $J$ substeps approximates the original duration $t$ with operator error at most $|t|^3 22^3/(12J^2)$. The complete original weights are $\|P_g U_{t,J} P_f Q_M\|_F^2/\operatorname{Tr} Q_M^2$. For measurements at 0 and $1/64$, $J = 4$ gives allowance $1331/6291456$, added to $\epsilon_{\mathrm{prep}}$, including subnormalized outcome states and the untouched reference. This is a compiled approximation to the same clock. The first flux measurement has its full backaction.

The ideal target supplies the finite Gibbs, thermal correlation and entropy identities for this actual $H$, with heat capacity $b^2 \operatorname{Var}(H)$. Both the target and $\rho_M$ are passive under cyclic source unitaries. Exact thermal correlation identities at $b$ are not asserted for $\rho_M$, and source passivity does not cover arbitrary operations consuming the purifying reference. The temperature and preparation service are declared; autonomous cooling and a spatial thermodynamic limit are separate constructions.

The spatial degree-zero Legendre mixed preparation has a concrete ground-state obstruction. On a periodic carrier with side at least two, fill all thirty negative-$\beta$ modes at each site while retaining its gauge density. The filled state is a gauge singlet because the native representation has determinant one. Both preparations have zero intersite coherence and identical electric and magnetic means. The onsite coefficient $25\beta/(17a)$ therefore lowers the total energy by $750V/(17a)$. The mixed state is not a finite ground; the comparison neither proves that this Slater alternative is the interacting ground nor excludes every continuum interpretation of the mixed family.

### 9.8 Gap-free low-energy preparation and signed detector records

The full spatial preparation retains sixty CAR modes per cell and a tensor product of native coordinate-word spaces, each of degree at most $\ell$ on one link. This cutoff differs from a single total-degree restriction. For $V$ cells, $D \le 2^{60V} 451^{3V\ell}$. Compressing the declared Wilson action gives an explicit finite Hamiltonian with certified spectral interval $[L, U]$. The finite Gibbs entropy identity and its log-dimension bound imply

$$\operatorname{Tr}(\rho_b H) - E_0 \le \frac{\log D}{b}, \qquad b > 0.$$

Indeed $S = \log Z + b\langle H\rangle \le \log D$ and $Z \ge e^{-bE_0}$. No positive gap or ground-energy oracle is required. The finite Lean bound reuses the existing entropy normalization and cardinality theorems.

A scaled positive Taylor polynomial in $b(U - H)/2$ prepares a canonical purification with half-trace error at most $\eta$. Its source density is stationary and has mean ground-energy excess

$$\epsilon \le \frac{\log D}{b} + (U - L)\eta.$$

Temperature and polynomial order are chosen from a finite tolerance. The joint preparation retains the conjugate gauge reference and selects no success branch. Its resources can be exponentially

large. Gauge invariance of this full-carrier density does not put it entirely in a source-only Gauss-singlet sector. Exact Wilson matrix entries and controlled smooth-action approximations have their separate compilation bounds; stationarity refers to the Hamiltonian actually used to prepare the polynomial state.

For every bounded probe $A$, stationarity and excess at most $\epsilon$ give the approximate ground inequality

$$\omega(A^*[H,A]) \geq -\|A\|^2\epsilon.$$

The positive term is $\operatorname{Tr}\rho A^*(H-E_0)A$; the other term is bounded using $[\rho,H]=0$. This finite premise supplies no convergence of states across physical spacings.

At fixed spacing and fixed per-link cutoff, the Wilson interaction has finite local factors and a volume-independent incident norm bound $J$. For a probe supported on $s_0$ cells, $\|\operatorname{ad}_H^n A\| \leq \|A\|(2J)^n \prod_{j=0}^{n-1}(s_0+3j)$, giving uniform local dynamics for small time and then all finite times by composition. Cooling growing cubes to excess $1/R$ and taking a local-state subsequence passes stationarity and the ground inequality to the quasi-local limit. Spatial averaging can give a translation-invariant ground state. This is a ground construction for the Wilson quantum-link action at fixed regulators; it selects no computable subsequence and proves no spacing or link-cutoff continuum limit.

The 208-state spectral control retains the unchanged charge-six Hamiltonian and its complete compensated CAR addition/removal instrument. An exact charge conjugation preserves $H$ and reverses the source charge, pairing the $N=1$ and $N=3$ spectra. Rational inertia comparisons bound its $N=2$ ground and the conjugate charged thresholds. These finite source energies have no physical particle or GeV assignment.

Write $p_i$ for the stationary preparation weights and $w_{ij}$ for original spectral-event weights, with $w_{ij}\geq 0$ and $\sum_j w_{ij}\leq p_i$. For every $\delta>0$,

$$\sum_{E_j-E_i\leq-\delta} w_{ij} \leq \frac{\sum_i p_i(E_i-E_0)}{\delta} \leq \frac{\epsilon}{\delta}.$$

Every downward event requires $E_i-E_0\geq\delta$. Summing this inequality proves the bound for complete contraction instruments without a channel-count factor. The finite Lean proof retains raw weights rather than conditional success probabilities. Its zero-excess case has no negative-energy events. The physical spectrum condition also requires a common spatial state limit and energy-momentum translations.

### 9.9 The common quadratic gauge characteristic

The native gauge calculation uses the full fifteen-state representation with Hermitian generators, $\operatorname{Tr}(T_aT_b)=I_a\delta_{ab}$, $I_a=2$ for weak and color generators, and $I_Y=10/3$ for $Y=q_6/6$. The inherited electric Casimir uses the inverse trace metric. With $H_E=e_0C_{\mathrm{tr}}/(2a)$ and $H_B=-\kappa_0\sum_p \operatorname{ReTr}U_p/(15a)$, the ordered plaquette expansion gives, with dimensionless momentum $k_i=ap_i$,

$$H_B^{(2)}=\frac{\kappa_0}{30a}\sum_{p,a} I_aF_{p,a}^2, \qquad \omega_g^2(k)=\frac{e_0\kappa_0}{15a^2}4\sum_i \sin^2(k_i/2).$$

The electric inverse metric cancels $I_a$ in Hamilton's equations. All twelve gauge directions share this quadratic speed. The lattice curl retains one longitudinal null and two transverse modes at nonzero momentum, with its zero-momentum flat coordinates treated separately.

The Wilson matter symbol has unit principal speed. The declared relation $e_0\kappa_0=15$ matches the gauge characteristic on the same clock; $e_0=1$, $\kappa_0=1/7$ instead gives squared gauge speed

1/105. This is an allowed architecture choice without measured-mass calibration. Rescaling the hypercharge coordinate changes electric and magnetic coefficients inversely and leaves the characteristic unchanged. A common quadratic cone does not establish Lorentz covariance of the interacting state, identify physical photon or gluon states, or derive a running coupling or chiral spectrum.

### 9.10 Spatial strain and pressure from the chosen action

The coframe response declares a constant spatial matrix $V$ with $\det V > 0$, $G = V^T V$ and $v = \det V$. On the finite-range Wilson source, the Dirac coefficient is $V^{-T}$, the Wilson Laplacian coefficient is $G^{-1}$, and electric and based-plaquette Gram coefficients are $G/v$. The magnetic subtraction retains the original flat potential and energy zero. Products are formed before coordinate-word compression. Every term is gauge invariant; the six symmetric strain derivatives retain the complete Fock space. Constant frame rotations have their explicit Dirac spin lift, without an inferred Lorentz boost.

These constant coframes retain a uniform causal estimate. The three outgoing link factors belong to their tail cell, so the positive electric quadratic form, including cross terms, remains onsite. Its self-adjoint generators therefore admit the same interaction-picture argument. Put $C = G^{-1}$ and $A = G/v$. With $\kappa(a) \le \kappa_0/a$, a full-Fock interaction majorant is

$$W(V) = 120\lambda \sum_i \max\{\sqrt{C_{ii}}, C_{ii}\} + 1080\lambda^2 \sum_{i<j} |C_{ij}| + \kappa_0\lambda^2 \left( 16 \sum_i A_{ii} + 64 \sum_{i<j} |A_{ij}| \right).$$

It reduces to the preceding Wilson bound at $V = I$. The conservative cross coefficient also covers collapsed periodic cells. The rational certificate uses $\sqrt{C_{ii}} \le \max\{1, C_{ii}\}$. On a compact family $\sigma_{\min}(V) \ge m > 0$, $\|V\|_{\mathrm{op}} \le L$, its supremum is finite. The declared coframe distance is at most $L$ times the coordinate path length, giving speed bound $2L \sup_V W(V)/\log\lambda$. The lower singular bound also controls growing physical-region support counts. Periodic carriers use quotient distance across their seams. This joins the mechanical and causal constructions; it supplies neither proper geometry nor convergence of their interacting states.

Write $H_{\mathrm{reg}}$ for the Wilson Laplacian energy. At fixed bare regulator length, isotropic strain $V(s) = e^s I$ gives

$$H(s) = e^{-s}(H_E + H_D + H_B) + e^{-2s} H_{\mathrm{reg}}, \qquad K_{\mathrm{iso}} = -H_E - H_D - H_B - 2H_{\mathrm{reg}}.$$

The Wilson term's extra coefficient is necessary. Along isolated driving, $d\langle H\rangle/dt = \langle \partial_t H\rangle$. For separately prepared Gibbs targets at fixed $b$, the same source instead gives $F' = \langle K\rangle$ and $S' = -b^2 \operatorname{Cov}(H, K)$. These different experiments are not identified by assuming thermalization during unitary evolution. The declared volume fixes the mechanical pressure $p = -\langle K_{\mathrm{iso}}\rangle/(3v)$.

The 208-state control shares exactly the thermal preparation's Hamiltonian, target, prepared state and clock. Its directional reduction sets the magnetic coefficient to zero and has $K_{\parallel} = H_E - H_D - 2H_{\mathrm{reg}}$ and $K_{\mathrm{iso}} = -H_E - H_D - 2H_{\mathrm{reg}}$. Each has norm at most 28, so effects $(I \pm K/28)/2$ and labels $\pm 28$ give a complete mean-reading instrument. They do not assert that $K$ has only two eigenvalues. The pressure error at declared $v = 1$ is at most $56\epsilon_{\mathrm{prep}}/3$. This control is an algebraic reduction with transverse classical links fixed; it is not a normalizable reducing subspace of the full three-dimensional quantum-link field.

Flat records do not determine the metric coupling: adding $\xi \log(\det V) I$ changes no flat Hamiltonian or fixed-flat record, but shifts strain by $\xi \operatorname{Tr}(S) I$ and pressure by $-\xi/v$. Declaring one complete

family fixes this response without a uniqueness theorem. Identifying its coframe and stress with the observers' proper geometry, and supplying the same-state regional modular and area estimates, are additional Einstein premises. This finite mechanical pressure is not a cosmological equation of state.

### 9.11 Finite holonomy and shared event meaning

An explicit link-register extension on the same icosahedral incidence illustrates defect creation with work accounting. Give each edge a sign $Z_e$ and each face the holonomy $B_f = \prod_{e\in\partial f} Z_e$. An edge flip toggles its two incident face signs. Complete reversible controls exchange a work quantum $2J$ for two negative faces, move one endpoint without changing the cost, and return the work upon annihilation. They preserve $H_{\mathrm{cost}} = J\sum_f (1-B_f)/2 + 2Jn$ on every input. For a region $R$, internal edges cancel and $Q_R = \prod_{f\in R} B_f = \prod_{e\in\partial R} Z_e$. Updates with even intersection with its boundary preserve $Q_R$; if $Q_R = -1$, those updates cannot make the entire region flat. This protects regional parity under the specified operations. Gauge invariance alone supplies no permanent individual defect, and a sphere has no nontrivial flat global $\mathbb{Z}_2$ holonomy. The declared link extension is distinct from the native protected-memory program and is not assigned a physical species or mass.

An explicit coupled construction makes this protected parity operationally visible. For either actual regional $Q_R$, or the separate ordered-memory $Q$ commuting with every native repair branch, declare

$$H_Q = \frac{3I - Q}{2} \otimes V, \qquad V = W(Y) + W(Y)^\dagger, \quad Y = \mathrm{diag}(1,2,3), \quad d = 0.$$

The full 307-state scalar sector and all native raw records are retained. At $t = \pi/(6\sqrt{56})$, a scalar vacuum gives occupied probabilities $1/4$ and $3/4$ in the positive and negative protected sectors. The geometric birth experiment has original joint no-pair/empty, no-pair/occupied, pair/empty and pair/occupied weights $(27, 9, 16, 48)/100$. Protected transport commutes with the coupled action. The strengths one and two are declared coefficients, not consequences of protection or a measured mass hierarchy.

Birth commutes with $H_{\mathrm{cost}}$ but not $H_Q$. Its scalar mean energy stays zero on the vacuum input while its second moment changes from $G$ to $73G/25$. A separate two-time energy experiment records controller work $0, +\sqrt{G}, -\sqrt{G}$ with weights $9/25, 8/25, 8/25$. That initial energy measurement disturbs the scalar preparation and is not the occupation experiment above. The uncoupled defect work counter does not supply the whole coupled experiment's energy balance. Neither protected memory nor regional parity is identified with scalar occupation, spacetime spin or a physical particle species.

The observer endpoint is also an explicit finite calculation. A complete four-bin instrument writes a pointer; controlled permutations copy only its orthogonal record labels. Two observers use different local codes and dictionaries to compare event meanings. For a chosen phase-analyzer experiment the original probabilities are $(1/2, 1/4, 0, 1/4)$. A deliberately swapped dictionary causes disagreement with probability $3/4$; correcting it changes no detector probability. Failure records and untouched quantum references are retained. Compatible event functions on a finite cover glue to a common function, and deterministic interpretation cannot increase total variation between complete event laws. These are simulation-generated observables. Smooth spacetime and field equations are effective mathematical descriptions with their own domains and approximation bounds; the program generates no additional objective continuum state. Neither agreement nor finite gluing proves the interacting relativistic limit.

### 9.12 Energy conservation and the geometric premises

The actual action response uses a Hermitian partition $H = \sum_x h_x$ and currents $J_{xy} = i[h_x, h_y]$. Then $\partial_t h_x(t) = -\sum_y J_{xy}(t)$, with exact cancellation of internal regional pairs. Integrated currents supply the finite coarse-energy theorem's premise. Time-dependent coefficients $f_x(t)$ in $H_f = \sum_x f_x h_x$ give work $\partial_t \langle H_f \rangle = \sum_x \dot{f}_x \langle h_x \rangle$. The full quantum-flux control has electric, onsite and hopping energy means $(9, 2, 0)$ and initial rates $(18, 0, -18)$.

On the scalar source with $d = 1$, the preparation $(\Omega + ib)/\sqrt{2}$ has exact component exchange and a complete current detector with ranks $(1, 1, 305)$. Changing $d$ at fixed state and clock gives $\partial_d p_{\text{occupied}}(\pi/30) = 2\sqrt{14}/3375$. This is response to an action coefficient; geometric stress requires an independently specified metric coupling. The regional Einstein consumer additionally requires actual curvature, proper clocks, the same-state modular diamond kernel, fixed-proper-volume area response and controlled small-ball errors. Energy conservation alone supplies neither the missing stress components nor these geometric identities. The existing matched-regional theorem needs only ten future timelike probes, with sharp decoder norm $17/8$. For a common proper radius $\ell$, it bounds the component Einstein residual by $255(e_{\text{bulk}} + \eta e_{\text{area}} + e_{\text{stationarity}})/(32\pi\eta\ell^4)$. Its tensor inversion is reused; the task is to realize the three regional estimates on one geometric state, with errors $o(\ell^4)$ and one positive spacetime-independent limiting $\eta$. The separate free-Weyl modular and stress construction cannot supply those inputs for an interacting state without an identified map.

The actual regional restriction of the faithful $Q^* = 0$ scalar Gibbs state supplies a distinct finite modular calculation. One family's six modes have a 22-dimensional positive support, with spectral ranks $(1, 6, 2, 13)$ and 42 retained zero outcomes. Its modular Hamiltonian includes $\log(p_{\text{dark}}/p_{\text{pair}})B_f$, where $B_f$ projects onto two pair states. For the first family the coefficient is $\log(13440/41873)$. A gauge-invariant pair/dark probe detects this induced interaction; no polynomial of local number reproduces it. The source-admitted tangent $B_f/2 - D_f/13$ has that entropy slope and an explicit relative-entropy remainder bound. The tracial regional conditional expectation exists but cannot preserve the interacting Gibbs state because it removes nonzero cross-center coherence. These are internal regions. Their modular operators are not a spatial diamond kernel, and global modular time does not restrict to their algebras as the same Hamiltonian evolution.

## Reproduction and mathematical scope

Every statement in this paper has a public executable certificate, proof text or formal module in the project repository [1], including exact characteristic polynomials, rational and integer enclosures, complete line tables, adversarial controls and the source-pinned multiloop replay. They distinguish exact algebra and rational interval proofs from floating spectral checks and empirical comparisons. A source hash identifies a calculation; its mathematical scope is the stated theorem and domain.

## References


[1] B. Müller et al., *Observer Patch Holography: formal proofs and executable receipts*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/main/model

[2] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, and D. A. Friedman, *Observers Are All You Need*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/observers_are_all_you_need.pdf

[3] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/reality_as_consensus_protocol.pdf

[4] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[5] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[6] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Memory and Observable Fields*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/executable_observer_construction.pdf

[7] Y. Koide, *New view of quark and lepton mass hierarchy*, Physical Review D **28**, 252 (1983).

[8] C. A. Brannen, *The Lepton Masses*, preprint, 2006. http://brannenworks.com/MASSES2.pdf

[9] H. Georgi and S. L. Glashow, *Unity of all elementary-particle forces*, Physical Review Letters **32**, 438–441 (1974).

[10] E. H. Lieb, *Two theorems on the Hubbard model*, Physical Review Letters **62**, 1201–1204 (1989).

[11] C. N. Yang, *$\eta$ pairing and off-diagonal long-range order in a Hubbard model*, Physical Review Letters **63**, 2144–2147 (1989).

[12] S. P. Martin and D. G. Robertson, *Standard Model parameters in the tadpole-free pure* $\overline{\text{MS}}$ *scheme*, Physical Review D **100**, 073004 (2019), arXiv:1907.02500.

[13] S. Antusch, K. Hinze, and S. Saad, *Updated Running Quark and Lepton Parameters at Various Scales*, arXiv:2510.01312, 2025. https://arxiv.org/abs/2510.01312

[14] Y. Aoki et al. (Flavour Lattice Averaging Group), *FLAG Review 2024*, arXiv:2411.04268, 2024, Tables 11–12. https://arxiv.org/abs/2411.04268

[15] Particle Data Group, *Resonances*, Review of Particle Physics, 2025 update. https://pdg.lbl.gov/2025/reviews/rpp2025-rev-resonances.pdf

[16] Particle Data Group, *Review of Particle Physics: Summary Tables*, 2026. https://pdg.lbl.gov/2026/tables/rpp2026-sum-leptons.pdf; https://pdg.lbl.gov/2026/tables/rpp2026-sum-quarks.pdf; https://pdg.lbl.gov/2026/tables/rpp2026-sum-gauge-higgs-bosons.pdf.

[17] D. Buchholz, *Gauss' law and the infraparticle problem*, Physics Letters B **174**, 331–334 (1986).

[18] H. P. Robertson, *The Uncertainty Principle*, Physical Review **34**, 163–164 (1929). https://doi.org/10.1103/PhysRev.34.163

[19] D. Petz, *Sufficient subalgebras and the relative entropy of states of a von Neumann algebra*, Communications in Mathematical Physics **105**, 123–131 (1986). https://doi.org/10.1007/BF01212345

[20] P. Duch and W. Dybalski, *Infrared problem in quantum electrodynamics*, arXiv:2307.06114.

[21] M. Lüscher, *Abelian chiral gauge theories on the lattice with exact gauge invariance*, Nuclear Physics B **549**, 295–334 (1999), arXiv:hep-lat/9811032.

[22] M. Lüscher, *Lattice regularization of chiral gauge theories to all orders of perturbation theory*, Journal of High Energy Physics **06** (2000) 028, arXiv:hep-lat/0006014.

[23] D. Kadoh and Y. Kikukawa, *A simple construction of fermion measure term in $U(1)$ chiral lattice gauge theories with exact gauge invariance*, Journal of High Energy Physics **02** (2008) 063, arXiv:0709.3658.

[24] Jarlskog, C. (1985). "Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation." *Physical Review Letters* **55**, 1039–1042. https://doi.org/10.1103/PhysRevLett.55.1039

[25] Peskin, M. E., and Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley. https://doi.org/10.1201/9780429503559

[26] Weinberg, S. (1967). "A model of leptons." *Physical Review Letters* **19**, 1264–1266. https://doi.org/10.1103/PhysRevLett.19.1264

[27] Y. Filmus, *Orthogonal basis for functions over a slice of the Boolean hypercube*, Electronic Journal of Combinatorics **23**(1), P1.23 (2016). https://doi.org/10.37236/4567

[28] D. A. Yarotsky, *Quasi-particles in weak perturbations of non-interacting quantum lattice systems*, arXiv:math-ph/0411042v1 (2004). https://arxiv.org/abs/math-ph/0411042v1

[29] S. Bachmann, W. Dybalski and P. Naaijkens, *Lieb–Robinson bounds, Arveson spectrum and Haag–Ruelle scattering theory for gapped quantum spin systems*, arXiv:1412.2970v2 (2015). https://arxiv.org/abs/1412.2970v2

[30] B. Müller et al., *OPH basis and conditional theorem-premise registry*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/claims/axiom_registry.yaml

## AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.